19 Draw a labelled diagram to show the structure of a simple type of hot-cathode tube, designed for sending a narrow beam of electrons on to a fluorescent screen. Show the electrical supplies to the tube, and explain how they act to give the electron beam. State any one way in which the beam could, if initially horizontal be deflected upwards.

(SC/GCE)

20 Draw a labelled diagram of a cathode ray oscilloscope showing the essential parts. What are the functions of

(i) the hot filament
(ii) the anode
(iii) fluorescent screen
(iv) Deflector plates.

State one way in which cathode rays differ from electromagnetic waves.

21 (a) Explain how neon signs are produced.
(b) State two factors on which the colour of light from a fluorescent tube depends.

(SSCE)

22 (a) Explain electrolysis.
(b) Classify the following substances as electrolytes and non-electrolytes; Sugar solution, paraffin, salt solution and grape fruit juice.

(SSCE)

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6 MAGNETIC FIELD

The concept of magnetic field was introduced in Book 1 Chapter 6. There we defined

The **magnetic field** as a region around a magnet in which the influence of the magnet can be felt or detected

In that chapter we mapped out some field patterns using a compass needle and iron fillings. We also discussed magnetic lines of force. The student should quickly review that chapter before getting into this.

At the end of this chapter the student should be able to:

1 Plot the magnetic field around
   (i) a bar magnet
   (ii) a straight conductor carrying current
   (iii) a solenoid
2 Make a magnet from a soft iron bar
3 Make an electromagnet
4 Describe the working principles of an electric bell and a telephone earpiece
5 Locate the earth's magnetic north-south direction
6 Explain the magnetic force on a moving charge
7 State the relation between magnetic force and the motion of a charge in a magnetic field.

6.1 Concept of Magnetic Field

The area around a magnet in which it can attract or repel objects or in which a magnetic force can be detected is called the magnetic field of the magnet.

Although this field cannot be seen, its presence can be demonstrated and mapped out using a compass needle. Such a needle when placed near the magnet will swing around and settle in a definite direction showing that a sort of force is present. Magnetic field is therefore an example of a *force field* and the force is called a *magnetic force*. *Magnetic field* is a *vector field*, i.e., such a field has both magnitude and direction. The compass needle is a sensitive instrument for finding the direction of the magnetic field. The direction of the field or a line of magnetic force at any point is taken as the direction in which a N-pole placed at that point will tend to move.

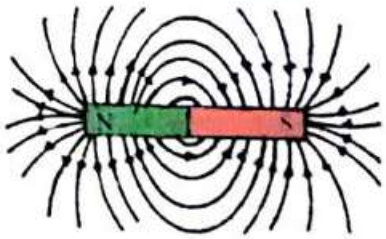
In Book 1, Chapter 6, we demonstrated the use of a compass needle and iron filings in mapping out the pattern of the magnetic field. In that chapter, we also discussed the concept of *Magnetic Lines of Force*.

Magnetic Lines of Force are imaginary lines along which a free-North pole would tend to move if placed in the field. A line of force may also be considered as a line such that the tangent to it at any point gives the direction of the field at that point.

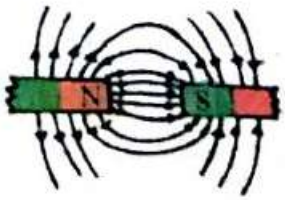
These magnetic lines of force do not really exist, hence we defined them above as *imaginary lines*. They are but useful devices or concepts we use to describe different magnetic phenomena.

6.2 Patterns of Magnetic Field

Magnetic field patterns can conveniently be observed using iron filings. The magnet is placed on paper and the iron filings are sprinkled lightly on the paper around the magnet. The paper is now tapped gently and the iron filings will be seen to turn and settle in definite directions. The pattern of the magnetic field and for other magnetic arrangements are as depicted in fig. 6.1



a. Bar magnet



b. Unlike poles

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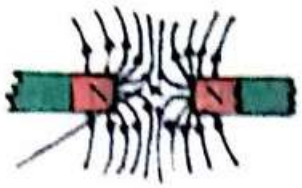


Fig. 6.1 Patterns of magnetic field

In fig 6.1c, there are no field lines, around P between the two N-poles facing each other. Thus there is no magnetic force at P which is known as a *neutral point*.

Magnetic field around a straight conductor carrying current.

Activity 6.1

1 Pass a thick insulated copper wire vertically through a hole in a cardboard sheet as shown in fig. 6.2. Sprinkle some iron filings uniformly on the cardboard around the vertical wire. Connect the ends of the wire to a battery. Switch on the current and tap the cardboard gently. Notice how the filings arrange themselves.

2 Switch off the current and place some compass needles around the wire. Note the directions to which the compass needles point. Switch on the current and note the swing of the needles and how they now point.

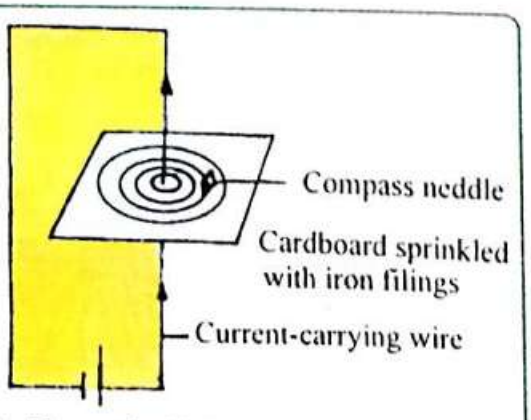


Fig. 6.2 Magnetic field around a straight current-carrying wire

It will be noticed that when current is switched on and the cardboard is gently tapped, the filings arrange themselves in a series of concentric circles about the wire as centre. Also as soon as the current is switched on, the needles will swing around and form a circle with the wire as centre. The field pattern as shown in fig. 6.2 consists of concentric circles with the wire as centre. The direction of the field depends on the direction of flow of the current. Such a direction can always be obtained by applying the *Right Hand Grip rule* or *Clenched Fist rule*.

If the straight wire is grasped with the right hand so that the thumb points in the direction of the current, then the direction in which the fingers are curled, indicate the direction of the magnetic field (fig. 6.3).

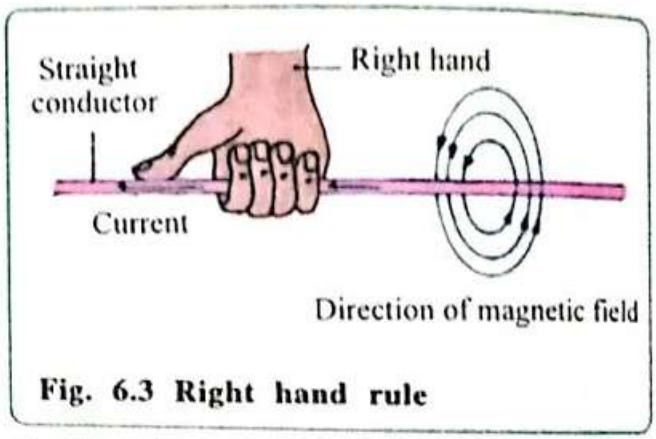


Fig. 6.3 Right hand rule

An alternative way of finding the field line direction is to use the *Maxwell's Corkscrew Rule*:

If a right-handed corkscrew is turned so that its tip travels along the direction of current, the direction of rotation of the corkscrew gives the direction of the magnetic field or magnetic lines of force

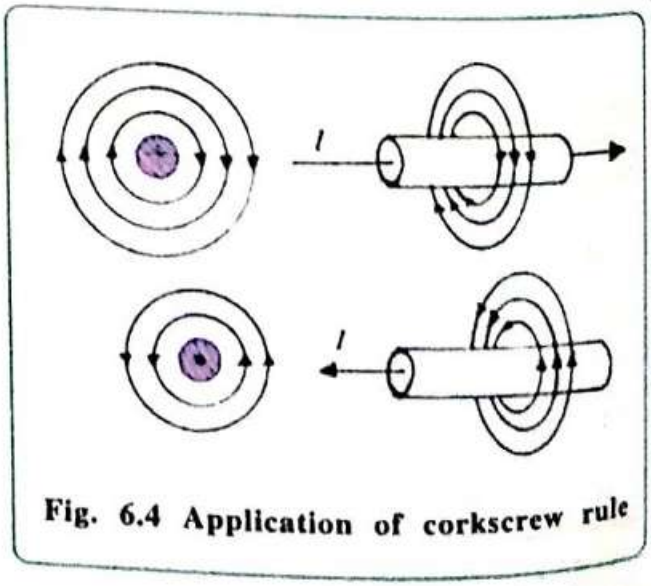


Fig. 6.4 Application of corkscrew rule

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In fig. 6.4a the cross (X) represents a current carrying conductor directed perpendicularly down the plane of the paper. In fig. 6.4b the dot ( ) represents a current carrying conductor directed perpendicularly upwards, out of the plane of the paper. The directions of the magnetic line of force are as shown.

Magnetic Field around a Solenoid

A solenoid is a long cylindrical coil of wire whose turns are usually wound close together as shown in fig 6.5a.

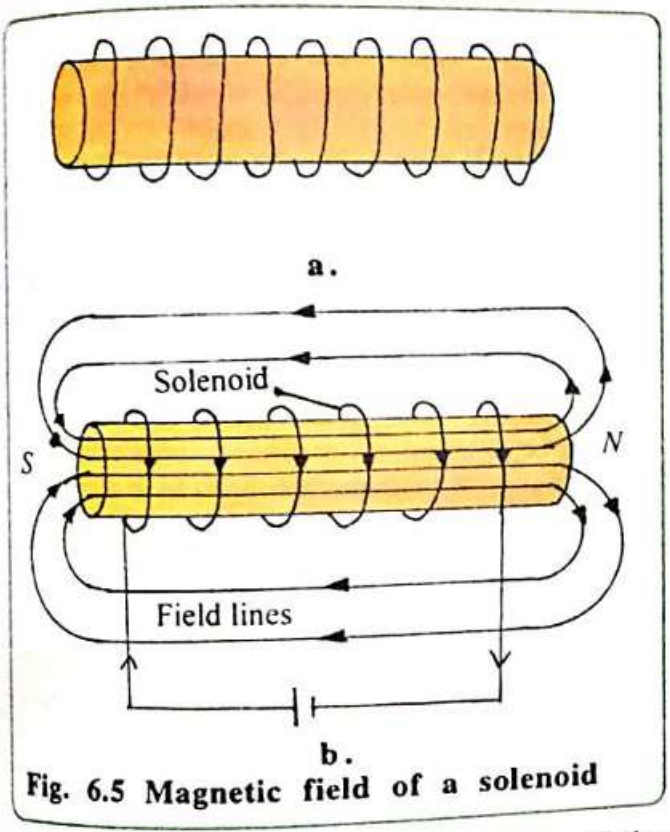


Fig. 6.5 Magnetic field of a solenoid

When current is passed along the turns of the solenoid, there is a magnetic field inside and around the solenoid as shown in fig. 6.5b. The field round the solenoid is of a similar pattern to that round a bar magnet. The lines of force outside the coil are continuations of those inside it. These lines of force emerge from one end N of the solenoid and return via the other end S. As can be observed with a compass needle, the end from which the lines of force emerge behaves like a N-pole of a bar magnet, while the end by which they return behaves like a S-pole.

Magnetic and non-magnetic substances

Substances that have magnetic properties, i.e., they can attract or be attracted by even weak magnets, are known as *magnetic substances*. Magnetic materials are materials that are capable of being magnetized. Examples are Iron, Steel, Cobalt, Nickel, Manganese. These are also known as ferromagnetic substances.

Other substances such as copper, brass, wood, glass, are not attracted by magnets. They have very feeble magnetic properties. Such substances are known as non-magnetic substances.

Magnets made from soft-iron lose their magnetic properties easily. They are therefore known as temporary magnets. Other magnets made from steel do not lose their magnetism easily. These are known as permanent magnets.

To distinguish between a magnet, a magnetic substance and a non-magnetic substance, we first suspend each specimen from its centre by means of a thread. The one that settles in a North-South direction is the magnet. We can then use this magnet to distinguish between the magnetic substance and the non-magnetic substance, by bringing one end of the known magnet near both ends of each of the two specimens. The specimen which is not affected by the magnet is the non-magnetic substance. The specimen which exhibits attraction at both ends is the magnetic substance.

6.3 Methods of Making Magnets

1 Electrical Method

The best and quickest method of making magnets, which is also the industrial method, is by the electrical method. Here we utilise the magnetic effect of an electric current. The material to be magnetized is inserted inside a solenoid, the ends of the coils of which are connected to a circuit as shown in fig. 6.6.

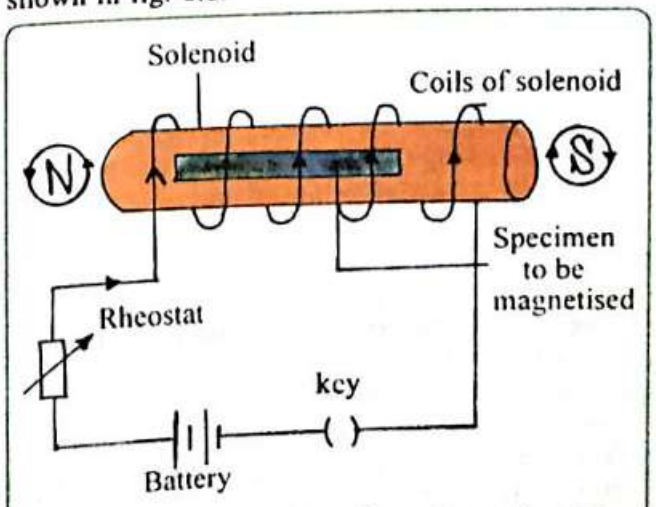


Fig. 6.6 Magnetization by electric method

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The current is switched on for a few seconds and then switched off. On testing, the specimen will be found to be magnetized. The polarity of the magnet so formed, depends on the direction of flow of the current in the solenoid. If the current flows anticlockwise in the coil when one end is viewed, then the specimen at that end has a North pole. If the current flows clockwise the end of the specimen is South-pole.

2 By Contact Method

a. The Single Touch Method

The specimen (e.g. a soft-iron bar) is placed on the bench. The needle is then stroked along its length with one end of a bar magnet. The magnet is kept in an inclined position as shown in fig. 6.7 and dragged along the specimen from one end, *A*, to the other, *B*, and lifted well away from the specimen and brought again to A. The process is repeated several times.

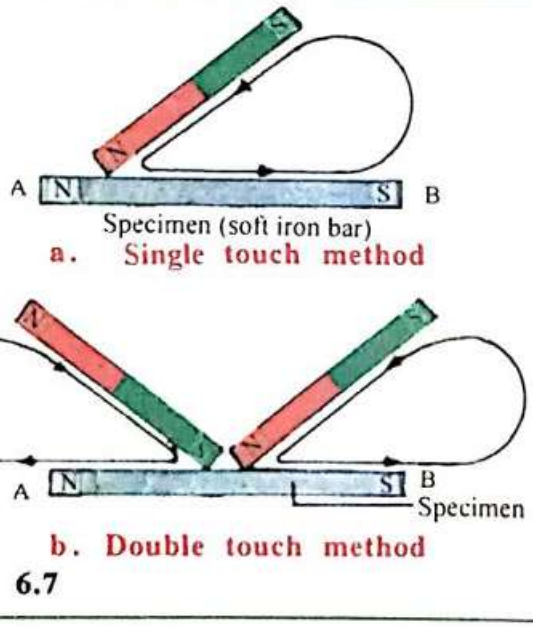


Fig. 6.7

At the end of the stroking process, it will be observed that the end *A* will have the same polarity as that of the stroking pole, and the end *B* will have an opposite polarity as that of the stroking pole. For example, if the stroking pole is N-pole as in fig. 6.7a, the end *A* is N-pole and the end *B* is the S-pole.

b. Divided Touch

The process is the same as in (a) but in this case each half of the specimen is stroked repeatedly in opposite direction by the opposite poles S and N of two bar magnets. The stroking is done simultaneously outwards from the centre of the specimen, not from its ends (fig. 6.7b).

At the end of the repeated stroking, each end of the specimen will be found to have an opposite polarity to that of the stroking pole. Thus in fig. 6.7b) end *A* will be a N-pole and end *B* will be a S-pole.

The divided touch method produces a stronger magnet in a shorter period than the single touch method.

Activity 6.2

Obtain a soft iron bar, place it on your laboratory desk and perform the single method of magnetization using the south pole of a bar magnet. Test the polarity of each end of the needle.

Repeat the double touch method of magnetization on another soft iron bar using similar poles of two bar magnets to stroke from the middle of the specimen and outwards to each end. Test whether the needle has become a magnet by suspending it from a thread and noting whether or not it settles in a North-South direction. Test also the polarity of each end of the bar.

Demagnetization

This is a process whereby a magnet is made to lose its magnetism. Demagnetization can be achieved by:

(i) Electrical method

The magnet is placed in a solenoid through which an alternating current (a.c) is flowing. The solenoid is placed with its axis pointing in an East-West direction. After a few seconds, the magnet is slowly withdrawn out of the solenoid to a long distance away. The rapid reversal in direction of the alternating current causes the material to reverse in polarity several times in a second. This process demagnetizes the magnet.

Electrical method is the most efficient way of demagnetizing a magnet.

(ii) Mechanical method

Another method of demagnetizing magnets is to hammer them hard when they are pointing in an East-West direction.

(iii) Heating method

When magnets are strongly heated, they lose their magnetism.

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Magnetic Properties of Iron and Steel

Soft iron is pure iron while steel is an alloy of iron and carbon. Steel is a much harder and stronger material than soft iron. Steel and iron have two different magnetic properties.

1 Iron is more easily magnetized than steel but it also loses its magnetism more quickly than steel.

2 Iron produces a stronger magnet than steel. Under the same magnetizing force, soft iron is magnetized to a greater degree than steel.

Because of the above differences (a) Steel is used for making permanent magnets such as compass needles (b) Iron is used for making temporary magnets such as electromagnets where strong magnetism is required only for a short time. Hence the core of an electromagnet is made of soft iron. (c) Iron nails are often used for laboratory experiments on magnetization and demagnetization because they are very easily magnetized and demagnetized.

Magnetic Screening or Shielding

Because iron is very easily magnetized, it greatly affects the magnetic field lines in its neighbourhood. Magnetic lines of force therefore tend to crowd into a piece of soft-iron ring as shown in fig. 6.8.

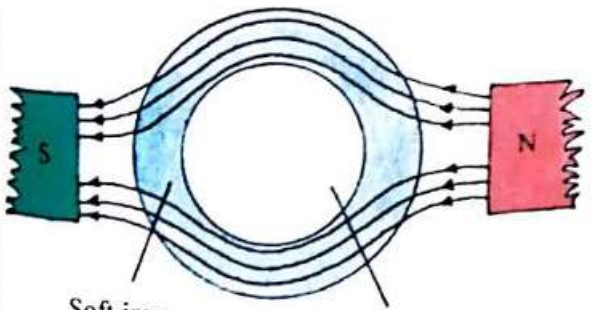


Fig. 6.8 Magnetic screening

The lines pass through the ring and none crosses the space inside. Thus the space inside the ring is shielded or screened from the effect of the magnetic field. This phenomenon is used in shielding delicate measuring instruments from external magnetic fields by placing them in soft-iron boxes. It is also used in protecting cathode-ray tubes from the magnetic fields outside them.

6.4 Electromagnets and the Application of Electromagnetic Fields

An electromagnet consists of a soft iron core in a current carrying solenoid. The magnetic field strength due to such a solenoid is increased by the presence of the soft iron core or rod within the solenoid. An electromagnet is a temporary magnet.

An electromagnet can be made by winding the coils or solenoids in opposite directions round the two arms P and Q of a U-shaped soft-iron core or bar (fig. 6.9).

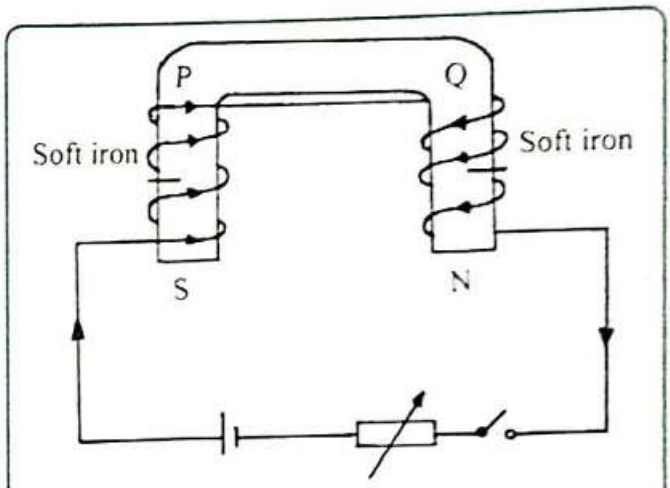


Fig. 6.9 Electromagnet

When current flows from a battery round the solenoid, it is observed that the end of P acts as a S-pole and the end of Q acts like a N-pole. Such an arrangement produces a strong magnetic field. However, magnetization and demagnetization occur rapidly when the current is switched on and off. Hence *electromagnets are temporary magnets.*

Electromagnets find many common applications in industry:

1 In producing intense magnetic fields such as those required in generators and electric motors.

2 They are used for lifting and transporting heavy pieces of iron and steel, e.g. steel plates, guiders, scrap iron, etc.

3 They can be used to separate iron from mixtures containing non-magnetic substances.

4 They are used in the construction of such electromagnetic devices as the Electric bell and the telephone earpiece.

The Electric Bell

The component parts of the electric bell are as shown in fig. 6.10

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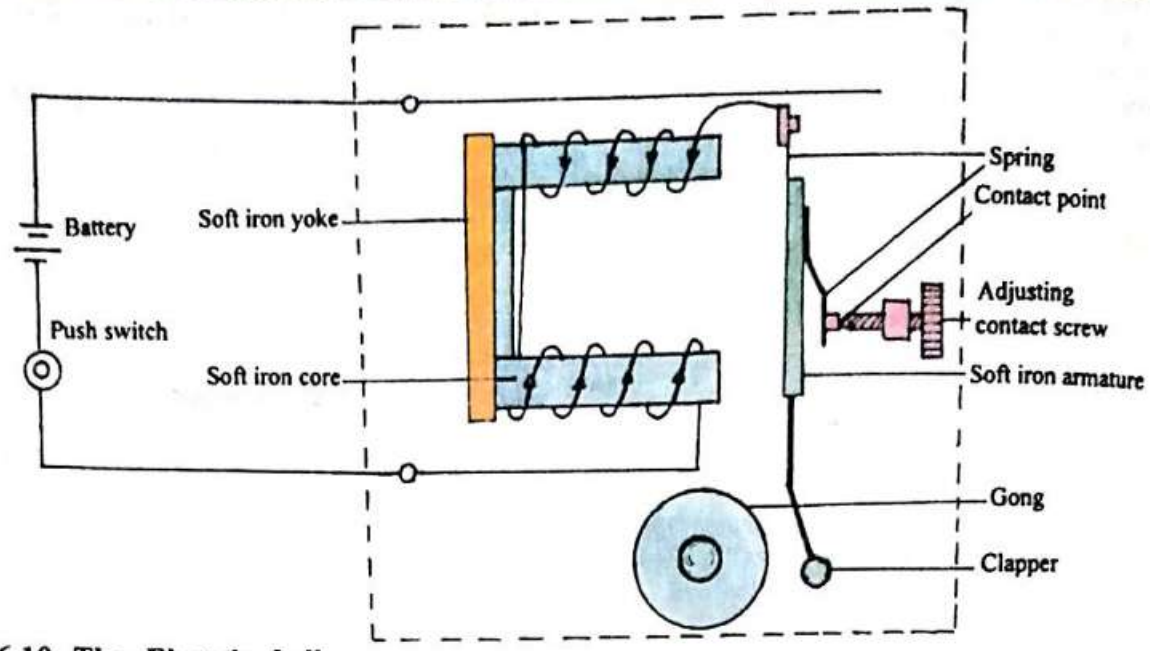


Fig. 6.10 The Electric bell

Working Principle

When the bell-push is pressed, the circuit of the system is completed and current flows round the electromagnet. The soft iron core becomes magnetized and the electromagnet attracts the soft iron armature attached to a spring. This makes the clapper to strike the gong. The circuit is broken as soon as the spring moving along with the soft iron armature breaks contact at *C*. As soon as this happens, the electromagnet loses its magnetism and therefore its attracting force on the armature. This armature now flies back due to the force of the spring, to remake contact once again at *C*. The circuit is thus reestablished and current then flows through the electromagnet once again. The sequence of events is repeated rapidly and the bell continues to ring as long as the bell push is pressed.

The instrument works on a 'make-and-break' device. The making of the contact between spring and screw at *C* causes the clapper to move and strike the gong and the consequent breaking of the contact at *C* causes the clapper to leave the gong.

Telephone Earpiece

The telephone earpiece is a device which converts the varying electric energy from the microphone into varying sound energy. The component parts consist of (i) the speech coils wound in opposite directions round two soft iron pole pieces, *A* and *B*, placed respectively at the *S* and *N* ends of a permanent strong magnet, *M*. (ii) A soft iron springy diaphram made of magnetic alloy and located in front of the iron cores.

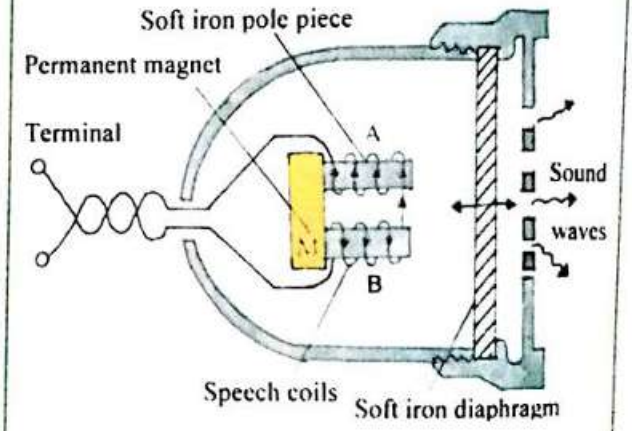


Fig. 6.11 Telephone earpiece

Working Principle

When a caller speaks into the microphone at the other end of the communication line, the varying sound energy from his speech, is converted into a varying speech current which is transmitted to the earpiece at the receiver's end. The variation in the electric current due to changes in the speech energy causes a variation in the magnetism of the electromagnet. This varying magnetism of the electromagnet causes vibrations of the diaphram placed in front of it. Vibrations of the diaphram creates sound waves of the same frequency in the air as the speech current. Hence the speaker at the other end is heard.

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6.5 The Earth's Magnetic Field

A bar magnet or a magnetic needle when suspended freely about its centre and away from any other magnet always comes to rest with its axis pointing roughly in a North-South direction. This is considered to be due to the presence of the earth's magnetic field. The magnetic field of the earth is believed to be due to the presence of an imaginary bar magnet situated at the centre of the earth and inclined at a small angle to the earth's axis of rotation (see fig. 6.12). This imaginary bar magnet has its South pole near the north geographic pole and its North pole near the South geographic pole.

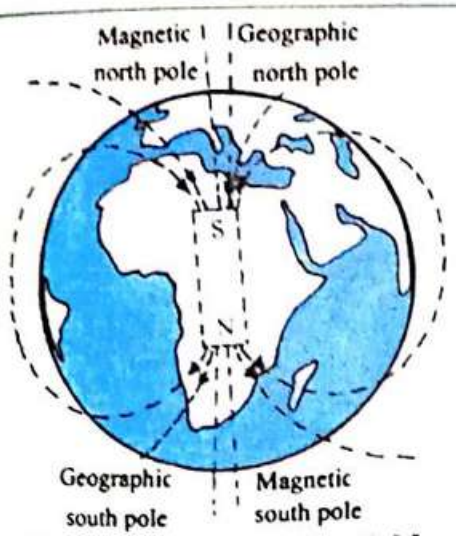


Fig. 6.12 Earth's magnetic field

Magnetic elements of a place

The magnetic elements of a place are the Angle of Declination, the Angle of Dip and the Horizontal Component of the Earth's magnetic field in that place. These three completely determine the magnetic field of the earth at each place.

Angle of Declination or Variation

At any place in the earth's surface, the magnetic North is not usually in the same direction as the Geographic North. It is therefore necessary to differentiate between two planes, the magnetic meridian and the geographic meridian.

The **Magnetic Meridian** at any place is a vertical plane passing through the magnetic axis of a freely suspended magnet at rest under the influence of the Earth's magnetic field.

The **Geographic Meridian** is the vertical plane containing the geographic north and south pole of the Earth.

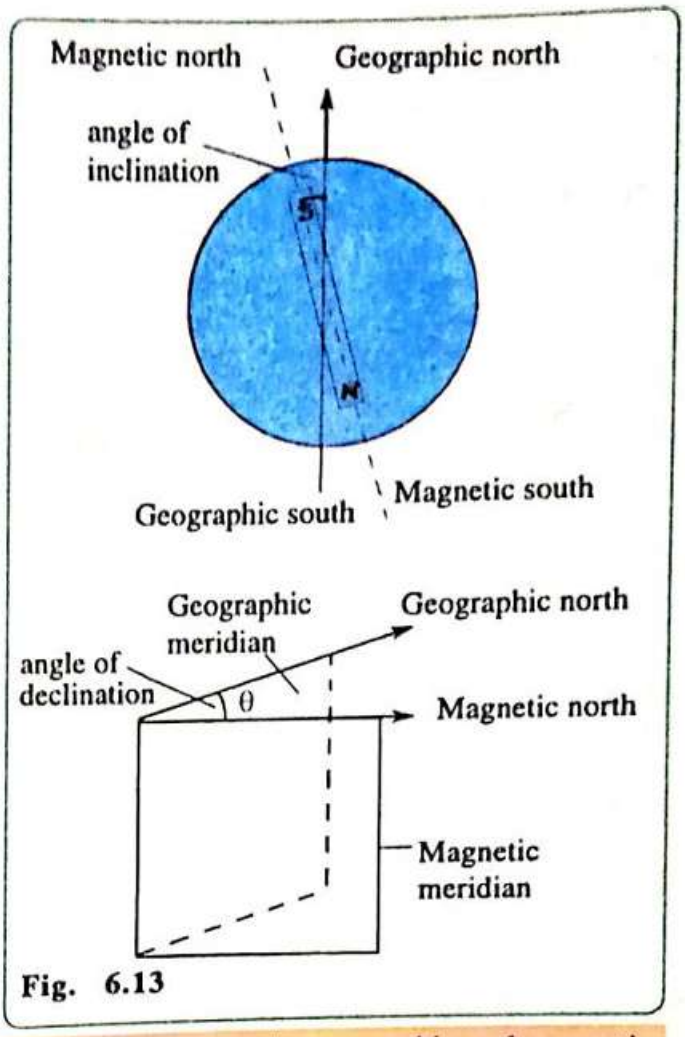


Fig. 6.13

The angle between the geographic and magnetic meridians at a place is called the **Angle of Declination** at that place.

The angle of declination is also the angle between the direction of the magnetic north and the direction of the geographic north at any place. The compass needle points to the direction of magnetic north. Mariners called the declination, *variation of the compass*. The declination is said to be east or west depending upon whether the north pole of the needle is east or west of the geographic meridian or geographic north.

Since mariners use the compass needle for determining their direction, it is very necessary for them to know the magnetic declination at different places on the earth through which they travel. With the knowledge of the declination at any place and the reading on their mariners' compass, they can calculate their true bearing. For example, if the mariner's compass gives a bearing of N 20° W or 20° West of North at a place where the declination is 15° West of North, then the true declination is

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35° i.e. (20 + 15)° West of North as shown in fig 6.14.

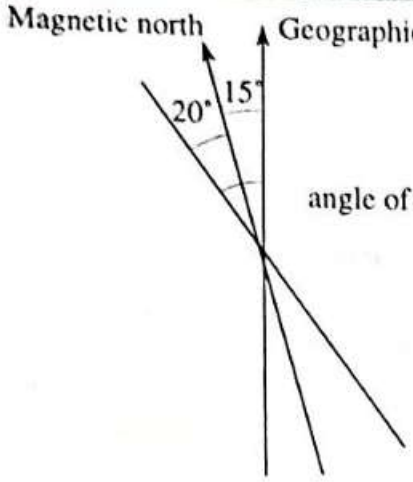


Fig. 6.14 Angle of true bearing = 20° + 15° = 35°

Angle of Dip or Inclination (I)

A freely suspended magnet will not only come to rest pointing approximately in a North-South direction, but will also be found to tilt (or dip) downwards at some angle to the horizontal.

The **angle of dip** (I) is the angle between the direction of the earth's resultant magnetic field and the horizontal

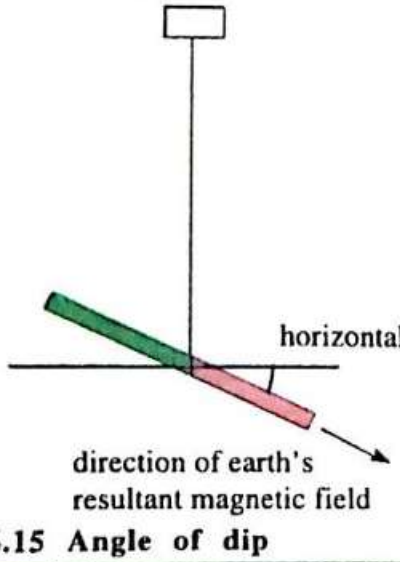


Fig. 6.15 Angle of dip

The angle of dip at any place may be measured with an instrument called the dip circle (see fig. 6.16).

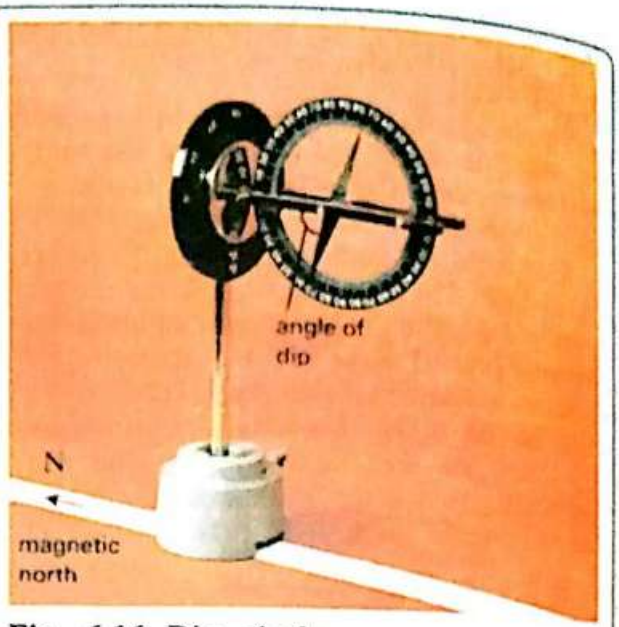


Fig. 6.16 Dip circle

The angle of dip also varies all over the earth's surface from 0° at the magnetic equator to 90° at the magnetic poles of the earth.

We can differentiate between a magnetized and an unmagnetized needle by suspending each at its midpoint. The unmagnetized needle will balance horizontally but the magnetized needle will balance pointing downwards.

The Horizontal and Vertical Components of the Earth's Magnetic Field

The earth's magnetic field at a place is a vector quantity. It can be resolved into its Horizontal and Vertical Components.

The **Horizontal Component** (H) of the earth's field at a point is the magnetic force acting in a horizontal direction.

The **Vertical Component** (V) of the earth's field at a point is the magnetic force acting in a vertical direction.

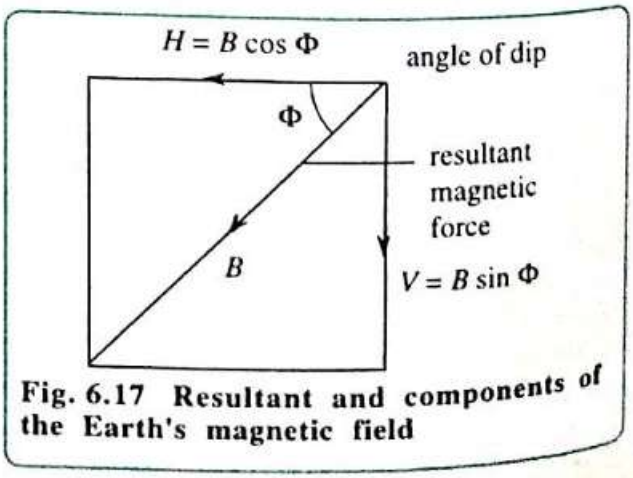


Fig. 6.17 Resultant and components of the Earth's magnetic field

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It is easy to see from fig. 6.17 that if the angle of dip at a place is Φ, then

$$\tan \Phi = \frac{V}{H} \qquad 6.1$$

Bar Magnet in Earth's Field. Neutral Point

The earth magnetic field consists of parallel lines pointing Northwards as shown in fig. 6.18. It is a uniform field, i.e., a field whose strength and direction are constant.

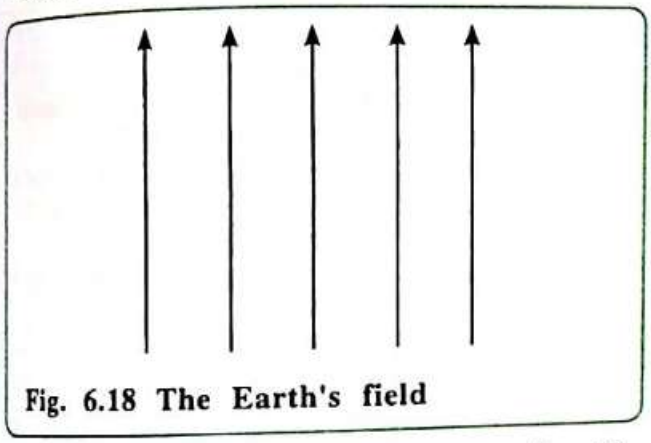


Fig. 6.18 The Earth's field

The magnetic field due to a bar magnet placed in the earth's magnetic field is as shown in fig. 6.19.

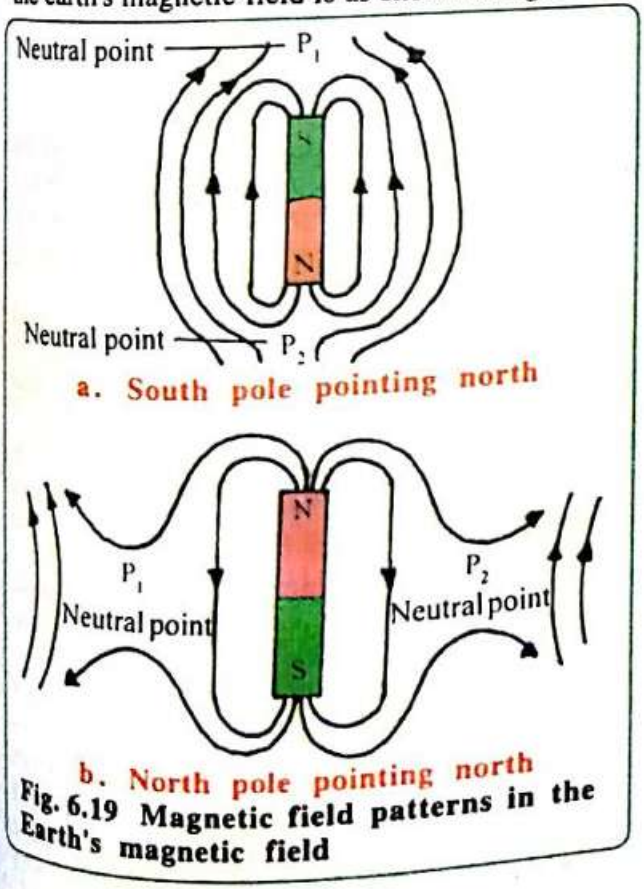


Fig. 6.19 Magnetic field patterns in the Earth's magnetic field

An important feature of the field is the presence of Neutral Points. These are points where the earth's magnetic field neutralizes completely the field due to the bar magnet. *At the neutral points the magnetic force due to the earth's field is equal and opposite to the magnetic force due to the field of the bar magnet.* The net magnetic force at such points is therefore zero. The position of the neutral point depends on whether the north pole of the bar magnet is placed pointing north or pointing south.

6.6 Magnetic Force on a Charge moving in a Magnetic Field

A magnetic field exerts a force on a charge moving in the field. We recollect that charges in motion constitute an electric current. For example, the motion of electrons along conducting wires, the rotation of electrons about the nuclei of atoms and the motion of electrons in a cathode ray tube, each constitutes a flow of an electric current.

The strength of magnetic field can be represented by a vector quantity B called the **flux density** or **magnetic induction**. If θ is the angle between the magnetic field direction and the direction of motion of the charge, then the force, F on the charge is given by:

$$F = qVB\sin\theta \qquad 6.2$$

Here q is the charge in Coulombs, V is the average velocity of the charge in metres per second, B is the flux density or magnetic induction whose unit is the *Tesla* (T), [1 Tesla = 1 Weber per m^2] and F is the force in Newtons. The basic important result of equation 6.2 has been confirmed by experiment.

The expression $VB\sin\theta$ in equation 6.2 can be represented by the expression $V \times B$ (pronounced V cross B). $V \times B$ is the cross product of V and B, both of which are vector quantities.

$$V \times B = VB\sin\theta \qquad 6.3$$

where θ is the angle between V and B. Hence equation 6.2 becomes

$$F = qVB\sin\theta = q(V \times B) \qquad 6.4$$

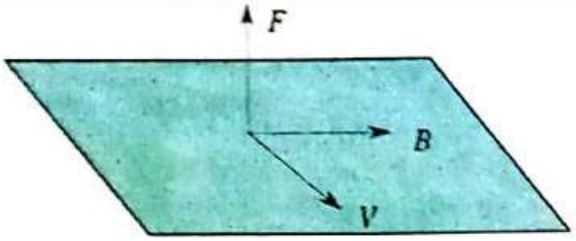
When V and B are in the same direction, the angle θ equals zero and from equation 6.4 the force equals

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zero because $\sin\theta = \sin 0^\circ = 0$. When V and B are at right angles, then $\sin\theta = \sin 90^\circ = 1$ and F is given by

$$F = qVB \qquad 6.5$$

Hence a charge moving in the direction parallel to a magnetic field experiences no force. A charge at rest in a magnetic field also experiences no force because V in equation 6.4 is zero. The vector $F = V \times B$ is perpendicular to the plane containing V and B.



$F = V \times B$

Fig. 6.20

The magnitude of the force is given by equation 6.2.

Example 6.1

Find the magnetic force experienced by an electron projected into a magnetic field of flux density 10 Tesla with a velocity of 5×10^{7} ms^{-1} and in a direction (i) 90° (ii) 60° and (iii) parallel to the magnetic field. Assume charge on an electron = 1.6×10^{-19} C.

Solution

$F = q(V \times B) = qVB\sin\theta$

(i) When $\theta = 90^\circ$

$F = 1.6\times10^{-19}$ (C) $\times 5\times10^{7}$ (ms^{-1}) $\times 10$ (T) $\times \sin 90^\circ$

$= 1.6\times10^{-19}\times5\times10^{7}\times10\times1$

$= \underline{8.0\times10^{-11}\text{ Newtons}}$

(ii) When $\theta = 60^\circ$

$F = 1.6\times10^{-19}$ (C) $\times 5\times10^{7}$ (ms^{-1}) $\times 10$ (T) $\times \sin 60^\circ$

$= 1.6\times10^{-19}\times5\times10^{7}\times10\times0.8660$

$= \underline{6.9\times10^{-11}\text{ Newtons}}$

(iii) When V and B are parallel, i.e., $\theta = 0$, $\sin\theta = 0$

Hence $F = qVB\sin 0$

$= 0.$

Because of the force experienced by a charge moving in a magnetic field, a magnetic field can be used to deflect or focus an electron beam. Hence we can now appreciate the effect of a bar magnet brought near the electron beam in a cathode ray tube (fig. 5.13). Magnetic deflection and focusing are made use of in cathode ray tubes used for television and in cathode ray oscilloscope.

The path of a charged particle moving perpendicular to a uniform magnetic field is a circle.

Summary

1 Magnetic field is the region round a magnet where a magnetic force is experienced.

2 Magnetic lines of force are imaginary lines along which a free north pole would tend to move if placed in the magnetic field.

3 Magnets are made by the electrical method and the contact methods.

4 Soft-iron is more easily magnetized than steel. It also loses its magnetism more easily than steel. For this reason steel is used for making permanent magnets but soft iron is used for manufacturing temporary magnets or electromagnets where a strong magnetism is required only for a short time.

5 Electromagnets are used in the construction of electric bells and telephone earpieces.

6 Magnetic elements of a place are (1) angle of declination, (2) angle of dip (3) Horizontal component of the earth's field. Angle of dip increases from zero at the magnetic equator to 90° at the magnetic poles.

7 Mariners' compass points to the direction of magnetic north. Mariners therefore need to know the declination at their location to enable them obtain their true geographical bearing.

8 *The magnetic meridian* at any place is the vertical plane containing the magnetic axis of a freely suspended magnet at rest under the action of the Earth's field.

9 *The geographic meridian* is the vertical plane containing the geographic north and south poles of the earth.

10 The magnetic force on a charged particle moving across a magnetic field is given by $F = qV \times B = qVB \sin\theta$. The force is greatest when the charged particle moves perpendicular to B ($\theta = 90^\circ$). The force is zero if the particle moves parallel to the field lines ($\theta = 0^\circ$). The

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direction of the magnetic force is perpendicular to the magnetic field B and to the velocity, V, of the particle.

Exercise 6

1 Which, if any, of the diagrams I - IV in fig. 6.21 correctly shows the direction of the magnetic field around a wire carrying a current passing perpendicularly through the centre of the circle RS, in the direction shown

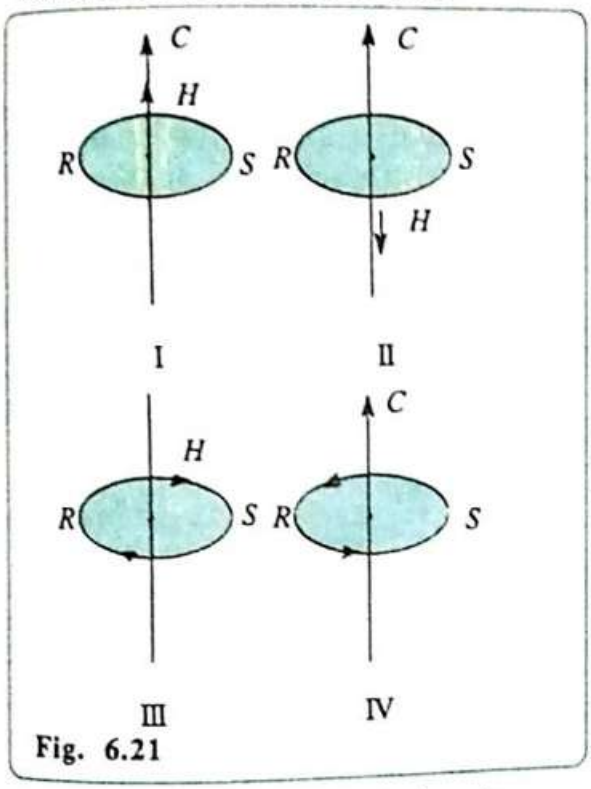


Fig. 6.21

C is direction of current flow in the wire
H is the direction of the magnetic field
a. I b. II c. III d. IV
e. none of the above?

(SC/GCE)

2 Which of the following statements is *not* correct?

A A magnetic field is a region in which a magnetic force can be detected.
B A line of force is a line along which a magnetic north pole would tend to move if it were free.
C Magnetic fields are scalar quantities
D Neutral points are obtained where the Earth's magnetic field is exactly equal and opposite to that due to a magnet.
E The dip angle is the angle between the horizontal and the Earth's total magnetic field.

(SC/GCE)

3 The magnetic meridian at a place is the

A plane at right angles to the direction of the Earth's magnetic flux.
B plane containing the longitude of the place and the Earth's axis.
C horizontal plane containing the latitude of the place.
D vertical plane containing the magnetic axis of a freely suspended magnet under the action of the Earth's field.
E direction in which a dip angle will set when placed at that place.

(SC/GCE)

4 The magnetic force on a charge moving in a magnetic field is given by

A $F = Vq \times B$ B $F = qV \times B$
C $F = BV \times q$ D $F = B \times qV$

5 The force on a charge travelling in a magnetic field is maximum when charge is

A moving parallel to the field
B moving perpendicular
C moving at 60° to the field
D moving at 45° to the field
E stationary in the field.

6 The angle of dip is

A 0° along the geographic equator and 90° at the geographic poles.
B 90° along the equator and 0° at the geographic poles
C 0° along the magnetic equator and 90° at the magnetic poles
D 90° along the magnetic equator and 0° at the magnetic poles
E 90° along the magnetic equator and 0° at the geographic poles.

(SC/GCE)

7 Which of the following statements is CORRECT about the earth's magnetic field?

A The angle of dip is the angle which a freely suspended magnet makes with the vertical.
B The angle of declination is the angle between the magnetic meridian and the geographic meridian.
C The angle of declination is the angle which a magnetic compass makes with the magnetic meridian.
D The angle of inclination is the difference between the angle of dip and the angle of declination.

(JAMB)

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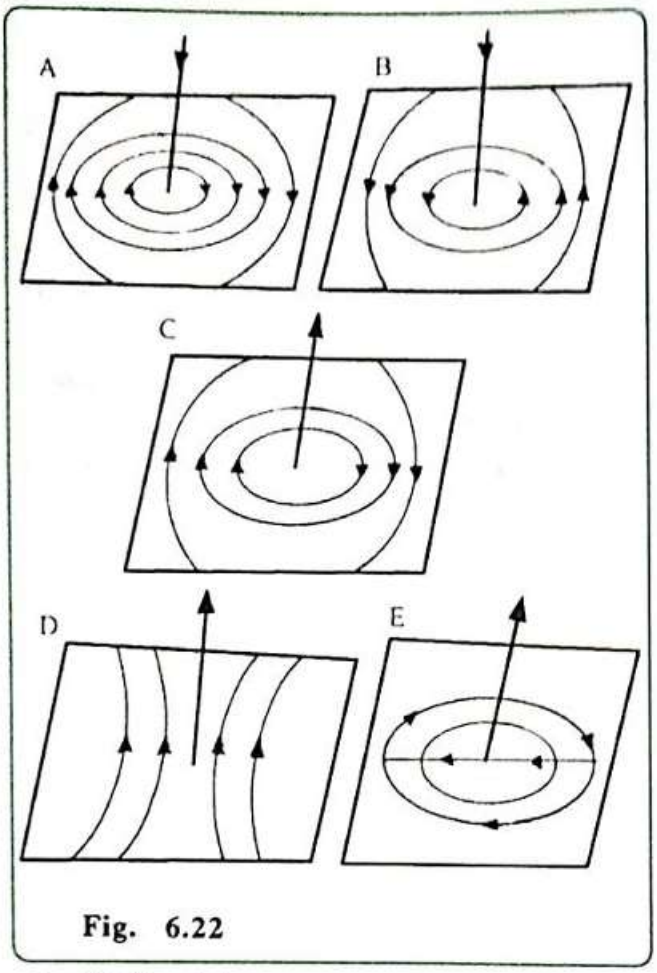


Fig. 6.22

8 Which of the following diagrams (fig. 6.22) gives the correct magnetic field pattern around a current carrying straight conductor?

9 Which of the following materials should be used for the core in order to produce the strongest magnetic effect at J in the circuit illustrated in the diagram (fig. 6.23)?

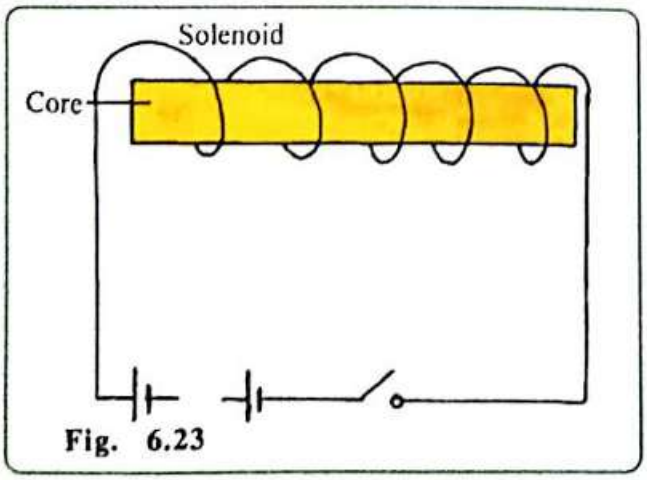


Fig. 6.23

A Copper B Iron C Brass
D Plastic E Steel

(SSCE)

10 If the angle of declination in a place is 10°, calculate the true geographic bearing if the compass needle reads N 40° E.
A N 50° E B N 40° E
C N 30° E D N 35° E
E N 25° E

(JAMB)

11 Draw the magnetic field around (i) a bar magnet (ii) a straight conductor carrying current (iii) a solenoid.

12 Define (a) magnetic field (b) magnetic lines of force.
Draw the lines of force due to a bar magnet placed in the Earth's magnetic field with the north pole pointing north.

13 Describe three methods of magnetizing a steel or an iron bar. Which of the methods is the most effective. Discuss the nature of the poles formed by each of the three methods described above.

14 Distinguish between magnetic and non-magnetic materials. Give 3 examples of each.

15 What *three* methods would you use to differentiate between two steel rods, one of which is magnetized and the other unmagnetized.

16 What are the differences between the magnetic properties of steel and iron? Which of them is more suitable for making (i) permanent magnets (ii) electromagnets (iii) magnetic screens. Give reasons for your choice.

17 State the material that is usually used to construct the core of the electromagnet in an electric bell. Give two reasons for your answer. Draw a labelled diagram showing the essential parts of an electric bell and describe how it works.

18 A bar magnet is placed on a table so that it lies in the magnetic meridian with (i) its N-poles pointing south and then (ii) its N-pole pointing north. In each case there are two points near the magnet where there is no horizontal magnetic field. Draw diagrams to show these points and explain why the magnetic field at these points is zero.

19 Write down the expression for the force exerted by a magnetic field on a moving charge. Define the terms used in the expression. Calculate the force on an airplane which has acquired a net change of 100 C and moves with a velocity of 300 ms^{-1} perpendicular to the Earth's magnetic field of 5×10^{-5} T.

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20 Find the magnetic force experienced by an electron of charge 1.6×10^{-19} C projected into a magnetic field of flux density 10 T, with a velocity of 3×10^{7} ms^{-1}, in a direction
(i) Parallel to the field
(ii) at right angle to the field
(iii) at 30° to the field.

21 Define (i) angle of dip (ii) angle of declination (iii) geographic meridian (iv) magnetic meridina.

A mariner's compass gives a bearing 15° West of North at a place where the declination is 19° West of North. What is the true bearing of the place. If the declination had been 19° East of North, what would have been the true bearing?

22 (a) Explain what is meant by a magnetic field.
(b) (i) Describe an experiment to show that a magnetic field exists around a straight wire carrying a current.
(ii) Draw a labeled diagram showing the pattern and direction of the magnetic field produced around the wire. (Neglect the earth's magnetic fields).
(c) Sketch the magnetic field due to two straight parallel wires carrying current in the same direction. Indicate the neutral point in the field.
(d) Explain with the aid of a labeled diagram, how a delicate magnetic material could be protected from the earth's magnetic field.
(e) A charge of 1.6×10^{-19}C enters a magnetic field of flux density 2.0T with a velocity of 2.5×10^{7} ms^{-1} at an angle of 30° with the field. Calculate the magnitude of the force exerted on the charge by the field.

(SSCE)

23 An electron moves with a speed of 2.00×10^{7} ms^{-1} in an orbit in a uniform magnetic field of 1.20×10^{-3}T. Calculate the radius of the orbit. (Mass of an electron = 9.11×10^{-31} Kg, charge on an electron = 1.61×10^{-19}C)

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7 ELECTROMAGNETIC FIELD

In the preceding chapters we have studied electric field and magnetic field as independent fields. This chapter will discuss the field representing the interaction of electric and magnetic forces.

At the end of the chapter the student should be able to:

1 Understand the concept of electromagnetic field.
2 Identify the direction of current, magnetic field and force in an electromagnetic field.
3 Explain the action of a loop of wire carrying current in a magnetic field.
4 Explain the basic working principle of the galvanometer and the electric motor.
5 State and explain Faraday's law of electromagnetic induction.
6 State and explain Lenz's law and the implications of Lenz's law.
7 Explain how the conservation principle is involved in Lenz's law with regards to energy.
8 Explain the principles underlying the production of direct and alternating current.
9 Explain the difference in the construction of d.c. and a.c. generators.
10 Explain the principle of a transformer and the induction coil.
11 State the uses of induction coils and transformers.
12 Explain why the cores of the induction coil and transformer are laminated.
13 Explain why it is preferred to have a high p.d. instead of high current transmission over a long distance.
14 Solve simple problems on power losses.

7.1 Concept of Electromagnetic field

In chapter 6 we showed that a current-carrying wire has a magnetic field around it consisting of concentric circles with the wire as their axis (fig. 6.2). Thus this magnetic field is directly traceable to the moving electricity in the wire.

In general a changing electric field is always accompanied by a magnetic field and conversely, a changing magnetic field is accompanied by an electric field. The joint interaction or interplay of electric and magnetic forces gives rise to an *electromagnetic field.*

Electromagnetic field is a field representing the joint interaction of electric and magnetic forces.

Electromagnetic field is considered to have its own objective existence in space just as the other fields we have studied.

Electromagnetic field exerts forces on charged particles. The force on a charge q moving with a velocity v (less than the velocity of light) is given by

$$F = q(E + v \times B) \qquad 7.1$$

7.2 Interaction between Magnetic Field and Currents

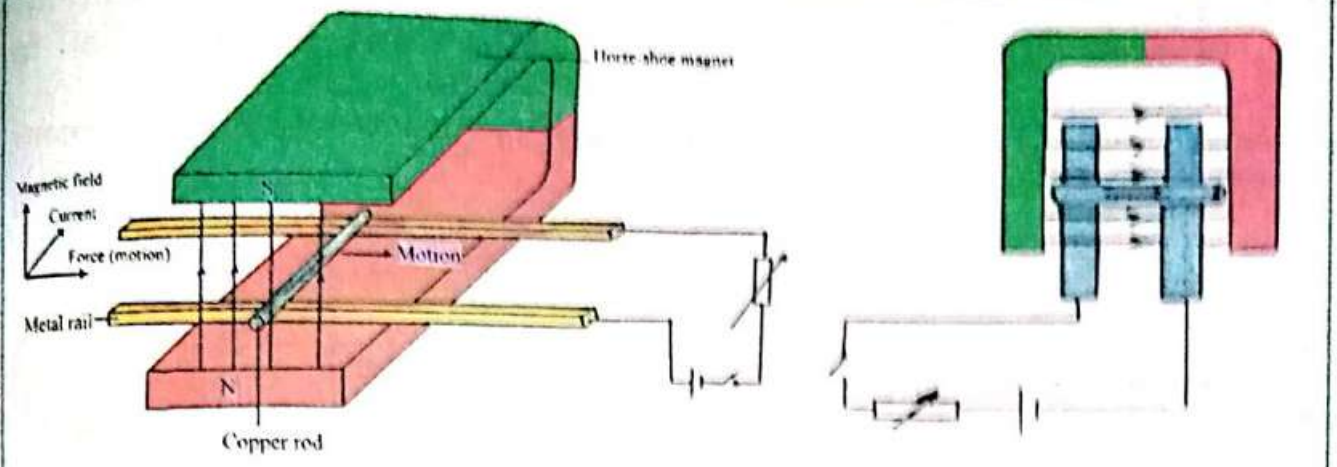
a. Force on a Current-Carrying Conductor in a Magnetic Field

A conductor carrying an electric current, when placed in a magnetic field experiences a mechanical force. We can demonstrate this by using a set-up as shown in fig 7.1a. Two metal rails are fixed on each side of a powerful horse-shoe magnet. A copper rod is placed across the rails. When we pass current through this copper rod, it is observed that the copper rod rolls along the rails, towards the right. If by adjusting the rheostat, we cause more current to flow through the rod, we will observe that the rod moves faster. Thus the force on the rod increases when the current increases.

If we reverse the direction of the current flow by reversing the connections at the battery terminals, the rod will be observed to move towards the left, opposite to the previous direction of motion.

If we turn the magnet such that the magnetic field is parallel to the length of the rod, as shown in fig 7.1b it will be observed that the current carrying rod remains stationary no matter the amount of current passed through it. There is therefore no force on the rod.

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a. Force on a current carrying conductor in a magnetic field

b. No force acts on the conductor

Fig. 7.1

The above observations lead us to the following conclusions.

1 A mechanical force acts on a current-carrying conductor whenever the conductor is placed in a magnetic field, at an angle to this field.

2 The direction of the force is always perpendicular to the direction of the current and the magnetic field.

3 The magnitude of the force produced is proportional to the current and the field strength. It increases when the current increases and when the field strength increases.

4 The magnitude of the force decreases when the angle between the conductor and the field decreases from 90°, and it becomes zero when the conductor lies parallel to the field.

Thus the force is directly proportional to the current, I, the field strength, B, and the angle between the conductor and the field.

Direction of Force

The direction of force on a current carrying conductor placed perpendicular to the magnetic field is given by *Fleming's left-hand rule* which is stated as follows:

If the thumb, forefinger and middle finger are held mutually at right angles to one another with the *Fore*-finger pointing in the direction of magnetic *Field*, and the *seCond* finger in the direction of *Current*, then the *thuMb* will point in the direction of *Motion* or force producing motion.

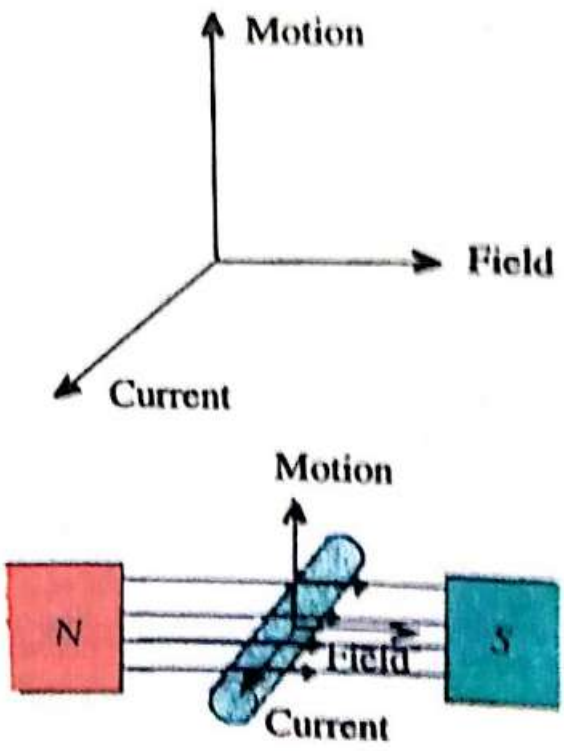


Fig. 7.2 Fleming's left hand rule

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7.3 Force between Conductors Carrying Current

If two parallel current-carrying conductors are arranged side by side, it is clear that each conductor is in the magnetic field of the other. We have shown that a conducting wire in a magnetic field is subject to a force. Hence each of the two conductors must exert a force on the other since each conductor would be lying in the magnetic field of the other. The direction of current flow in each wire determines whether the force acting between the conductors would be attractive or repulsive.

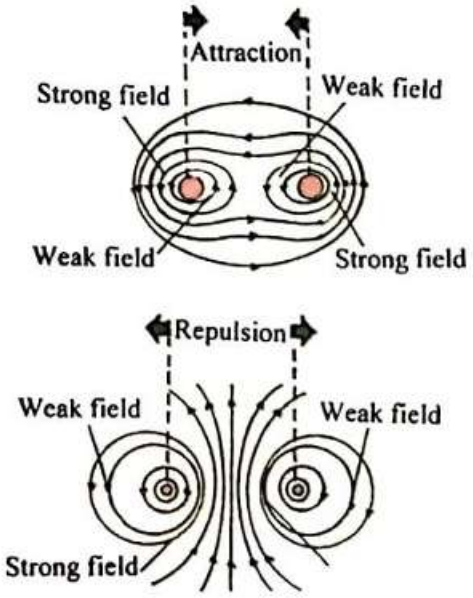


Fig. 7.3 Attraction and repulsion between two parallel current carrying conductors

If the currents flow in the same direction as shown in fig 7.3a the pattern of magnetic field which can be depicted using iron filings and compass needle, is as shown in the diagram. The force between two such conductors is an attractive force, causing the two conductors to move towards each other.

If the current flows in opposite direction in the two conductors, the field pattern is as shown in fig 7.3b, and the force between the conductors is a repulsive force.

7.4 Motion of a Current-Carrying Loop or Coil in a Magnetic Field

Fig 7.4 shows a coil carrying current in a magnetic field.

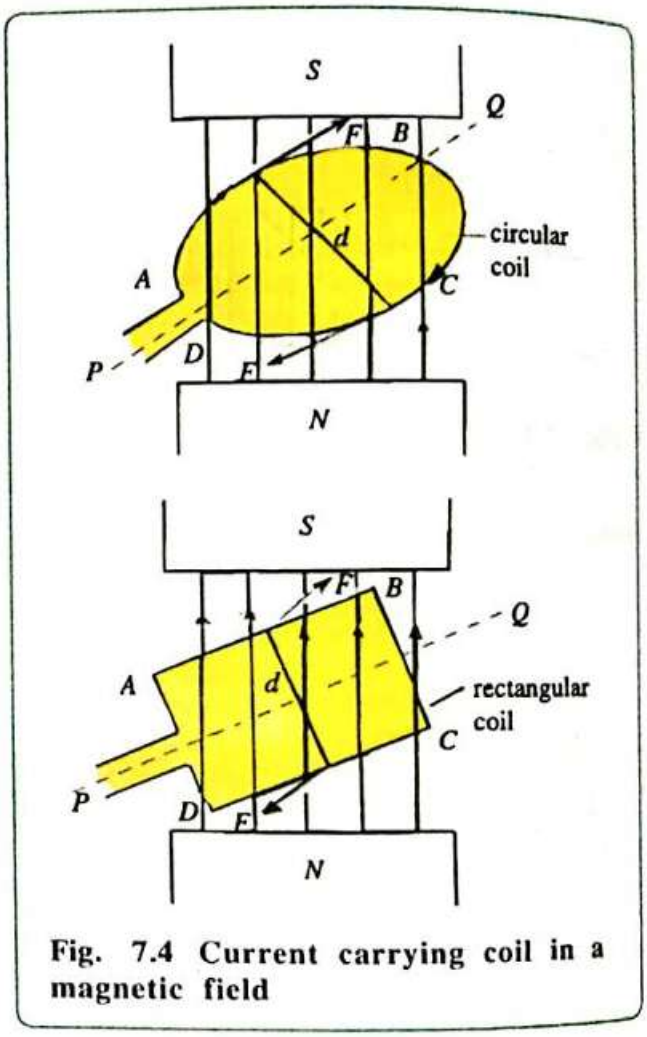


Fig. 7.4 Current carrying coil in a magnetic field

When current flows round the loop as shown, the side AB of the coil experiences a force acting upwards (outwards from the page) and the side CD experiences an equal force acting downwards (into the page) (We can obtain the direction of motion using Fleming's left-hand rule). These two equal but oppositely directed forces separated by the distance, d, constitute a couple whose moment or torque produces a turning effect or torsion tending to rotate the coil about the axis PQ. This torque causes the coil to rotate in a clockwise direction. If the direction of either the current or the magnetic field is reversed, the coil will rotate in an anti-clockwise direction, in accordance with Fleming's left-hand rule.

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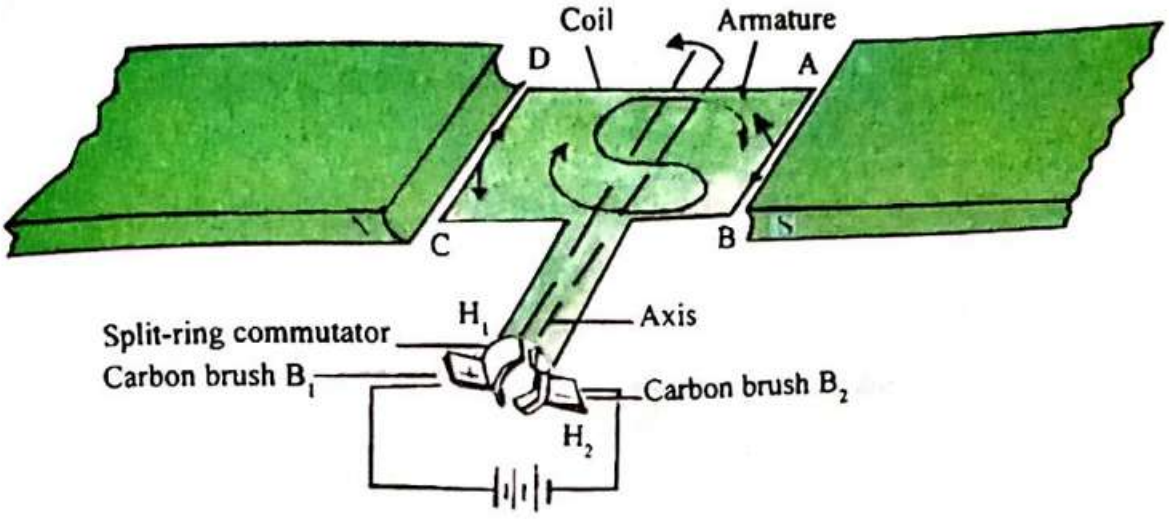


Fig. 7.5 The electric motor

7.5 Applications of Electromagnetic Field

1 The Electric Motor

Structure

The electric motor is a device for converting electrical energy into mechanical energy. It consists of (i) a rectangular coil $ABCD$ of insulated wire, known as the armature which can turn about a fixed axis. (ii) A powerful magnetic field in which the armature turns is provided by two curved pole pieces of a powerful magnet . (iii) A commutator consisting of a split copper ring, two halves of which are insulated from each other. Each end of the coil is attached to each half of the split copper ring. (iv) Two carbon brushes B_1, B_2, which are made to press lightly against either side of the *split-ring commutator*.

Working Principle

Let us suppose that the coil is initially in a horizontal position as shown in fig 7.5. Let us also assume that current flows from the battery through the coil in a clockwise direction as shown. Under this condition, an upward force acts on the arm AB of the coil, and a downward force acts on CD. The two equal and oppositely directed forces constitute a couple whose torque causes the coil to rotate in a clockwise direction. The split rings to which the ends of the coil are attached rotate along with the coil. The momentum of the moving coil carries it past the vertical position.

As soon as the coil passes the vertical position, the two halves of the split ring commutator automatically change contact from one brush to the other. Commutator half H_1 is now in contact with brush B_2 and H_2 is in contact with B_1. This reverses the direction of current flow in the coil and hence the direction of forces on the coil. AB now experiences a downward force and CD an upward force.

The coil therefore rotates through half a turn more where the current again reverses by means of the sliding brushes and the split-ring commutator. A repetition of this reversing process at each half-turn gives rise to a continuous rotation of the coil as long as the current flows through it.

Practical direct current electric motors

Practical electric motors have:

1 Strong magnetic fields produced by electromagnets.
2 Several coils of wire wound in the slots of a laminated soft-iron cylinder.
3 High currents. All of these add up to produce a powerful rotation.

Energy losses in electric motors can arise from (a) Work done against friction in the bearings and commutator (b) Eddy currents formed in soft-iron

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cylinder. This is reduced by laminating the cylinder.
(c) Heat losses (or I^2R) in the windings.

2 Moving Coil Galvanometer

This galvanometer is one of the most sensitive and accurate methods for detecting or measuring extremely small currents or potential differences.

Structure

It consists essentially of:

(i) A light rectangular vertical coil $ABCD$, pivoted in jewelled bearings P and Q, such that it can move in a vertical plane. The coil is made of several turns of insulated wire wound upon a rectangular aluminium frame.

(ii) Two curved pole pieces, N and S, of a horse-shoe magnet, and a soft-iron core or cylinder, inserted between the pole pieces. This arrangement produces a radial or uniform magnetic field in the air gap between the soft-iron core and the pole pieces. The coil moves in this radial field. The radial field is necessary if the instrument is to have a uniform scale.

(iii) Two spiral non-magnetic control springs of phosphor bronze, each of which is attached to the jewelled bearings or spindle. Current enters or leaves the rectangular coil through these spiral springs. The springs also provide the control couple.

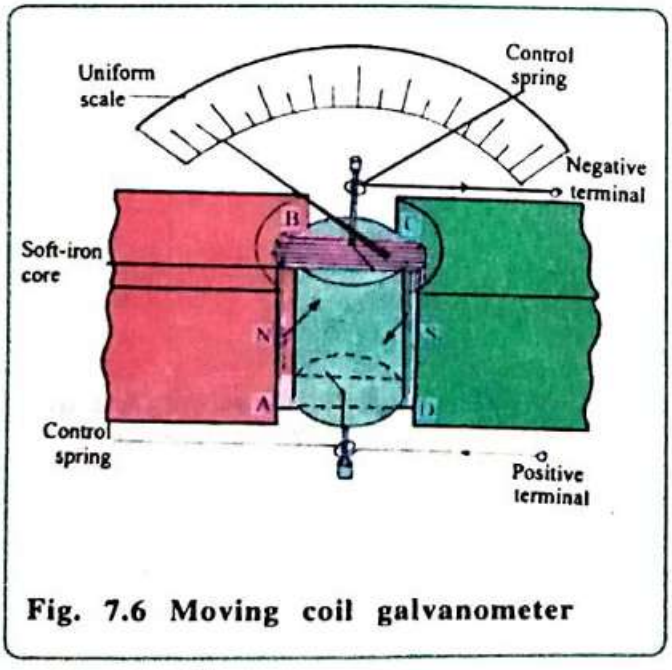


Fig. 7.6 Moving coil galvanometer

(iv) A pointer attached to the axle moves over a uniform scale.

Working Principle

The current to be measured is passed into the galvanometer from a terminal through one spring, leaves through the other. When current flows in the coil, forces act on the arms AB and CD as shown. The two forces constitute a couple whose torque tends to rotate the coil in a clockwise direction across the radial magnetic field provided by the magnet and the soft-iron core. The movement of the coil is opposed by the spiral springs, and the coil will come to rest in a position where the couple due to the current is exactly balanced by the control couple due to the springs. The angle of rotation of the coil is thus controlled by the springs.

The angle through which the coil rotates is recorded by the pointer attached to the coil. This angle is proportional to the size of the forces, which in turn is proportional to the current. The scale reading of the pointer will therefore give a measure of the current. Due to this proportional variation, the scale of the instrument is a uniform scale, i.e., equal divisions on the scale denote equal steps of current. The uniform scale is as a result of the radial magnetic field.

The *sensitivity* of the galvanometer is its ability to detect small currents while its *accuracy* is its ability to indicate current values very close to the true value. To increase the sensitivity of the galvanometer (i.e. to obtain a larger deflection for the same current):

(i) the magnetic field is made stronger.
(ii) the number of turns in the rectangular coil is increased.
(iii) the area of the coil is increased.
(iv) the springs should be made of thinner wire to enable them to twist more easily. Such weak springs give only a small control couple.

Advantages of moving coil galvanometer

(i) It has a uniform scale in which equal divisions represent equal steps of currents or p.d.
(ii) The high sensitivity enables very small currents to be detected or measured.
(iii) Because of its strong magnetic field, no extra measures need to be taken to shield the instrument from stray magnetic fields.

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(iv) By fitting on a suitable resistance the instruments can be made to read many different ranges of current or p.d.

Disadvantage

It cannot measure alternating current (A.C) because of the rapidly changing direction of the current.

7.6 Electromagnetic Induction

Michael Faraday (1791-1867) made the great discovery in 1831 that an electric current could be produced in a conductor under the influence of a varying magnetic field. In one of his experiments, he wound two coils, *A* and *C* on a steel ring *R*, as shown in fig. 7.7 and observed that when the switch *S* was closed, a deflection was obtained on the galvanometer *G*, but when *S* was opened *G* showed a deflection in the opposite direction

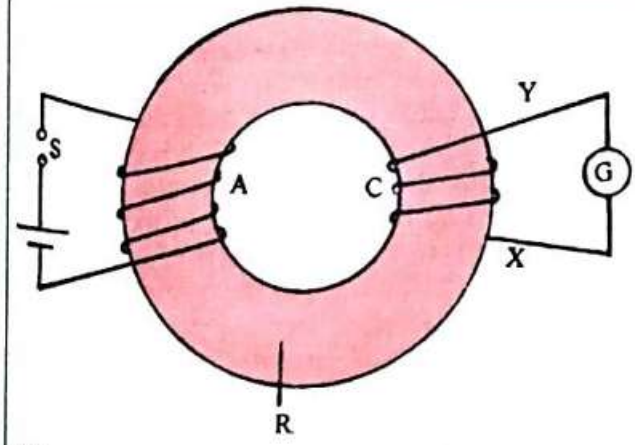


Fig. 7.7 Faraday's experiment

We can repeat some of Faraday's experiments in the following activity:

Activity 7.1

1 Produce a solenoid by winding several coils of insulated copper wire around a cardboard cylinder
2 Connect the solenoid to a centre-zero galvanometer as shown.
3 Move the *N*-pole of a bar magnet *X* horizontally towards the end of the coil. Note the direction of deflection of the galvanometer.
4 Move *X* away from the coil and note the new direction of deflection of the needle off the galvanometer.
5 Keep the magnet stationary near the coil. Does the galvanometer show any deflection?
6 Repeat steps 3 – 5, but now with the south pole of the magnet and note the direction of movement of the needle of galvanometer.

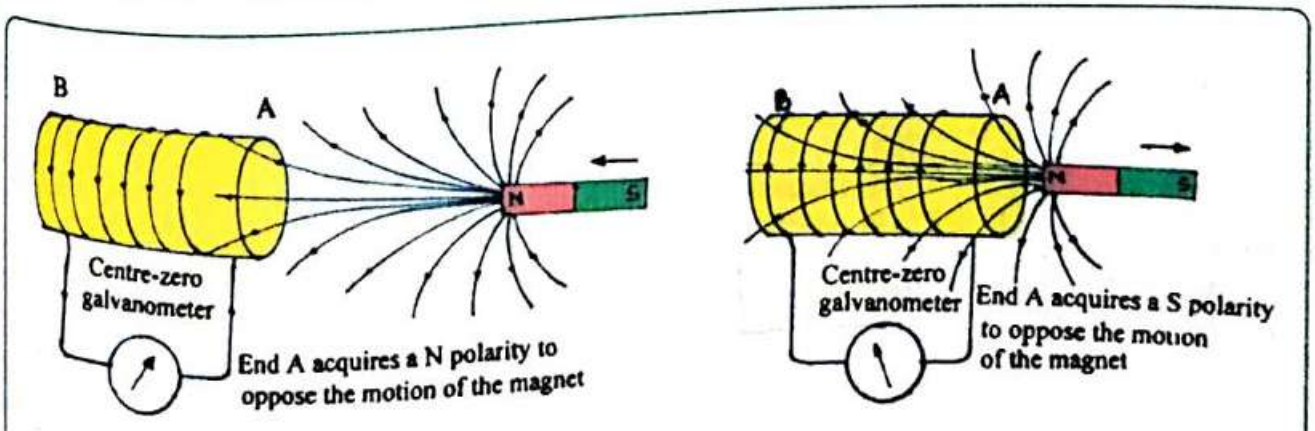


Fig. 7.8 Induced current

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7 Keep the magnet stationary and move coil towards the magnet, and then away from the magnet. In each case note the direction of deflection of galvanometer.

8 Repeat the above experiments but this time, increase the speed of the movements and note the effect on the galvanometer deflection.

The results of the above experiments show that:

(i) Current will flow in the galvanometer whenever there is relative motion between the coil and the magnet. There is such a relative motion when the magnet moves towards the coil or away from the coil, and when the magnet is stationary but the coil is moved. Such a current is called *induced current* and the phenomenon is called *electromagnetic induction*.

Electromagnetic Induction is the production of electric current or voltage in a conductor whenever there is a relative motion between the conductor and a magnetic field(or a mgnet).

(ii) The induced e.m.f. or current depends on (a) the speed of motion of the magnet or the coil. The faster the speed of motion, the larger the induced current. (b) the number of turns of the coil and (c) the presence of a soft-iron core inside the coil.

(iii) The direction of the induced current reverses when the direction of motion of the magnet or the coil is reversed.

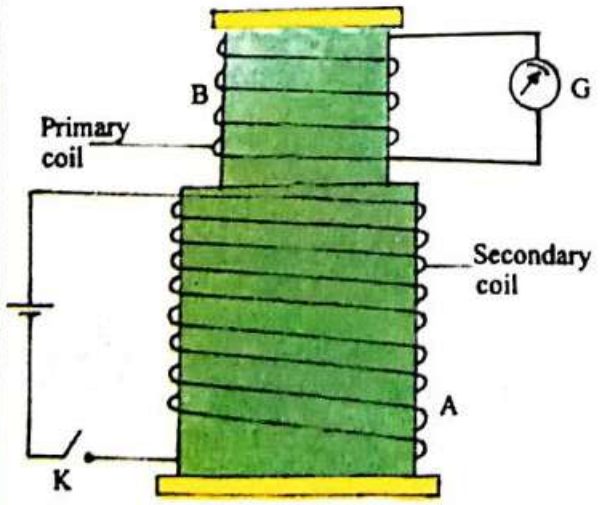


Fig. 7.9 Primary and secondary coils

Primary and Secondary Coils

Another method of producing and observing induced current is through the use of two coils close to each other, one of which is connected to a battery, and the other is connected to a galvanometer as shown in fig. 7.9.

Activity 7.2

1 Produce two solenoids *A* and *B* by winding several turns of insulated copper wire around the same iron core in the form of a cylinder. Connect *A*, to a battery and a tapping key, *K* and connect *B* to a galvanometer as shown in fig. 7.9. Place *A* close to *B*.

2 Cause a current to flow momentarily through the coils of *A* by depressing the key *K*, and observe the deflection of the galvanometer. Place a soft iron inside *A* and *B* and depress *K* again. Note the greater deflection of the galvanometer.

3 Cause a current to flow continuously through *A* by depressing and holding down the key *K*. Note that *G* shows no deflection at all while current flows continuously.

4 Release the key and cut off current flowing through *A*. Note that *G* again shows a deflection but this time in opposite direction to that in (2). Note how soon this current dies down to zero.

The coil *A* connected to the battery and key, in which current is made to flow directly from the battery is called the *primary coil*. The coil *B* connected to the galvanometer, in which current is induced is called the *secondary coil*. We thus see from the above experiments that a current is induced in a secondary coil whenever the circuit connected to the primary coil is made or broken. The galvanometer in the secondary coil shows a momentary deflection when the current is started and another momentary deflection but in opposite direction when the current in the primary coil is cut off. No current is induced in the secondary coil when a steady current flows in the primary coil. Thus induced current flows in the secondary coil only when the current in the primary coil is changing. The changing current in the primary coil produces a changing magnetic field which produces an induced current in the secondary coil.

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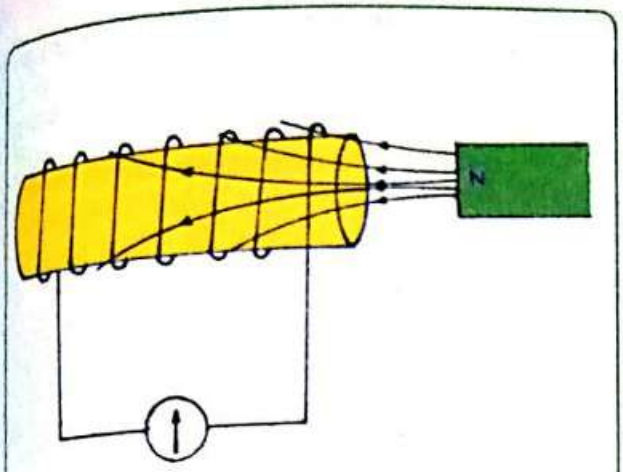


Fig. 7.10 Magnetic lines (or flux) linking a solenoid

When a magnet is near a solenoid as in fig 7.10, we say that magnetic lines of force (or flux) which spread out from the magnet link the coils of the solenoid. These magnetic lines of force move whenever the magnet moves. From the results of Activity 7.1 and 7.2, we say that an induced current is produced in a coil whenever there is a change in the magnetic lines of force (or flux) linked with the coil.

7.7 Induced E.M.F. in a Straight Conductor, Fleming's Right Hand Rule

An induced e.m.f. is produced in a straight conductor when the conductor moves across the lines of force of a uniform magnetic field. The direction of the induced e.m.f. is given by *Fleming's Right Hand Rule.*

If the thumb, fore-finger and middle finger of the right hand are held mutually at right angles to each other, with the Fore-finger pointing in the direction of the field, and the thuMb pointing in the direction of Motion. then the mIddle finger will point in the Induced e.m.f or current, I.

The induced e.m.f. is a maximum when the conductor moves at right angles to the lines of magnetic field and to the conductors own length. It is zero when the conductor moves parallel to the lines of magnetic field. The faster the speed of the conductor, the greater the induced e.m.f.

Numerically the induced e.m.f. in a straight conductor of length, l, moving with a velocity, v, perpendicular to a field of magnetic induction B is given by

$$\text{E.m.f.} = Blv \qquad 7.1$$

volt $\text{Weber.m}^{-2} \times \text{m} \times \text{ms}^{-1}$

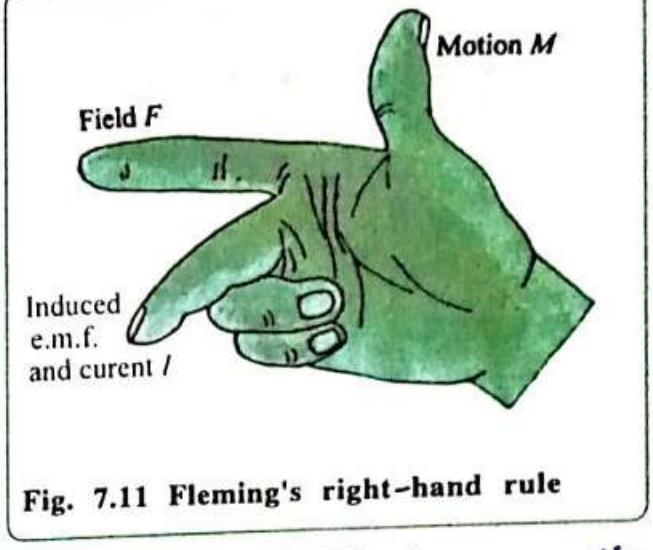


Fig. 7.11 Fleming's right-hand rule

7.8 Laws of Electromagnetic Induction

There are two laws of electromagnetic induction:

1 *Faraday's law* and
2 *Lenz's law.*

Faraday's law states that whenever there is a change in the magnetic lines of force (e.m.f) is induced, the strength of which is proportional to the rate of change of the flux linked with the citcuit.

The magnetic flux or field lines linking a coil depend on (i) the magnetic field strength, (ii) the number of turns of the coil (iii) the area of each turn. Thus to obtain a large induced e.m.f., we cause a coil of many turns to move at a high speed across a strong magnetic field. Faraday's law gives the magnitude of the induced e.m.f.

Lenz's law states that the induced e.m.f. is in such a direction as to oppose the motion or change producing it.

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Thus Lenz's law gives the direction of the induced current or induced e.m.f.

Consider a *N*-pole of a magnet moving towards the face of a coil whose ends are joined to a galvanometer (fig. 7.12). In fig. 7.12(i) the face of the coil will acquire a North polarity. This follows from Lenz's law. The North polarity on this face will oppose the movement of the magnet by repulsion of like poles. Hence current in the coil will flow anticlockwise as shown.

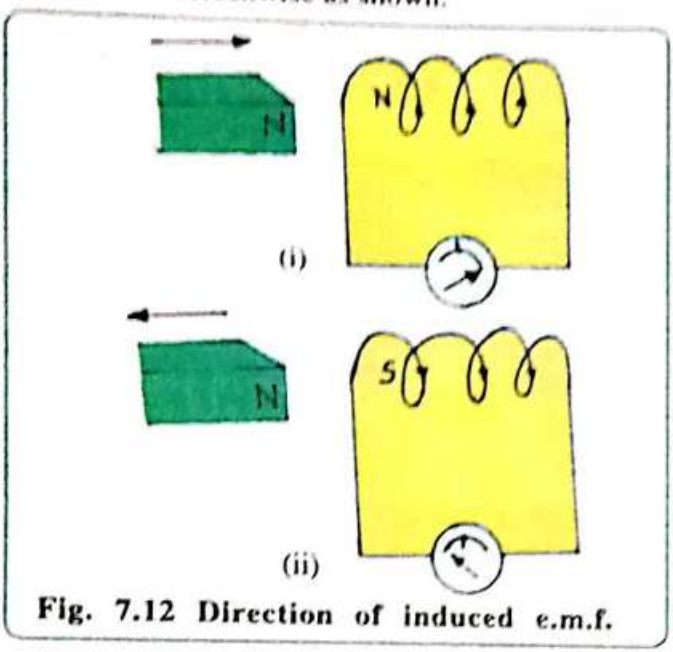


Fig. 7.12 Direction of induced e.m.f.

If the magnet moves away from the coil (as in fig. 7.12ii), the face of the coil will acquire a south-polarity to oppose this motion by the attraction of unlike poles. Current in the coil will therefore flow in a clockwise direction.

Lenz's law as an example of conservation of energy principle

Lenz did not establish his law as a result of experimental observation but from the *conservation of energy principles*, which states that energy can neither be created nor destroyed but can be transformed from one form to another.

When the *N*-pole of the magnet is approaching the coil in fig. 7.12i the induced current in the coil is in such a direction as to produce a *N*-pole at the nearest face of the coil. Repulsion between the *N*-poles will tend to stop the movement of the magnet. We therefore have to apply an extra force continually on the magnet to maintain its motion against this repulsion. If this force, F, moves the magnet through a distance s, we have done a mechanical work equal to $F \times s$. It is this extra mechanical work that is transformed into electrical energy which appears in the form of induced current in the coil.

If the *N*-pole of the magnet were moving away from the coil the nearest face of the coil will acquire a *S*-polarity to oppose this motion by the attraction between the *N* and *S*-poles. To maintain the movement of the magnet, we again have to apply an extra force F continually through a distance, s. This extra mechanical work, $F \times s$ done against the attractive force of the poles appears in the form of induced current in the coil. Hence mechanical energy has been transformed into electrical energy.

7.9 Eddy Currents

When a solid metallic sheet (e.g. copper or aluminium sheet) swings in a magnetic field, it cuts through the magnetic flux or field lines. Consequently, according to Lenz's law, an e.m.f. is induced in the sheet which opposes this movement of the sheet. Hence the sheet quickly comes to rest and its motion is said to be *damped*. The induced e.m.f. that brings this about produces an induced current in the sheet. This induced current is known as *Eddy Current*, because it flows in circular paths or closed loops within the conducting material like eddy currents in water. Eddy currents generate heat in the material leading to wastage of energy in most electrical devices.



Fig. 7.13 Production of eddy currents

The oscillations of the solid sheet are much less damped if the sheet is *laminated* (i.e. cut into slots). This is because the eddy currents cannot flow across the gaps between the slots even though an e.m.f. is still induced to oppose motion.

Eddy Currents are desirable in moving coil instruments because of the resulting damping known as *electromagnetic damping*.

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7.10 Electrical Devices Using Electromagnetic Induction

Our discussions so far have shown that electricity can be generated by moving a conducting wire through a magnetic field. Electricity will be generated as long as there is relative motion between the conductor and the magnetic field. The generated electricity is actually a voltage called an *induced voltage* (or e.m.f.) and the method of generating such an induced voltage when a conductor cuts through the lines of magnetic field is called electromagnetic induction. This induced voltage will cause an induced current to flow if the ends of the conductor are connected through a closed circuit. Also reversing the direction of motion of the conductor relative to the magnetic field reverses the direction or polarity of the induced e.m.f. or current.

A machine that converts mechanical energy into electrical energy or electrical energy into mechanical energy is called a *Dynamo*. When it changes mechanical energy into electrical energy, a dynamo is called a *Generator*; but when it changes electrical energy into mechanical energy, it is called a *Motor*.

a. Generators

There are two classes of generators; the Alternating Current (A.C.) generator, and the Direct Current (D.C.) generator.

(i) Simple Alternating Current (A.C.) Generator

Structure The A.C. generator consists of

a. *an armature* — a rectangular coil consisting of a large number of turns of insulated wire wound on a laminated soft iron core.

b. A *magnetic field* created by the curved poles of a horse-shoe magnet or an electromagnet.

c. Two copper *slip rings* R_1, R_2, to which the ends of the rectangular coil are connected and which rotate with the armature.

d. Two stationary *carbon brushes* which are made to press lightly against the slip rings.

Principle of Operation

When the armature (coil) *ABCD* rotates at a steady speed about a fixed axis in the magnetic field of the electromagnet, it cuts through the magnetic lines of force. As a result e.m.f. is produced in the terminals of the coil by electromagnetic induction.

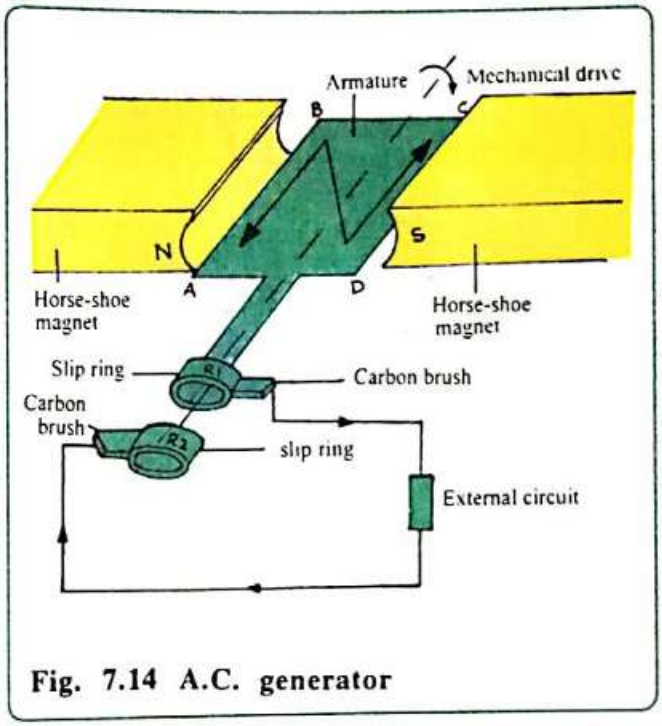
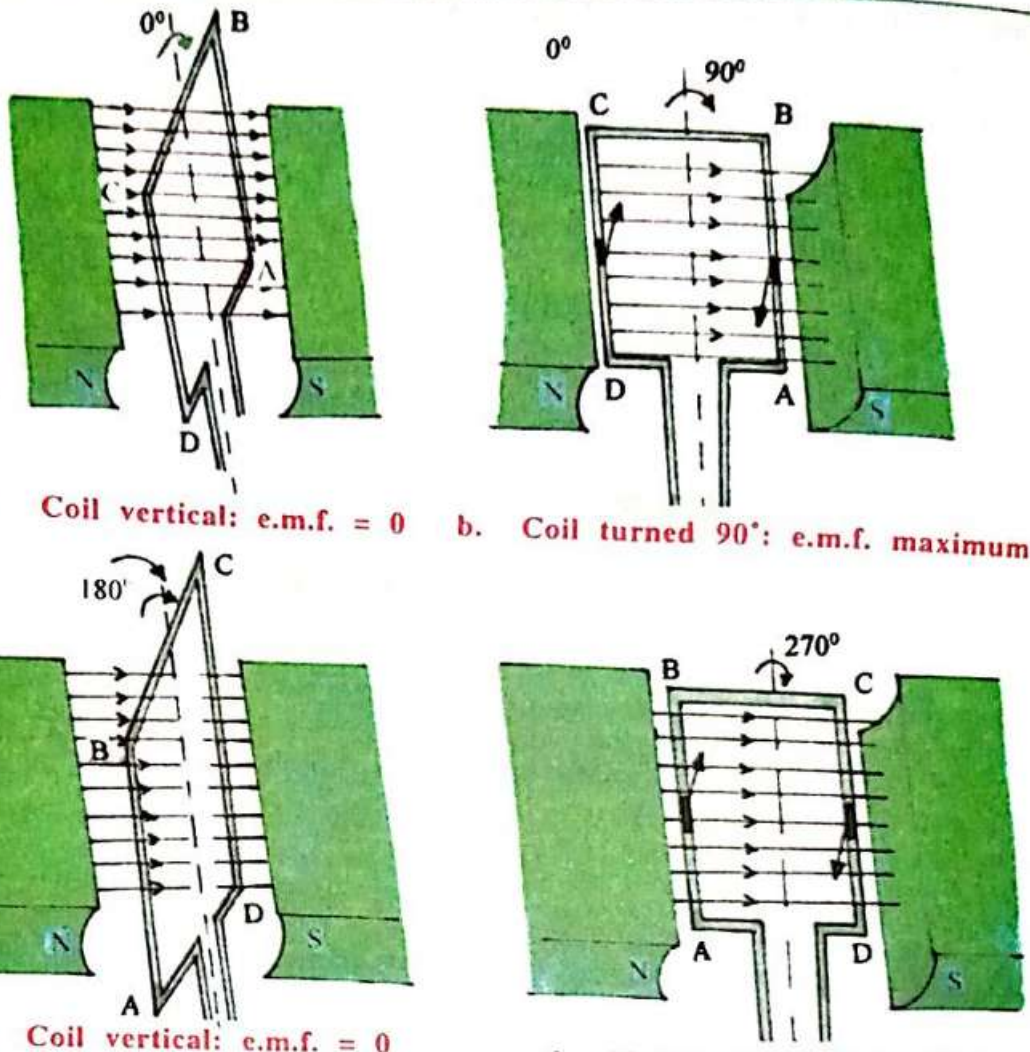


Fig. 7.14 A.C. generator

Let us consider a complete rotation of the armature *ABCD* as shown in fig 7.15(a-d). In (a) the plane of coil is vertical. *AB* and *DC* are moving parallel to or along the lines of force. No cutting of flux is taking place, hence no e.m.f. is induced at this instant. Though the number of lines linking the coil is maximum, the rate of change of flux is zero. In (b) the plane of the coil is parallel to the field. Though no lines of force link the coil, the rate of change of flux is maximum at this position. Sides *AB* and *DC* are moving at right angles to the field. E.m.f is induced along the sides as shown in accordance with the Fleming's right hand rule. In between (a) and (b) the induced e.m.f. changes continually from zero at (a) to a maximum or peak value at (b). Further rotation of the coil decreases the value of the induced e.m.f. until at last it becomes zero when the coil is again in the vertical position as in (c). During the second half of the rotation the induced e.m.f. follows the same pattern as that in the first half, except that the e.m.f. is induced in the coil in the reverse direction due to the direction of motion of *AB* and *CD* during this half of the continued clockwise rotation of the coil.

The e.m.f. in the coil thus reverses each time the coil passes the vertical. There are two such reversals in each complete revolution of the coil.

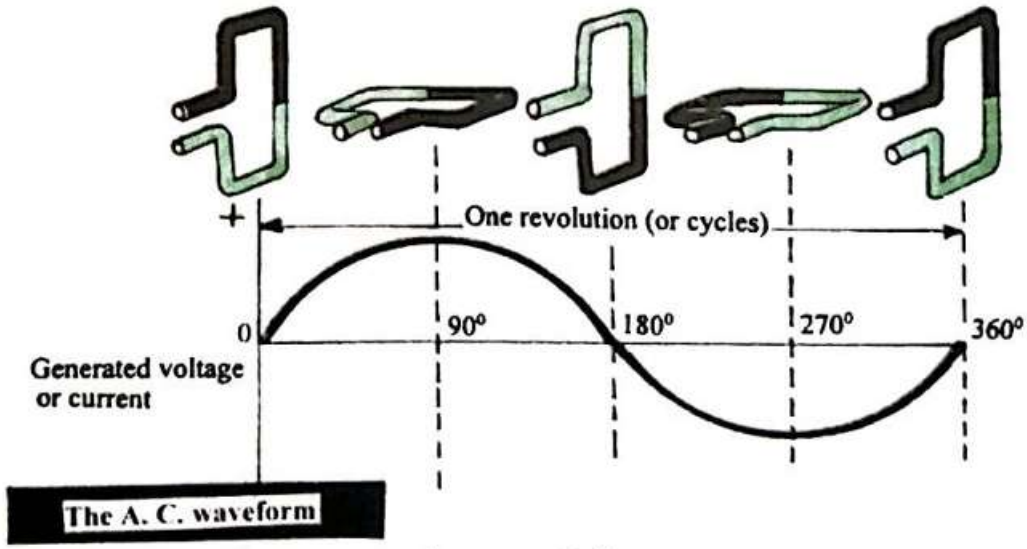
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a. Coil vertical: e.m.f. = 0 b. Coil turned 90°: e.m.f. maximum

c. Coil vertical: e.m.f. = 0 d. Maximum reversed e.m.f.

Fig. 7.15 Action of simple a.c. generator



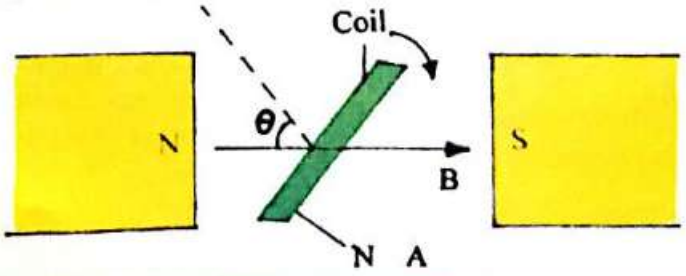


Fig. 7.16a

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An alternating voltage whose waveform is as shown in fig 7.16 is obtained at the terminals by means of carbon brushes and slip rings.

E.m.f. from a generator: $E = E_0 \sin \omega t$

We wish to show that the e.m.f. generated by the a.c. generator and whose waveform is as shown in fig 7.16a can be represented by the equation $E = E_0 \sin \omega t$.

Let the coil of area A rotate in a magnetic field of strength B. Let the normal to the coil make an angle θ, with the field as shown in fig. 7.16b.

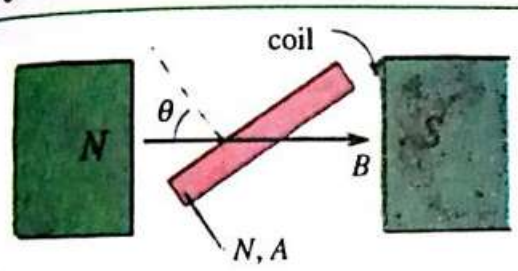


Fig. 7.16b Coil inclined in magnetic field, B

Flux Φ through the coil is then given by

$$\Phi = AB\cos\theta \qquad 7.2$$

If there are N number of turns in the coil then flux is given by:

$$\Phi = NAB\cos\theta \qquad 7.3$$

Let the coil turn with a steady angular velocity ω given by

$$\omega = \frac{d\theta}{dt} \qquad 7.4$$

Induced e.m.f. is given by

$$E = -\frac{d\Phi}{dt} \qquad 7.5$$

The above equation follows from Faraday's and Lenz's laws.

Hence

$$E = -\frac{d\Phi}{dt} = \frac{d}{dt}(NAB\cos\theta)$$

$$= -NAB\frac{d}{dt}(\cos\theta)$$

$$= NAB\sin\theta\frac{d\theta}{dt} \qquad 7.6$$

But from equation 7.4, $\omega = \frac{d\theta}{dt}$

$$\therefore \quad E = NAB\omega\sin\theta \qquad 7.7$$

When $\theta = 90°$, equation 7.7 becomes

$$E = NAB\omega = E_0 \qquad 7.8$$

where E_0 is the maximum value of the induced e.m.f., which occurs when the coil is parallel to field direction.

Putting equation 7.8 in equation 7.7 we have

$$E = E_0 \sin\theta. \text{ But } \omega = \frac{\theta}{t} \text{ or } \theta = \omega t.$$

$$\therefore \quad E = E_0 \sin \omega t \qquad 7.9$$

Alternating Current

If the alternating e.m.f. produced by the rotation of the armature is applied to an external resistance, an alternating current (A.C) is obtained. This A.C. is conducted to the external circuit or resistance by means of two carbon brushes B_1 and B_2 which are made to press lightly against the slip rings R_1 and R_2.

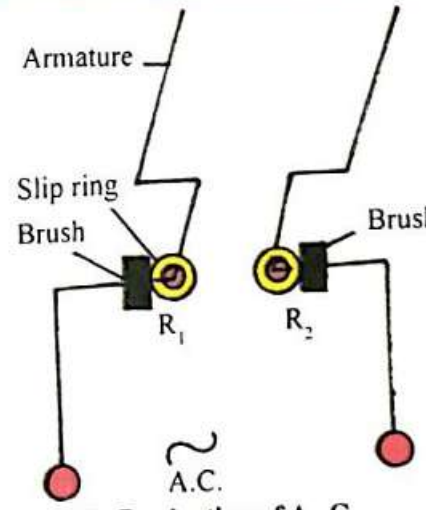


Fig. 7.17 Production of A . C.

An alternating current (a.c) is a current which changes continually in direction while passing through a conductor

Measurement of A.C.

An alternating current can be measured by

1 the Hot-wire ammeter — using the heating effect of current,
2 the moving iron ammeter — using the magnetic effect of current and
3 a moving coil galvanometer to which a rectifier is attached to convert the a.c. to a d.c.

Hot-wire Ammeter

The principle of operation of the Hot-wire ammeter is based on the heating effect of electric current.

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The working components of the instrument are as shown in fig 7.18.

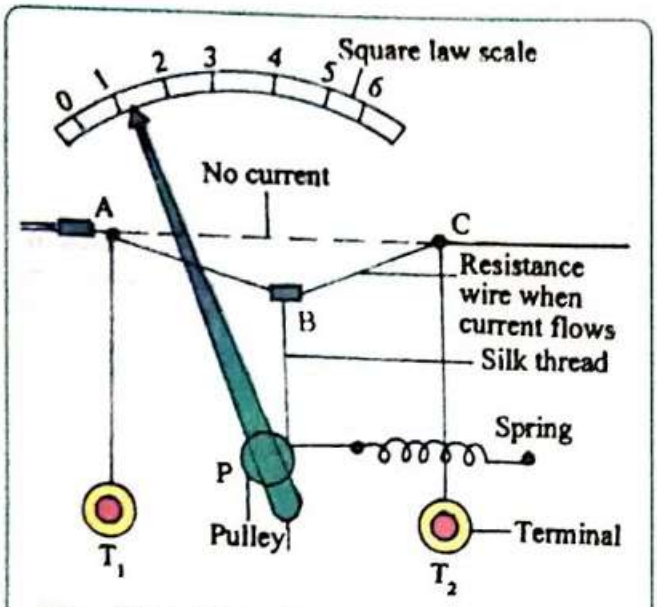


Fig. 7.18 Hot-wire ammeter

The current to be measured flows through a fine resistance wire AC which is connected to the terminals T_1 and T_2 of the instrument. A silk thread connected to the midpoint B of the resistance wire passes round the pulley P to a spring S.

When a current flows along the wire, it heats up and expands. The wire thus becomes slack and this slackness allows the silk thread to be pulled round the pulley by the action of the spring. So the pointer attached to the pulley, P, moves over a scale.

The scale is calibrated by passing currents of known values through the wire. The scale is non-uniform-scale, i.e. the markings are not uniformly spaced. This is because the heat produced in the wire is proportional to the square of the current ($H = I^2Rt$). The scale is therefore a 'square-law scale'.

Since the heating effect of current does not depend on the direction of current, the Hot-wire ammeter is suitable for measuring an alternating current and a direct current.

Moving Iron Ammeter

The ammeter consists of a solenoid inside which are two pieces of soft iron — a fixed piece and a moving piece. The pointer which moves over a non-linear scale is attached to the moving piece of soft iron.

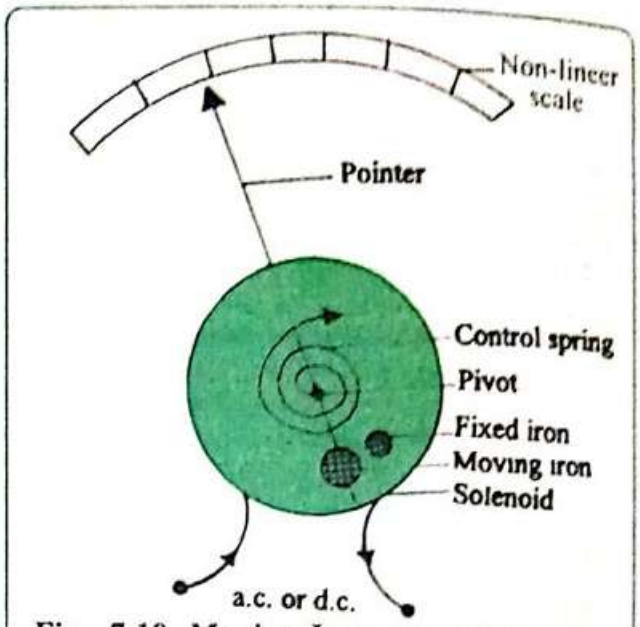


Fig. 7.19 Moving-Iron ammeter

When current passes through the instrument, both the moving and the fixed pieces of soft iron become magnetized with like poles facing each other. The movable soft iron piece is thus repelled from the fixed soft iron piece and the pointer attached to the moving piece rotates across the scale. The angle of rotation of the pointer is a measure of the current. The pointer will come to rest when the force of repulsion is balanced by the opposing force due to the control spring.

The advantages of this instrument are (i) it is simple and cheap to make. (ii) It can measure both d.c. and a.c. since both soft iron pieces are always magnetized with like poles facing each other whichever the direction of the current. The instrument is however less accurate than a moving coil galvanometer and it has a non-linear or non-uniform scale. (iii) A moving coil galvanometer, on its own, cannot measure a.c., but when a rectifier is attached to it, this converts the a.c. to d.c. which can then be measured by the galvanometer.

Direct Current (D.C.) Generator

A direct current (d.c) is one which always flows in one direction even though it may vary in value.

An a.c. generator can be made to produce a d.c. by replacing the two slip rings with a single split-ring or commutator as in electric motor. A split-ring commutator is a slip-ring that has been split into two segments which are insulated from each other. The ends of the coil are connected one to each split ring or commutator segment as shown in fig. 7.20.

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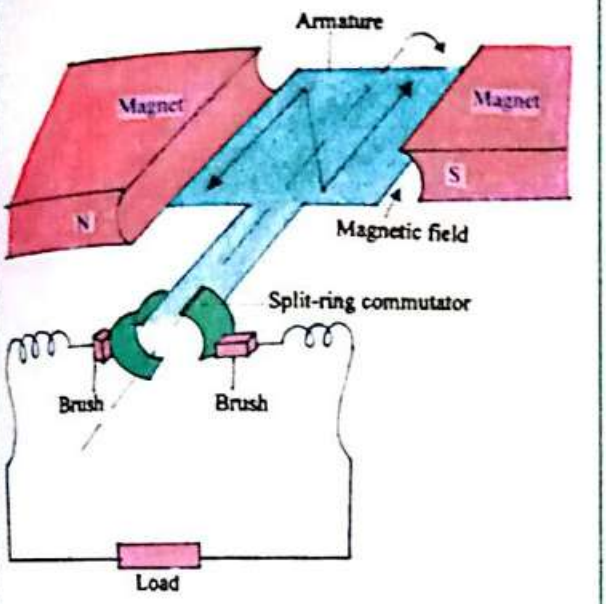


Fig. 7.20 Direct current generator

The split ring or commutator is a current reverser. When the armature coil is rotated, the commutator automatically switches each end of the coil from one brush to the other each time the coil completes one-half a revolution. As the current reverses in the coil after each half revolution, the connections between the coil and the brushes are reversed through this action of the commutator.

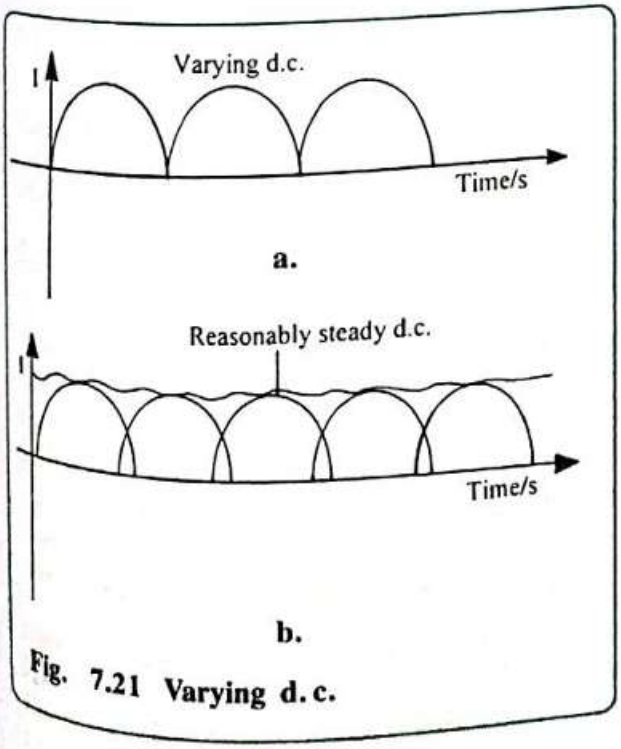


Fig. 7.21 Varying d.c.

This reversal occurs at the moment when the coil is vertical (i.e. when current is zero). The induced current appears in an external circuit always in the same direction as shown in fig. 7.21. Such a voltage is however a varying or pulsating direct voltage. It is not similar to constant voltage obtained from a battery.

To obtain a reasonably steady direct voltage or current, the armature is constructed with several coils mounted at equal angles to each other, and the commutator is also divided into the same number of segments as the coil, with each coil connected to two segments. The output voltage is as shown in fig. 7.21b. It is nearly maximum at all times, and has only a slight ripple variation.

Practical Generators

To produce larger e.m.f.'s in practical a.c. or d.c. generators,

1 The armature is constructed with a large number of turns in the coil.
2 The coil is wound on a soft iron core so as to increase the magnetic flux through the coil.
3 The strength of the magnetic field is made as high as possible.
4 The armature is made to rotate at a fast rate.

Comparison of d.c. generator and an electric motor

If we compare the d.c generator discussed above and the electric motor discussed on p. 420, we will notice that both machines are identical in construction. If a battery is connected to a simple d.c. generator, it will run as an electric motor. Conversely if the armature of the electric motor is made to rotate, it will act as a d.c. generator delivering current at the terminals. However when used in this way the electric motor produces an e.m.f. in opposition to that applied to it. This is called a back e.m.f.

7.11 The Transformer

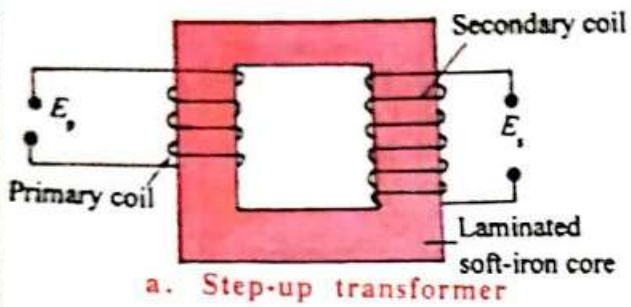
A transformer is an electrical device for changing the size of an a.c. voltage. It acts to increase or decrease the e.m.f. of an alternating current.

Structure

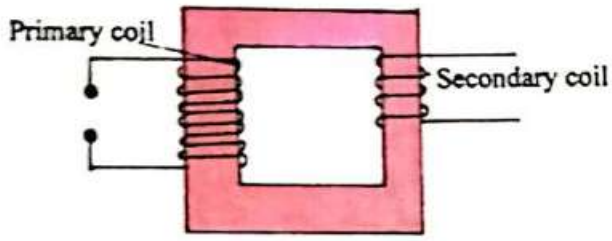
In its simplest form, it consists of two separate sets of coils, the *Primary coil* and the *Secondary coil*. The primary coil is the input winding of turns of wire and the secondary coil is the output winding. The coils are wound round a soft-iron core. The soft-iron core acts to increase and

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concentrate the magnetic flux within the core. It is also laminated, i.e., it consists of sheets of soft-iron insulated from each other instead of a solid block of iron. This lamination reduces loss of energy in the form of heat due to *eddy currents* induced in the core.



a. Step-up transformer



b. Step-down transformer

Fig. 7.22 Transformer

Principle of operation

When an alternating e.m.f. or a.c. voltage (E_p) is applied at the terminal of the primary coil (P) an alternating magnetic flux is produced in the iron core which links or threads the secondary coil (S). An alternating e.m.f. (E_s) of the same frequency as that of E_p is induced in the secondary coil by *mutual inductance*.

Mutual Inductance is the flow of induced current or voltage in a coil due to an alternating or varying current in a neighbouring coil.

The same flux or lines of force, link each turn of the primary and secondary coils. Hence the total flux linking the two coils is proportional to their number of turns, N_p in the primary coil and N_s in the secondary coil. The induced e.m.f. in the secondary coil (E_s) depends on the e.m.f. in the primary coil and on the ratio of the number of turns in each coil. It can be shown that

$$\frac{E_s}{E_p} = \frac{N_s}{N_p} \qquad 7..10$$

or $$\frac{\text{Secondary E.m.f. } (E_s)}{\text{Primary E.m.f. } (E_p)} = \frac{\text{No. of turns in the Secondary coil}}{\text{No. of turns in the Primary coil}}$$

The above relation is a very important transformer law.

In an ideal transformer, with a 100% efficiency, the power developed in the secondary coil is equal to the power developed in the primary coil. This follows from the law of conservation of energy.

Thus

$$I_s \times E_s = I_p \times E_p \qquad 7.11$$

where I_s, I_p, are the secondary and primary currents. Hence

$$\frac{E_s}{E_p} = \frac{I_p}{I_s} \qquad 7.12$$

But from equation 7.10 we have

$$\frac{E_s}{E_p} = \frac{N_s}{N_p}$$

$$\therefore \quad \frac{E_s}{E_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s} \qquad 7.13$$

Hence *the currents in the two coils are inversely proportional to the number of turns in the coils but the voltages are directly proportional to the number of turns in the coils.*

To use a transformer to increase an applied voltage, i.e. to make E_s greater than E_p, we find from equation 7.10 that Ns must be made greater than N_p. Such a transformer which increases or *steps up* the applied or primary voltage is called a *Step-Up transformer*. However we see from equation 7.13 that when $N_s > N_p$, then $I_s < I_p$. Hence in a step-up transformer, the primary current is greater than the secondary current but the primary voltage is less than the secondary voltage.

Another type of transformer decreases or *steps down* the primary voltage. It is called the *step-down transformer*. In this transformer $E_p > E_s$

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and hence $N_p > N_s$ and $I_p < I_s$. Thus we see that in a step down transformer, there are more turns in the primary coil than in the secondary coil and more current flows in the secondary circuit than in the primary.

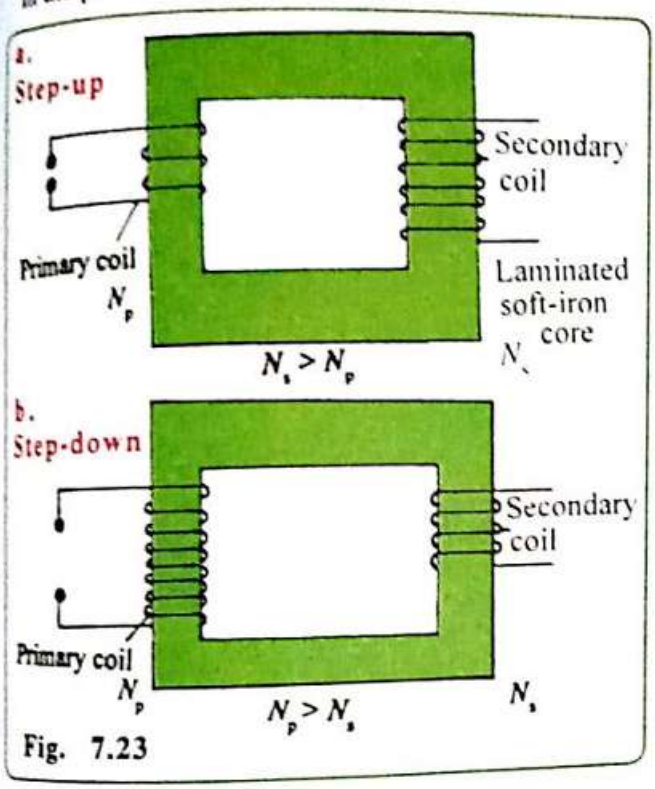


Fig. 7.23

The ratio $\frac{N_s}{N_p}$ is called the *turns ratio* of the transformer. It is more than 1 for a step-up transformer and less than 1 for a step-down transformer.

Energy losses in practical transformers

The relationships discussed above apply to an ideal transformer where there are no energy losses, and the transformer efficiency is 100%. In practical transformers however, there are energy losses due to:

1 **Eddy Currents** Such currents produced by the varying flux cutting the iron core, reduce efficiency because they consume power from the primary. Energy is lost in the form of heat in the iron core due to eddy current. Such loss can be reduced by laminating the core. Laminations of core reduces eddy currents by breaking up their path of flow.

2 **Hysteresis Loss** is wasted energy due to reversing the magnetization of core. The core is made to go through a cycle of magnetization during each alternating cycle of the primary current. Hysteresis loss is reduced by the use of special alloys in the core of the primary coil, or by the use of soft iron cores.

3 **I^2R (or Heat) Loss** Because the primary and secondary coils have resistance, some energy is lost in the form of heat (I^2R) in the coils. This heat loss can be reduced by using thick wires, or low resistance coils.

4 **Leakage of Magnetic Flux** Some energy is lost due to leakage of magnetic flux. This arises because not all the lines of induction due to current in the primary coil pass entirely through the iron core to the secondary coil. This loss is reduced by special forms of coil winding, or by efficient core design.

It is because of the above losses that the efficiency of practical transformers is less than 100%.

The efficiency of a transformer is defined by

$$\text{Efficiency} = \frac{\text{Output Power}}{\text{Input Power}} \times 100\%$$
$$= \frac{\text{Power in the secondary coils}}{\text{Power in the primary coils}} \times 100\%$$

Example 7.1

Find the turns ratio in a transformer which delivers a voltage of 120 volts in the secondary coil from a primary voltage of 60 volts.

Solution

$$\text{Turns ratio} = \frac{N_s}{N_p}$$

N_s = No of turns in the secondary coil

N_p = No of turns in the primary coil

$$\frac{E_s}{E_p} = \frac{N_s}{N_p}$$

$$\frac{120}{60} = \frac{N_s}{N_p} = 2$$

Turns ratio = 2

Example 7.2

A transformer has 500 turns in the primary coil and 300 turns in the secondary coil. If the primary coil is connected to a 220 V mains, what voltage will

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be obtained from the secondary coil? What type of transformer is this?

Solution

$$\frac{E_s}{E_p} = \frac{N_s}{N_p}$$

$$E_s = E_p \times \frac{N_s}{N_p}$$

$$= 220 \times \frac{300}{500}$$

$$= 132 \text{ volts}$$

It is a step-down transformer because secondary voltage is less than primary voltage (132 < 220) and also turns ratio is less than 1.

Example 7.3

a. Draw a labelled diagram to explain the working of a transformer which can produce 24 V from a 240 V supply.

b. If the efficiency of the transformer is 80% and the current in the secondary is 10 A, calculate the current in the 240 V supply.

c. Give two reasons which explain why the efficiency of the transformer cannot be 100%.

(SC/GCE)

Solution

a. The turns ratio $\frac{N_s}{N_p} = \frac{24}{240} = \frac{1}{10}$

In the diagram, there should be more turns in the primary than in the secondary coil and the ratio $\frac{N_p}{N_s}$ should be 10,

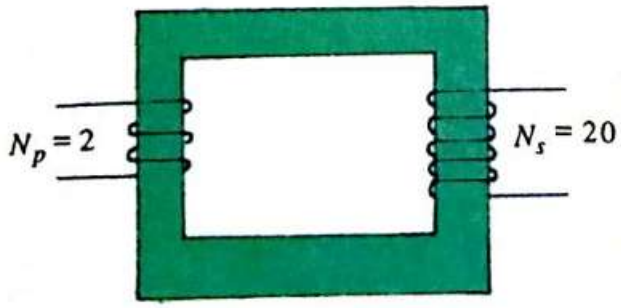


Fig. 7.24

b. Efficiency is 80%

i.e. $\frac{\text{Power in the secondary coils}}{\text{Power in the primary coil}} = 80\%$

$$\frac{I_s E_s}{I_p E_p} = \frac{80}{100}$$

But $\frac{E_s}{E_p} = \frac{24}{240}$

$$\therefore \quad \frac{24 I_s}{240 I_p} = \frac{80}{100}$$

$I_s = 10$ A (Given)

$$\therefore \quad I_p = \frac{100}{80} \times \frac{24}{240} \times 10$$

$$I_p = \frac{100}{80}$$

$$= \underline{1.25 \text{ A}}$$

c. See text under 'energy loss in the transformer'.

Example 7.4

A transformer supplies 15 V from a 220 V mains. If the transformer takes 0.7 A from the mains when used to light three lamps connected in parallel and each rated 15 V, 40 W, calculate

(i) the efficiency of the transformer

(ii) the cost of using it for 24 hrs at 30 k per kWh.

Solution

Primary or Input Power = $I_p V_p = 0.7 \times 220$

$= 154$ W

Secondary or Output Power = $I_s V_s = (I_s \times 15)$ W

From Power = IV, we have that

$$I = \frac{\text{Power}}{\text{Voltage}} = \frac{P}{V}$$

I_s = Output current in each lamp $= \frac{40 \text{ A}}{15} = 2.67$ A

Total current taken by the 3 lamps in parallel

$= 3 \times 2.67 = 8$ A

Output power = $I_s \times 15 = 8 \times 15 = 120$ W

(or directly from total power = 3×40 W = 120 W)

$$\text{Efficiency} = \frac{\text{Output Power}}{\text{Input Power}} \times 100\%$$

$$= \frac{120}{154} \times 100\%$$

$$= \underline{78\%}$$

Power consumed = (0.7×220) W

$$= \frac{0.7 \times 220}{1000} \text{ kW}$$

Total Power consumed in 24 hrs

$$= \frac{0.7 \times 220}{1000} \times 24 \text{ kWh}$$

Cost at 30 k per kWh

$$= \left(\frac{0.7 \times 220}{1000} \times 24 \times \frac{30}{100}\right) \text{ Naira} = \underline{₦1.1}$$

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Fig. 7.25

7.12 Power Transmission

Power generated at power stations are distributed over large distances to consumers through metal cables. Power is given by $P = IV$, and can thus be transmitted either at low current and high voltage or at high current and low voltage.

Because the metal cables through which the power is transmitted have a certain amount of electrical resistance, transmitting power at high current will lead to loss of energy in the form of heat in the cables ($H \propto I^2R$). To avoid this, power is therefore transmitted at high voltage and low current. This is known as *High tension transmission*.

Low currents lead to low energy loss in form of I^2R. Also since lower currents require thinner cables, cost of cable materials is considerably reduced if power is transmitted with low currents and high voltage.

Step down transformers are used to reduce the high transmitted voltages to lower voltages required in homes and factories.

Advantage of A.C. over D.C. in Power Transmission

A.C. is preferred to D.C. in high tension transmission because the A.C. can easily be transformed to high voltages with very low energy loss. A.C. voltages can be stepped up or down by transformers while D.C. voltages cannot.

A transformer cannot of course be operated with a D.C. because as soon as the current has attained a steady value in the primary coil there is no change in the magnetic flux in the core and therefore no induced e.m.f. in the secondary coil. D.C. would require the use of motors and dynamos in order to produce a voltage change. This will bring about greater power losses than when A.C. is used. A transformer can be modified to step up a D.C. voltage by connecting a *make-and-break device* at the primary circuit.

7.13 The Induction Coil

This is an electrical device that is capable of producing a very high intermittent e.m.f. by electromagnetic induction from a low voltage d.c. source e.g. a battery.

Structure

It consists of (i) a primary coil, made up of a few turns of insulated thick copper wire, wound on a laminated soft iron core. The lamination of the core is to reduce loss of energy due to *eddy currents*. (ii) Secondary coil overlapping the primary coil. The secondary coil is made of a large number of turns of insulated thin copper wire. The ends of this coil are connected to an adjustable spark gap created by two adjustable brass rods. (iii) In front of the soft iron core of the primary coil is a *make-and-break-device* consisting of a soft iron armature carried on a light brass spring to which is attached a platinum contact with adjustable screw. The primary circuit is completed from this contact through a key and the battery which provides the current.

Principle of operation

When the primary circuit is completed through pressing the key, current flows in the primary coil and magnetizes the soft-iron core on which the coil is wound. The soft-iron armature is attracted by the magnetized core, thereby causing a break in the circuit at the point. As a result the magnetic field through the core and the primary coil collapses rapidly. The armature then flies back and remakes

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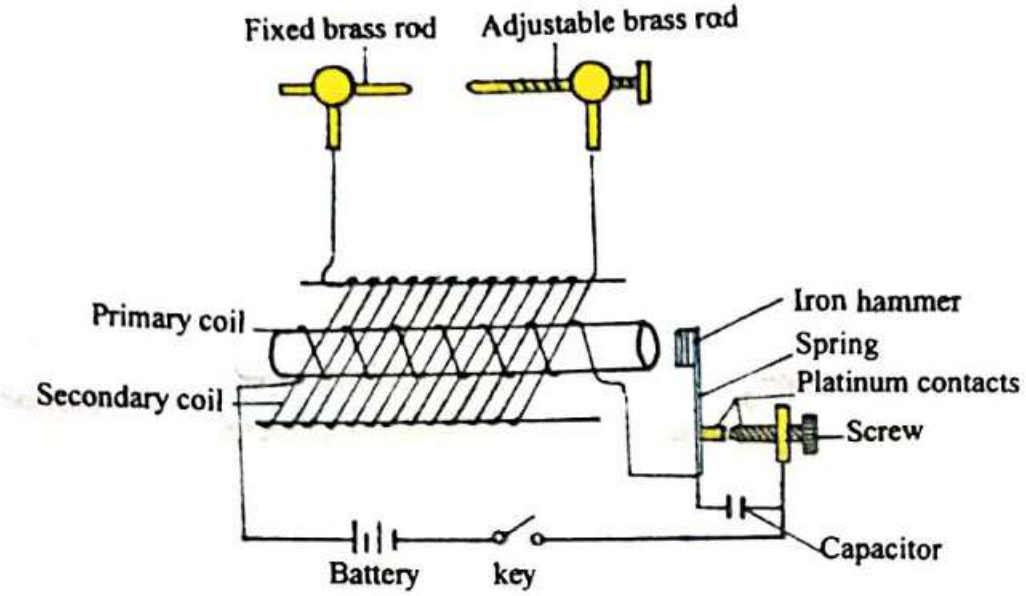


Fig. 7.26 Induction coil

contact with the adjustable screw through the spring action of the brass spring. The circuit is hence completed once again and the process described above is repeated rapidly.

This action of the *make-and-break device* causes rapid changes in the magnetic flux through the primary coil. This induces a rapidly changing e.m.f. in the secondary coil, and because the number of turns of the secondary coil is very large compared to those of the primary coil, a large e.m.f. is produced in the secondary coil. This e.m.f. is large enough to cause a spark across the air gap. The capacitor (C) connected between the armature and the adjustable screw serves to reduce the sparking produced across the gap of the *make-and-break device*, due to the back e.m.f. generated in the primary coil.

Uses

The induction coil is commonly used in motor car ignition systems and in the operation of X-ray tubes.

Summary

1 Electromagnetic field is the field due to the interaction of electric and magnetic forces.

2 A current-carrying conductor placed in a magnetic field experiences a force whose direction is given by Fleming's left-hand rule. The force is always directed perpendicular to the direction of current and the magnetic field. No force acts on the conductor if placed parallel to the magnetic field.

3 An electric motor is an application of electromagnetic field. It consists of a rectangular coil of insulated wire (the armature) that rotates in a magnetic field between two pole pieces of a horse-shoe magnet. A split-ring commutator reverses the direction of the current in the coil after each half a cycle so that the coil rotates continually.

4 Another application of electromagnetic field is the moving coil galvanometer, a very sensitive instrument for detecting and measuring small currents or p.d.'s. In the galvanometer, a rectangular coil mounted on a soft-iron core rotates in a radial magnetic field provided by two pole pieces of a permanent magnet. The instrument has a uniform scale due to the radial field. A strong magnetic field and a weak spring increases the sensitivity of the instrument.

5 *An induced e.m.f. (or current)* is always produced in a circuit whenever there is a change in the magnetic flux linking the circuit. To increase the induced e.m.f. in a coil rotating in a magnetic field we increase (i) the area of coil (ii) the number of turns in the coil (iii) the speed of rotation of the coil (iv) the strength of the magnetic field.

6 There are two laws of electromagnetic induction (i) *Faraday's law* states that

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whenever there is a change in the magnetic flux linking a circuit, an e.m.f. is induced which is directly proportional to the rate of change of the magnetic flux. (ii) *Lenz's law* states that an induced e.m.f. flows in such a direction as to oppose the motion producing it. Lenz's law illustrates the principle of conservation of energy. This is because the energy expended as induced current is derived from the work done when the magnet is moved.

7 In an A.C. generator, an insulated rectangular coil rotates steadily in a magnetic field. The a.c. is obtained through the action of slip-rings and brushes. To obtain a d.c. from the generator, the slip-rings are replaced by a split ring commutator.

8 In a transformer, an a.c. voltage at the primary coil can be changed to an a.c. voltage at the secondary coil. An important transformer equation is $\frac{E_s}{E_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$.

The transformer operates in the following way: an a.c. flowing in the primary coil creates a continuously varying magnetic flux through the core. This varying flux induces an alternating e.m.f. in the secondary coil.

Energy losses due to (i) Eddy currents (ii) Hysteresis (iii) Heat (in the form of I^2R) in the coil (iv) leakage of magnetic flux, reduce the efficiency of the transformer to less than 100%.

A transformer operates on an a.c. A d.c. in the primary does not produce the changing flux and therefore cannot induce an e.m.f. in the secondary.

9 It is preferable to transmit electrical power at a low current and a high voltage. This ensure little power loss due to heat (I^2R) in the cables. Lower currents also require cheaper cables.

That is, mechanical energy (the Kinetic energy due to the motion of the bar magnet) is changed to electrical energy in the form of induced current.

Exercise 7

1 The current produced by a simple dynamo is not steady because
- A a back e.m.f. opposes the induced voltage
- B eddy currents oppose the motion which induces them, and absorbs energy from the current.
- C the magnetic field produced by the magnet is not sufficiently uniform.
- D the induced current opposes the motion which causes it, in accordance with Lenz's law.
- E the rate of change of flux varies with the position of the coil of the dynamo.

(SC/GCE)

2 Induced current depends on the
1. Number of turns in the coil.
2. Strength of the magnet.
3. Speed with which the magnet is plunged into the coil.

Which of these is/are false?
- A I only
- B II only
- C II and III only
- D III only
- E None of the above

(SC/GCE)

3 To convert an alternating current dynamo into a direct current dynamo, the
- A number of turns in the coil is increased
- B strength of the field magnet is increased
- C slip rings are replaced with split ring commutator
- D coil is wound on a soft iron armature

(JAMB)

4 Which of the following operations will not lead to an increase in the induced e.m.f. in a coil of wire rotating between the poles of a magnet? Increasing the
- A area
- B strength of the magnet
- C gap between the poles of the magnet
- D number of turns in the coil
- E speed of rotation of the coil.

(SC/GCE)

5 Which of the following statements about a generator is not correct?
- A It can produce direct current.
- B It can produce alternating current.
- C It requires an external supply of energy to rotate the coil.
- D It requires an external supply of current to the coil.
- E It may require the use of a commutator.

(SC/GCE)

6 Which of the following devices would be used on its own in the working of a petrol-driven motor car engine for obtaining a high voltage from a low one?
- A Induction coil
- B A.C. dynamo
- C D.C. generator
- D The transformer
- E The electric motor

(SC/GCE)

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7 A transformer with 5500 turns in its primary is used between a 240 V a.c. supply and a 120 V kettle. Calculate the number of turns in the secondary.

A 11,000 B 2750 C 460
D 232 E 10

(SSCE)

8 What is the effect of using split-rings in a simple d.c. motor?

A The direction of rotation of the coil is reversed.
B The current in the coil flows in the same direction.
C The current in the coil becomes alternating.
D The direction of force on the coil is reversed.
E The contact with the voltage source is cut off.

(SSCE)

9 Which of the following reduce(s) the effect of the back e.m.f. generated in the primary coil of an induction coil?

I The capacitor in the circuit
II The make and break contact in the circuit
III The ratio of turns in the secondary to that in the primary

A I only B II only
C I and II only D II and III only
E I, II and III.

(SSCE)

10 If a current-carrying coil is mounted on a metal frame, the back e.m.f. induced in the coil causes

A Inductance B Eddy currents
C Electromagnetism D Dipole moment

(JAMB)

11 a. What do you understand by *electromagnetic induction* and *induced electromagnetic force.*

b. Describe the working of a simple a.c. generator. How can it be modified to provide d.c.

12 State the laws of electromagnetic induction. Describe an experiment to illustrate Lenz's law.

With the aid of a diagram describe the principle of operation of an induction coil. Mention two applications of this device.

13 a. State the laws of electromagnetic induction.

b. Describe a simple experiment to show how an induced e.m.f. can be produced.

c. State two factors on which the magnitude of the induced e.m.f. depends.

(SSCE part question)

14 With the aid of a diagram describe the action of a transformer. Why is the core laminated? Why is the efficiency of practical transformers less than 100%

15 A transformer designed to convert a 220 V a.c. mains to 10 V has 3300 turns on the primary coil. How many turns should be in the secondary coil to attain this conversion? Assume the transformer is 100% efficient.

16 What do you understand by *Eddy Currents*? How are they reduced in the transformer? Write down 3 other processes of energy loss in a transformer. How can such losses be reduced?

A transformer is designed to convert a 25 V supply to an output of 240 V. If the transformer is 90% efficient, calculate the current in primary winding when the output terminals are connected to a 240 V 80 W lamp.

17 Distinguish between a *step-up* and a *step-down transformer*. Give two reasons why it is preferred to transmit power over long distances using a high voltage and a low current?

A step down transformer operates a 2400 Volt line and supplies a current of 60 A. The turns ratio of the transformer is $\frac{1}{24}$. Calculate the secondary voltage, the primary current and the power output, if the transformer is 100% efficient.

18 Describe with the aid of a labelled diagram, any one form of a.c. generator. Explain why the output is an alternating one. State for the generator you have described (i) the circumstances which determine the instant at which a peak (maximum) voltage occurs (ii) the factors which affect the magnitude of the peak voltage.

(SC/GCE)

19 Draw a carefully labelled diagram to show the working parts of a *moving coil galvanometer.* Explain how it works and indicate how its sensitivity may be increased.

(SC/GCE)

20 Draw the magnetic lines of force due to two current-carrying straight wires close to each other,

a. When the current in both wires flow in the same direction.

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b. When the currents flow in opposite direction. Indicate in your diagrams the directions of current, lines of magnetic field and the forces acting on the wires.

21 Draw a labelled diagram of the structure of a simple d.c. electric motor. Mark clearly the direction of the supply current and that of the rotation of the armature. Discuss the functions of the various parts. Why is the efficiency of the electric motor less than 100%?

22 Explain the principle of operation of the transformer. How can it be modified to step up a d.c. voltage?

A transformer with 50 and 3000 turns is used to step down 300 V a.c. Calculate the output voltage. Explain why the measured value of the output voltage would be less than the calculated value. How could the difference be reduced in practice?

(SC/GCE)

23 a. Draw a simple labelled diagram illustrating the principle of a step-down transformer and explain how it works.

b. State three ways by which energy is lost in a transformer and how they can be minimized.

c. If a transformer is used to light a lamp rated at 60 W 220 V from a 440 V a.c. supply, calculate the:

(i) ratio of the number of turns of the primary coil to the secondary coil in the transformer;

(ii) current taken from the mains circuit if the efficiency of the transformer is 95%.

(SSCE)

24 a. Describe with the aid of a labelled diagram, the mode of operation of a moving-coil galvanometer.

b. State *three* methods by which the sensitivity of a galvanometer can be increased.

c. State *two* essential differences between a moving-coil galvanometer and a d.c. generator.

d. Explain the term *eddy current* and state *two* devices in which the currents are applied.

(SSCE)

25 (a) State the law of electro-magnetic induction.

(b) Explain how one of the laws illustrates the principle of conservation of energy.

(c) (i) Draw a labelled diagram of a simple d. c. electric motor and explain how it works

(ii) State two reasons why the efficiency of an electric motor is less than 100%.

(SSCE)

26 (a) State Faradays laws of electromagnetic induction.

(b) Draw a labelled diagram of an induction coil and explain how it works.

(c) How is the effect of eddy currents minimized in the coil?

(d) State the reasons why a capacitor should be included in the primary circuit of the coil.

(e) State three uses of an induction coil.

(SSCE)

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8 SIMPLE A.C. CIRCUITS

A.C. circuits are circuits through which an alternating current flows. Such circuits are used extensively in power transmission, radio and television, computer technology, telecommunication and in medicine.

Alternating currents are produced by time dependent alternating voltages given by the relation $E = E_0 \sin \omega t$ as was derived in chapter 7. Much of what we learned about d.c. circuits also apply to a.c. circuits. However, the effects of coils and capacitors are much different in a.c. than in d.c. applications. The effect of such voltages on resistors, capacitors and inductors will be discussed in the present chapter.

At the end of this chapter, the student should be able to:

1 Explain the peak and r.m.s. values of current and p.d.
2 Establish the phase relationship between current and p.d in an a.c. circuit.
3 Explain reactance and impedance.
4 Determine current in circuits containing
   a. resistance and inductance;
   b. resistance and capacitance; and
   c. resistance, inductance and capacitance.
5 Determine power in an a.c. circuit.

8.1 Nomenclature in A.C Circuits

An Alternating Current (A.C.) is one that varies sinusoidally or periodically, in such a way as to reverse its direction periodically. The commonest form of such an a.c. can be represented by

$$I = I_0 \sin 2\pi ft = I_0 \sin \omega t \qquad 8.1$$

where I is the *instantaneous current* at a time t, I_0 is the maximum *(or peak) value of current* or its amplitude; f is the *frequency* and $\omega (= 2\pi ft)$ is the *angular velocity*, (ωt) is the *phase angle* of the current. Alternating voltage is also represented by

$$V = V_0 \sin 2\pi ft = V_0 \sin \omega t \qquad 8.2$$

Here, V, V_0 are the *instantaneous* and *peak* (or maximum) values of the voltage or its amplitude. For example, if an a.c. voltage is represented by the relation $V = 4 \sin 900\pi t$, the peak voltage $V_0 = 4$ V and $2\pi ft = 900\pi t$ or $f = \frac{900}{2} = 450$ Hz. Then $\omega = 2\pi f = 900\pi$.

8.2 Peak, and r.m.s. Values of A.C.

An alternating current (or voltage) varies sinusoidally as shown in fig. 8.1 which is a sine waveform. The amplitude or peak value of the current I_0, is the maximum numerical value of the current.

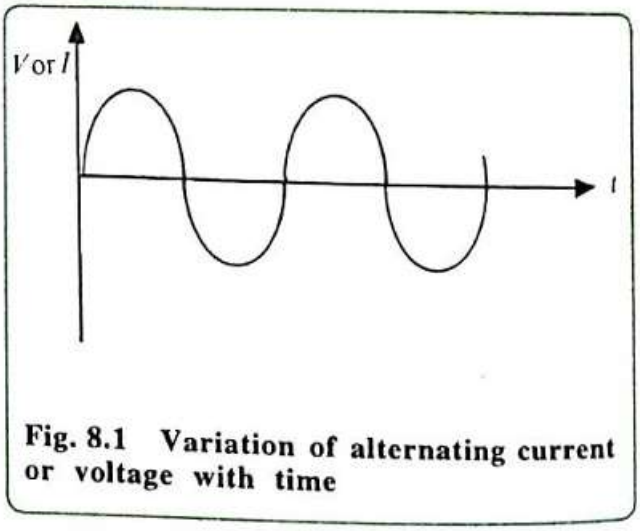


Fig. 8.1 Variation of alternating current or voltage with time

The root-mean-square (r.m.s.) value of the current is the effective value of the current.

Root - mean - square current is that steady current which will develop the same quantity of heat in the same time in the same resistance

The r.m.s. value for the current is given by

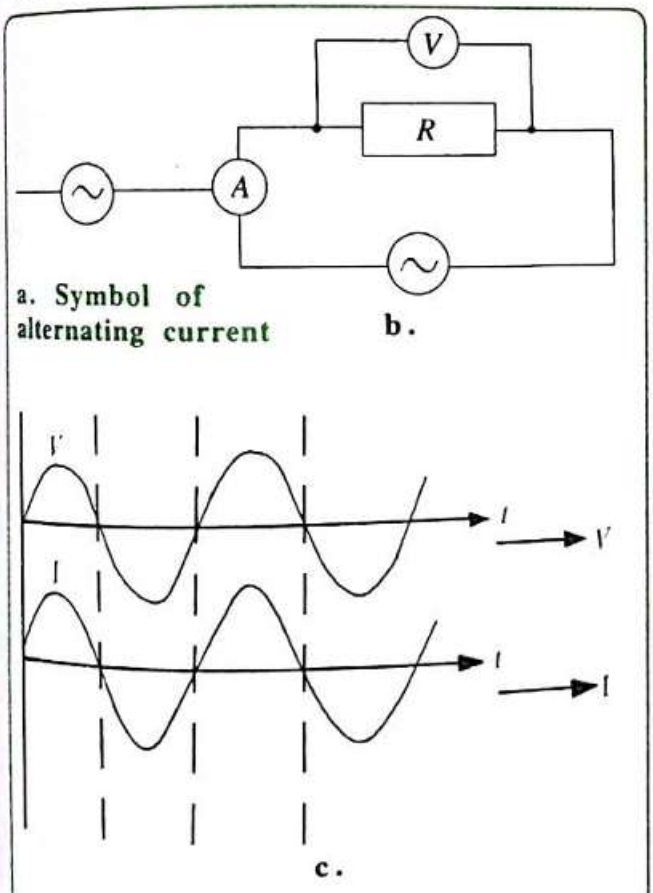
$$I_{r.m.s.} = \frac{I_0}{\sqrt{2}} = 0.707 I_0 \qquad 8.3$$

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The moving iron and hot-wire meters measure the average value of the square of the current called the mean-square current. They are however calibrated in such a way as to indicate the r.m.s. current directly. Thus most a.c. meters read the effective or r.m.s. values. The average value of an a.c. voltage or current is zero.

8.3 a. Resistance in A.C. Circuit

Fig 8.2 shows a simple a.c. circuit. The symbol of an a.c. voltage source is as shown in fig 8.2a. At any instant, the current through the resistor (R) is I and the voltage across it is V.



a. Symbol of alternating current b. c.

Fig. 8.2 Resistance in a.c. circuits, I and V in phase

From Ohm's law, we have that

$$V = IR$$

Thus the current is given by

$$I = \frac{V}{R}$$

If we put $V = V_0 \sin \omega t$, then the current is also given by

$$I = \frac{V}{R} = \frac{V_0 \sin \omega t}{R} \qquad 8.4$$

$$= I_0 \sin \omega t \qquad 8.5$$

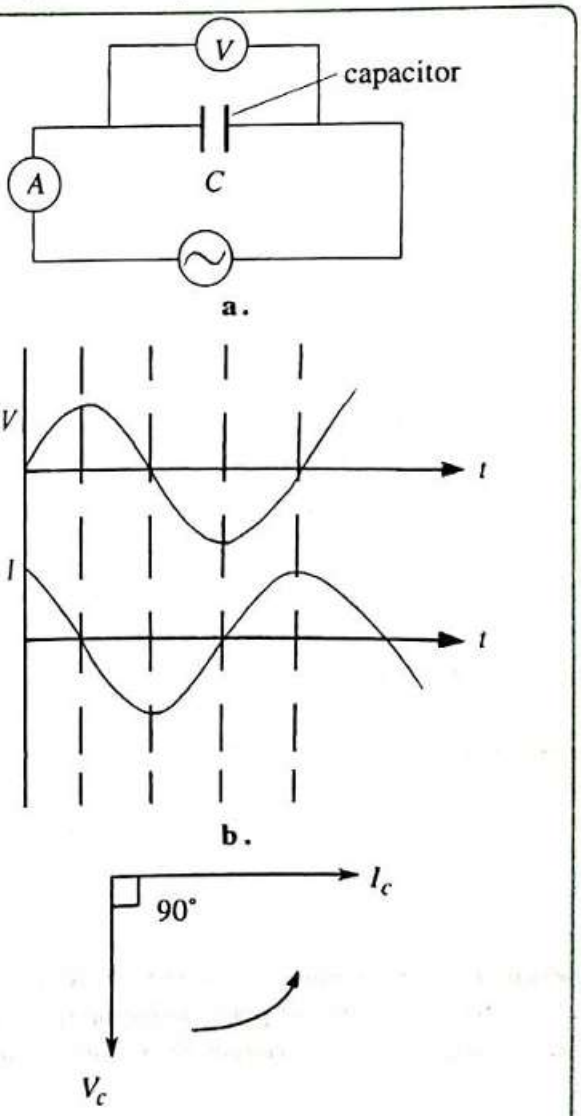
The voltmeter and ammeter connected in the circuit will read the r.m.s. values of voltage and current. Hence we can also write that

$$I_{\text{r.m.s.}} = \frac{V_{\text{r.m.s.}}}{R} \qquad 8.6$$

The voltage and the current are said to be in phase or in step with each other. *This means that both of them attain their maximum, zero and minimum values at the same instant in time* (fig. 8.2c).

b. Capacitance in an A.C. Circuit

Let us consider another simple circuit shown in fig 8.3a. where an a.c. voltage is connected in series with a capacitor of capacitance C (Farads).



a. b. c.

I_c leads V_c by $\frac{\pi}{2}$ radians or 90° or by $\frac{1}{4}$ cycle.

Fig. 8.3 Capacitance in an a.c. circuit

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Both experiment and theory show that the voltage (V) and current (I) are *out of phase* (not in step). The current is said to *lead* on the voltage and the voltage is said to *lag* on the current. The phase difference between the current and the voltage is 90° or ($\pi/2$) radians (fig. 8.3c).

Thus if $V = V_0 \sin \omega t$ then $I = I_0 \sin\left(\omega t + \frac{\pi}{2}\right)$.

The capacitor opposes the flow of current. *This opposition to the flow of a.c. offered by the capacitor is known as capacitive reactance, X_c.* This is given by the relation

$$X_c = \frac{1}{2\pi fC} \qquad 8.7$$

When an a.c. voltage of frequency f is applied to a capacitance, C, then

$$V = IX_c \qquad 8.8$$

In other words, an Ohm's law relation applies to a capacitor. In it, R is replaced by $X_c = \frac{1}{2\pi fC}$. Hence the unit of X_c is in ohms.

Example 8.1

A 2 μF capacitor is connected directly across a 150 V_{rms}, 60 Hz a.c. source. Find (a) the r.m.s. value of the current (b) the peak value of current.

Solution

From equation (8.7), $X_c = \frac{1}{2\pi fC}$

$$X_c = \frac{1}{2\pi \times 60 \times (2 \times 10^{-6})}\ \Omega$$

$$= \underline{1324.4\ \Omega}$$

From $V = IX_c$ we have

$$I_{r.m.s.} = \frac{V_{r.m.s.}}{X_c} = \frac{150}{1324.4}\ \text{A}$$

$$= \underline{0.113\ \text{A}}$$

Peak current, $I_0 = \sqrt{2} \times I_{r.m.s}$

$$= \underline{0.160\ \text{A}}$$

c. Inductance in A.C. Circuit

In fig 8.4a., an a.c. voltage is connected across an inductor L. The sinusoidal voltage causes a sinusoidal current to flow.

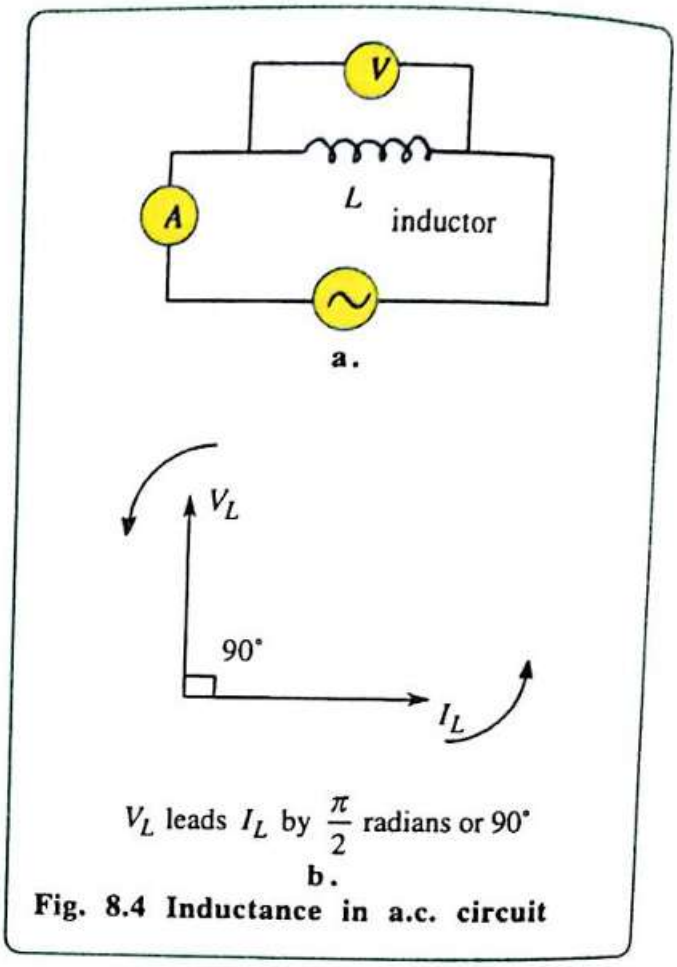


Fig. 8.4 Inductance in a.c. circuit

The induced e.m.f in the inductor L opposes the change in the current. As a result the current is delayed behind the voltage in the circuit. We say that the current I lags behind V by $\frac{\pi}{2}$ radians or 90° or by $\frac{1}{4}$ cycle. I and V have a phase difference of 90° (or $\frac{\pi}{2}$). Thus if $V = V_0 \sin \omega t$, then $I = I_0 \sin(\omega t - \pi/2)$. Like R and C, an inductor L opposes the flow of current; i.e. it has an impedance effect known as *inductive reactance*, X_L. Then

$$V = IX_L \qquad 8.9$$

where the unit of X_L is in ohms. Equation 8.9 is an ohm's law type of equation where the inductive

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reactance, X_L replaces the usual factor R. It can be shown that

$$X_L = 2\pi fL \qquad 8.10$$

where the unit of L is henrys (H), f is in hertz (Hz) and X_L is in ohms.

Reactance is the opposition to the flow of a.c. offered by a capacitor or an inductor or both.

Example 8.2

Find the impedance (inductive reactance) across an inductor of 0.2 H inductance when an a.c. voltage of 60 Hz is applied across it. If the voltage is given by $V = 150\sin 120\pi t$, calculate the r.m.s. and peak values of the current.

Solution

$X_L = 2\pi fL = 2\pi \times 60 \times L$ ohms

$= 120\pi L = 120 \times \pi \times 0.2$

$= \underline{75.40\ \Omega}$

If $V = 150\sin 120\pi t$, then we see that f = 60 Hz, V_0 = 150 volts (This follows from the relation, $V = V_0 \sin 2\pi ft$)

$V_{r.m.s} = 0.7V_0 = 0.7 \times 150 = 105$ volts

$$I_{r.m.s} = \frac{V_{r.m.s.}}{X_L} = \frac{105}{75.4}$$

$= \underline{1.39\ A}$

$$I_0 = \frac{V_0}{X_L} = \frac{150}{75.4}$$

$= \underline{1.99\ A}$

8.4 Series Circuit Containing Resistance (R), Inductance (L) and Capacitance (C)

Consider a circuit containing a Resistance (R), an inductance (L) and a Capacitance (C) in series, as shown in fig. 8.5.

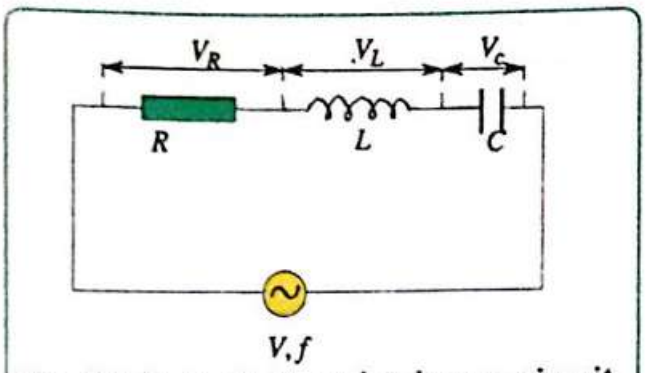


Fig. 8.5 R, L, C in series in a.c. circuit

If an alternating voltage $V = V_0 \sin 2\pi ft$ is put across the circuit, it is observed that a steady state current given by $I = I_0 \sin 2\pi ft$ will flow along the circuit. It can be shown that the maximum or peak value of the current is given by:

$$I_0 = \frac{V_0}{[R^2 + (X_L - X_c)^2]^{1/2}} = \frac{V_0}{\sqrt{R^2 + X^2}} \qquad 8.11$$

where $X = X_L - X_c$.

Putting $Z = [R^2 + (X_L - X_c)^2]^{1/2}$ equation 8.11 becomes

$$I_0 = \frac{V_0}{Z}$$

$$I_{r.m.s.} = \frac{V_{r.m.s.}}{Z}$$

Z is known as the *Impedance of the circuit.*

Impedance (Z) is the overall opposition of a mixed circuit containing a resistor, an inductor and/or a capacitor. It is measured in ohms.

From equation (8.7), $X_c = \frac{1}{\omega C} = \frac{1}{2\pi fC}$

From equation (8.10), $X_L = \omega L = 2\pi fL$

Hence *Impedance* (Z) can be written as

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$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$Z = \sqrt{R^2 + \left(2\pi f L - \frac{1}{2\pi f C}\right)^2} \qquad 8.12$$

Summarizing, we have that the voltages across R, L, C are

$$V_R = IR$$
$$V_L = IX_L$$
$$V_c = IX_c \qquad 8.13$$
$$V = IZ = I\left[R^2 + (X_L - X_c)^2\right]^{1/2}$$

Example 8.3

Find the r.m.s. value of an alternating current whose peak value is 5 amps.

Solution

$$I_{r.m.s} = \frac{I_0}{\sqrt{2}}$$
$$= 0.707 \times I_0$$
$$= 0.707 \times 5$$
$$= \underline{3.53 \text{ Amps}}$$

Example 8.4

In an a.c. circuit the peak value of the potential difference is 180 V. What is the instantaneous p.d. when it has reached $\frac{1}{8}$ th of a cycle?

(SSCE)

Solution

1 cycle covers 360°

$\frac{1}{8}$ th cycle covers $\frac{360^\circ}{8} = 45^\circ$

$E = E_0 \sin \omega t = E_0 \sin 45^\circ$

$= 180 \sin 45^\circ = 180 \times \frac{1}{\sqrt{2}}$

$= \underline{90\sqrt{2} \text{ Volts}}$

Example 8.5

a. A circuit consists of a resistor 500 ohms and a capacitor of 5 μF connected in series. If an alternating voltage of 10 V and frequency 50 Hz is applied across the series circuit, calculate
   (i) the Reactance of the Capacitor
   (ii) the current flowing in the circuit
   (iii) the voltage across the capacitor

(b) If the capacitor is replaced with an inductor of 150 mH, calculate the impedance and voltage across the inductor.

Solution

(i) $X_c = \frac{1}{2\pi f C}$

$= \frac{1}{2\pi \times 50 \times 5 \times 10^{-6}} = 636.62$ ohms

$= \underline{636.62 \text{ ohms}}$

(ii) $Z = \sqrt{R^2 + X_c^2}$ (since there is no L)

$= \sqrt{500^2 + 636.62^2}$

$= 809.5$ ohms

$I = \frac{V}{Z} = \frac{10}{809.5}$ amps

$= 12.35 \times 10^{-3}$ A

$= \underline{12.35 \text{ mA}}$

(iii) $V_c = IX_c = 12.35 \times 10^{-3} \times 636.62$

$= \underline{7.86 \text{ volts}}$

b. $X_L = \omega L = 2\pi f L$

$= 2\pi \times 50 \times 150 \times 10^{-3}\ \Omega$

$X_L = 47.12$ ohms

$Z = \sqrt{R^2 + X^2} = \underline{502.2 \text{ ohms}}$

$I = \frac{10}{502.2}$ amps

$= 19.9 \times 10^{-3}$ A $= \underline{19.9 \text{ mA}}$

$V = IX_L = 19.9 \times 10^{-3} \times 47.12$

$= 938 \times 10^{-3}$ V

$= \underline{938 \text{ mV}}$

8.5 Phase Relation between I and V in an A.C. Circuit. Vector Diagrams

When an alternating voltage is placed across a R, L, C, series circuit, the resulting alternating current (I) has the same frequency as the voltage (V) but the two differ in phase or are said to be *out of phase*.

Phase is the state of vibration of a periodically varying system at a particular time; ωt = phase angle.

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Two vibrating systems with the same frequency are said to be in phase if their maximum, zero, and minimum values occur at the same time: otherwise they are said to be out of phase.

For two systems vibrating *out of phase*, the phase difference is the angle between the two vectors representing the two alternating quantities.

If a varying voltage, $V = V_0 \sin \omega t$ is applied across an RLC circuit, the current is found to be *out of phase* by an angle ϕ, i.e.

$I = I_0 \sin(\omega t - \phi)$

Theory shows that the phase difference between the voltage and the current through an RLC series circuit is given by

$$\tan \phi = \frac{X}{R} \qquad 8.14$$

or $\phi = \tan^{-1} \frac{X}{R}$

where X is the reactance, $(X_L - X_c)$, and R is the resistance.

Let us consider the following cases:

(i) For a circuit containing only a resistance R, the a.c. voltage vibrates *in phase or in step* with the alternating current.
Thus $\phi = 0$ (fig 8.2d)

(ii) For a circuit containing only a capacitance, C, theory shows that V_c and I_c are out of phase by an angle $\frac{\pi}{2}$ radians (or 90°) (fig 8.3c).

The current I_c, leads on the voltage V_c by 90°. This means that the angle by which a particular phase of I_c is in advance of a similar phase in V_c is 90° or $\frac{\pi}{2}$ radians or $\frac{1}{4}$ cycle.

In consideration of time, I_c gets to its maximum, minimum or zero value ahead of V_c. If $V_c = V_0 \sin \omega t$, then

$I_c = I_0 \sin\left(\omega t + \frac{\pi}{2}\right)$.

(iii) If only an inductor L is connected to the a.c. voltage, theory shows that Current I_L, *lags* on the voltage V_L by $\frac{\pi}{2}$ radians (i.e. V_L leads I_L)

I_L lags on V_L by $\frac{\pi}{2}$ radians means that the interval of angle or time by which a specific phase of V_L is delayed with respect to the same phase of I_L is $\frac{\pi}{2}$.

(iv) In a circuit containing R, L, C the current is the same for all the components of the circuit, and is in phase with the voltage across R. Using V_R as the reference vector, we have the other voltage vectors as shown in fig. 8.6.

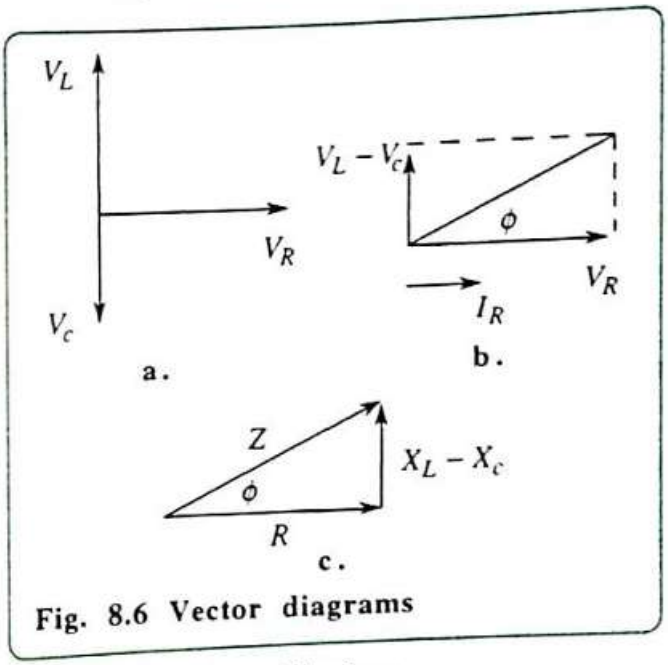


Fig. 8.6 Vector diagrams

The effective voltage V is given

$$V^2 = V_R^2 + (V_L - V_c)^2 \qquad 8.15$$

This is obtained by using Pythagoras theorem in fig. 8.6b.

The phase angle is given by

$$\tan\phi = \frac{V_L - V_c}{V_R} = \frac{X_L - X_c}{R} \qquad 8.16$$

If $X_L > X_c$, ϕ is positive and I lags V. If $X_L < X_c$, ϕ is negative and I leads V.

For R and L in series, we have

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$$V^2 = V_R^2 + V_L^2$$
$$I = \frac{V}{\sqrt{R^2 + X_L^2}} \qquad 8.17$$
$$\text{and } Z = \sqrt{R^2 + X_L^2}$$

Current, I, lags on the applied voltage V by ϕ given by

$$\tan\phi = \frac{V_L}{V_R} = \frac{X_L}{R} \qquad 8.18$$

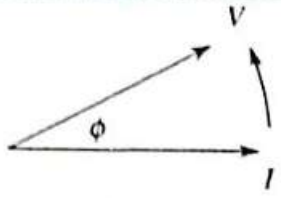


Fig. 8.7 I lags V, V leads I

For R and C in series, we have

$$V^2 = V_R^2 + V_c^2$$
$$I = \frac{V}{\sqrt{R^2 + X_c^2}} \qquad 8.19$$
$$Z = \sqrt{R^2 + X_c^2}$$

Current I leads on the voltage V by an angle ϕ given by

$$\tan\phi = \frac{V_c}{V_R} = \frac{X_c}{R} \qquad 8.20$$

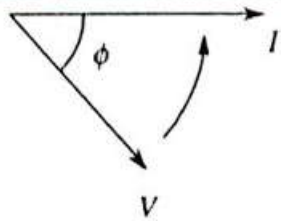


Fig. 8.8 V lags I, I leads V

8.6 Power in an A.C. Circuit

The *average power* in an a.c. circuit is given by

$$P = IV\cos\phi \qquad 8.21$$

where I, V, are the effective (r.m.s.) values of the current and voltage respectively and ϕ is the angle of lag or lead between them. The quantity $\cos\phi$ is known as the *Power Factor* of the device. The power factor can have any value between zero and unity for ϕ varying from 90° to 0°. For $\phi = 90°$ (or $\cos\phi = 0$) average power P, is zero. A power factor of zero means that the device is either a pure reactance, inductance or capacitance. Thus no power is dissipated in an inductance or capacitance. However if I is the r.m.s. value of the current in a circuit containing a resistance, R, then the power absorbed in the resistance is given by

$$P = I^2R \qquad 8.22$$

For an a.c. circuit, the instantaneous power is given by

$P = IV$ (instantaneous values.)

From fig. 8.6c., we see that power factor

$$\cos\phi = \frac{R}{Z} = \frac{\text{Resistance}}{\text{Impedance}} \qquad 8.23$$

Example 8.6

A series circuit consists of a resistance 600 ohms and an inductance of 5 henrys. An a.c. voltage of 15 volts (r.m.s.) and frequency 50 Hz is applied across the series circuit. Calculate

(i) the Current, I, flowing through the circuit
(ii) the voltage across the inductor
(iii) the phase angle between I and the applied voltage
(iv) the average power supplied
(v) the p.d. across the resistance.



Solution

The reactance in the inductor is given by

$$X_L = 2\pi fL = 2\pi \times 50 \times 5 = 500\pi \text{ ohms}$$
$$Z = \sqrt{R^2 + X_L^2} = \sqrt{600^2 + (500\pi)^2}$$
$$= 1.69 \times 10^3 \text{ ohms}$$

(i) $$I_{\text{r.m.s.}} = \frac{V_{\text{r.m.s.}}}{Z} = \frac{15}{1.69 \times 10^3}$$
$$= 8.88 \times 10^{-3} \text{ A}$$
$$= \underline{8.88 \text{ mA}}$$

(ii) Voltage across the inductor

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$V_L = IX_L = 8.88\times10^{-3}\times500\pi$
$= 4.44\pi$ volts
$= \underline{14.95 \text{ volts}}$

(iii) $\tan\phi = \dfrac{X_L}{R} = \dfrac{500\pi}{600} = 2.62$
$\phi = 69.1^\circ$

(iv) Power supplied $P_s = I^2R$
$= (8.88\times10^{-3})^2\times600$
$= \underline{4.73\times10^{-2}\ W}$

(v) p.d. across $R = V_R = IR = 8.88\times10^{-3}\times600$
$= \underline{5.33 \text{ ohms}}$

8.7 Resonance in RLC Series Circuit

The current, I, in an RLC series circuit is given by

$$I = \frac{V}{Z} = \frac{V}{\sqrt{R^2 + (X_L - X_c)^2}}$$

where $X_L = 2\pi fL$ and $X_c = \dfrac{1}{2\pi fC}$. The maximum current is obtained in the circuit when the impedance Z, in the above equation is minimum. This happens when

$$X_L = X_c \text{ or } 2\pi fL = \frac{1}{2\pi fC} \qquad 8.24$$

Resonance is said to occur in an a.c. series circuit when the maximum current is obtained from such a circuit.

The frequency at which this resonance occurs is called the resonance frequency (f_0). This is the frequency at which $X_L = X_c$ or $2\pi f_0 L = \dfrac{1}{2\pi f_0 C}$.

Hence solving the above equation we obtain that f_0 is given by

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \qquad 8.25$$

or since $\omega = 2\pi f$, we can write the condition of resonance as:

$$\omega_0 = \frac{1}{\sqrt{LC}} \qquad 8.26$$

The variation of current I, and frequency f, in an RLC series circuit is as shown in fig. 8.9.

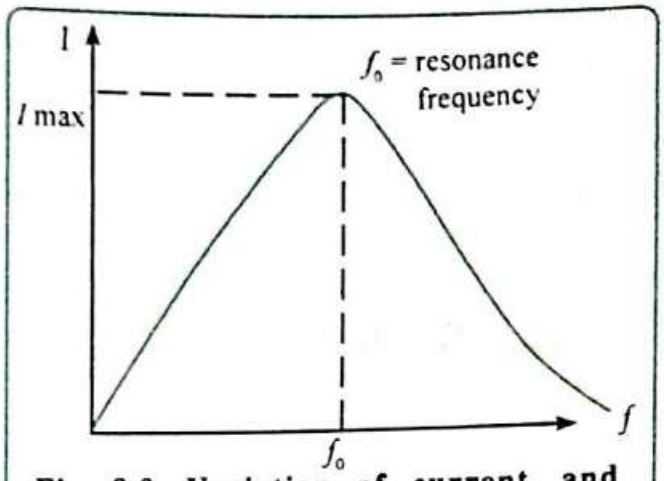


Fig. 8.9 Variation of current and frequency

From this diagram we see that at frequencies above or below the resonant frequency, f_0, the current is less than the maximum current, I_{max}. At $f = f_0$, the current is maximum.

Applications of Resonance

The resonant circuit finds applications in electronics. It is used to tune radios and TVs. Its great advantage is that it responds strongly to one particular frequency. The other frequencies have very little effect.

Hence such a resonant circuit can select one signal of a definite frequency from a jumble of other signals available to it. That resonance frequency corresponds to that of a particular incoming radio signal. When this happens, maximum current is obtained and the distant radio station is loudly and clearly heard.

Example 8.7

An a.c. voltage of amplitude 2.0 volts is connected to an RLC series circuit. If the resistance in the circuit is 5 ohms, and the inductance and capacitance are 3 mH and 0.05 μF respectively, calculate

(i) the resonance frequency f_0.

(ii) the maximum a.c. current at resonance.

Solution

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{3\times10^{-3}\times5\times10^{-8}}}$$
$$= \frac{1}{2\pi\sqrt{15\times10^{-11}}}$$

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$= 1299.545$ Hz

$= \underline{13 \text{ kHz}}$

At resonance $Z = R$ since $X_L = X_c$

$$I_0 = \frac{V_0}{R} = \frac{2}{5}$$

$= \underline{0.4 \text{ Amps}}$

Summary

1 In an a.c. voltage given by $E = E_0 \sin \omega t$, E is the instantaneous e.m.f., E_0 is the peak or maximum value or the amplitude, ω is the angular velocity, ωt is the phase angle of the alternating voltage.

2 The root-mean-square (r.m.s.) value of the current is that steady current which will develop the same amount of heat in the same time in the same resistance.

$$I_{\text{r.m.s.}} = 0.707 I_0 \quad \text{or} \quad \frac{I_0}{\sqrt{2}}$$

$$V_{\text{r.m.s.}} = 0.707 V_0 \quad \text{or} \quad \frac{V_0}{\sqrt{2}}$$

3 Most a.c. meters (e.g. Hot wire ammeter or moving iron ammeter) measure the effective or r.m.s. value of the current.

4 The average value of an a.c. voltage or current is zero. As a result, a.c. voltage is not suitable in application such as electroplating.

5 When an a.c. voltage V is applied to a resistor R, I is given by ohm's law, $V = IR$. The a.c. voltage and current are in phase or in step.

6 When an a.c. voltage is applied across a capacitor C, the current is given by the relation $V = IX_c$ where X_c is the reactance (capacitive reactance). $X_c = \frac{1}{2\pi fC}$. Its unit is ohms when f is in Hz and C is in farads. I leads V by 90° (or $\frac{\pi}{2}$ radians), or V lags I by 90° (or $\frac{\pi}{2}$ radians or $\frac{1}{4}$ cycle).

7 When an a.c. voltage is applied across an inductor L, then the current is given by the relation $V = IX_L$, where X_L is the inductive reactance; $X_L = 2\pi fL$. X_L is in ohms when f is in Hz and L is in Henrys. V leads I or I lags V by 90° (or $\frac{\pi}{2}$ radians).

8 The opposition offered to the passage of an alternating current by either the inductor or the capacitor or both is called the *reactance*.

9 When an a.c. voltage is applied to R, L, C in series, then voltage and current are related by $V = IZ$ where Z is known as the *impedance*. $Z = \sqrt{R^2 + (X_L - X_c)^2}$ and is measured in ohms. $X (= X_L - X_c)$ is known as reactance. *Impedance* is the overall opposition of a mixed circuit containing a resistor, and inductor and/or a capacitor. It is measured in ohms.
The voltage across the R, L, C in series combination is given by $V^2 = V_R^2 + (V_L - V_c)^2$.

10 No average power is lost in the inductor or capacitor. Power loss in the resistor is IV or I^2R. In general, the average power loss in a.c. circuit is $IV \cos\phi$, where $\cos\phi$ is known as the power factor, ϕ is the phase angle between I and V and I and V are the r.m.s. values. Power factor $\cos\phi$ is zero for inductors and capacitors and unity for a resistor.

11 Resonance frequency (f_0) is the frequency at which a maximum current is obtained in an RLC series circuit

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \quad \text{or} \quad \omega_0 = \frac{1}{\sqrt{LC}}.$$

Exercise 8

1

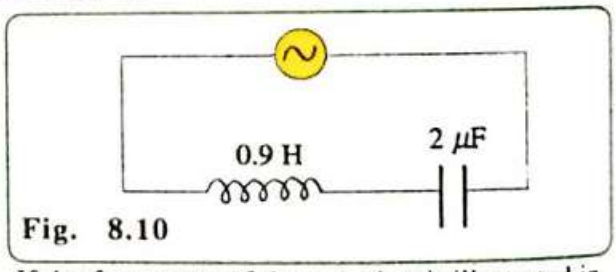


Fig. 8.10

If the frequency of the a.c. circuit illustrated in fig. 8.10 is $\frac{500}{\pi}$ Hz, what would be the reactance in the circuit?

A $\frac{0.0009}{\pi}\Omega$ B $400\ \Omega$

C $1030\ \Omega$ D $1400\ \Omega$

E $2500\ \Omega$.

(SSCE)

2 At what frequency will a 20 μF capacitor have a reactance of 500 ohms?

A $\frac{100}{\pi}$ B $\frac{50}{\pi}$ C 50

D 100π E $\frac{30}{\pi}$

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3 What is the peak value of a voltage whose r.m.s. value is 100 V

A 140 V B 70 V C 141.4 V
D 50 V E 80.60 V

4 In an RLC series a.c. circuit power is dissipated in

A Resistance only
B Reactance only
C Resistance and Reactance
D Resistance, inductance and capacitance
E None of the above.

5 Average power dissipated in RLC circuit is given by

A I^2X B I^2X_c C I^2R
D I^2V E $I^2(X_L - X_c)^2$

6 At what frequency will a 12.0 mH inductor have a reactance of 880 ohms?

A 1.17×10^2 Hz
B 1.71×10^4 Hz
C 1.17×10^4 Hz
D 2.88×10^{-4} Hz
E 60 Hz

7 The resonance frequency (f_0) in an RLC series circuit is given by

A $f_0 = 2\pi\sqrt{LC}$ B $f_0 = \dfrac{2\pi}{\sqrt{LC}}$

C $f_0 = \dfrac{1}{2\pi\sqrt{LC}}$ D $f_0 = \dfrac{1}{\sqrt{LC}}$

E $f_0 = \dfrac{1}{2\pi\sqrt{\dfrac{L}{C}}}$

8

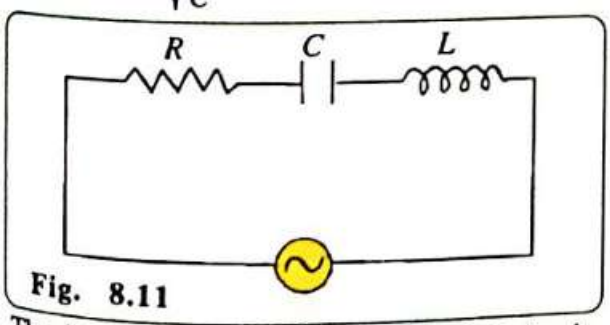


Fig. 8.11

The diagram above shows a series RLC circuit. At resonance, the current in the circuit is

A $\dfrac{V}{R}$ B $V\omega C$ C $\dfrac{V}{\omega L}$

D $\dfrac{V}{\omega CRC}$

(JAMB)

9 When the r.m.s. voltage of a source of electricity supply is given as 240 V, it means that the peak value of the supply is

A 240 V B 340 V C 480 V
D 57600 V

(JAMB)

10

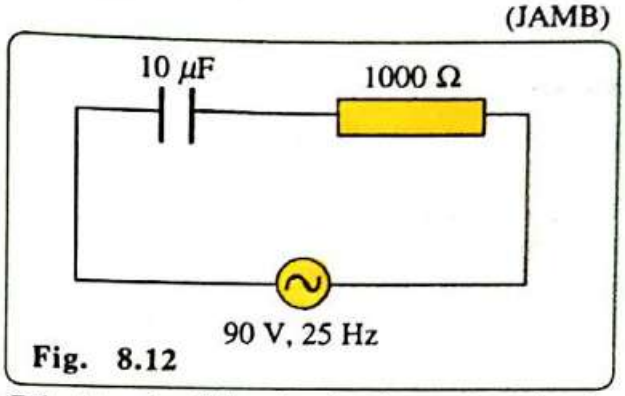


Fig. 8.12

Calculate the following in the series circuit shown above

(i) reactance of the capacitor
(ii) impedance of the circuit
(iii) current through the circuit
(iv) voltage across the capacitor
(v) average power used in the circuit

(SSCE)

11 a. State the laws of electromagnetic induction.

b. (i) Describe a simple experiment to show how an induced e.m.f can be produced.
(ii) State two factors on which the magnitude of the induced e.m.f. depends.

c. Explain what is meant by the r.m.s. value of an alternating current.

d. (i) If the alternating current is represented by $I = I_0 \sin \omega t$ state what the symbols I, I_0, ω, and ωt represent.
(ii) Calculate the instantaneous value of such a current, if in a circuit it has r.m.s. value of 15.0 A when its phase angle is 30°.

(SSCE)

12 a. Explain the terms *reactance* and *impedance* in an a.c. circuit.

b. A source of e.m.f 240 V and frequency 50 Hz is connected to a resistor, an inductor and a capacitor in series. When the current in the capacitor is 10 A, the potential difference across the resistor is 140 V and that across the inductor is 50 V. Draw the vector diagram of the potential differences across the inductor, the capacitor and the resistor. Calculate the:

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(i) potential difference across the capacitor

(ii) capacitance of the capacitor

(iii) inductance of the inductor

(SSCE)

13 Explain the meaning of *Peak value* and *root-mean-square value* of an alternating current. If the peak value of such a current is 5 amps, calculate its r.m.s. value.

14 An alternating voltage is represented by the equation $E = E_0 \sin\omega t$. Write down

(a) the r.m.s. value of the voltage

(b) the frequency of the voltage

(c) the phase angle of the voltage

15 Explain what is meant by the terms *Impedance, Phase angle* and *Reactance*, as applied to an a.c. circuit. Calculate the impedance and phase angle for an a.c. circuit having a 100 ohm resistance, $5\mu F$ capacitor and 1.5 Henry inductor in series, if an a.c. voltage of frequency 100 Hz is applied across the circuit.

16 Describe the action of an ammeter adequate for measuring an alternating current.

17 (a) At what frequency would a $10\mu F$ capacitor have a reactance of 5000 ohms?

(b) At what frequency would a 5-henry inductor have a reactance of 4000 ohms?

18 (a) Write down an expression for the effective power of an a.c. circuit; explain the meaning of *power factor*.

(b) A series circuit has a resistance of 100 ohms and an impedance of 200 ohms. What power is consumed in the circuit when a p.d. of 120 volts is impressed across it?

19 Draw a vector diagram of the relationship of I and V for an a.c. circuit containing

(a) a pure inductor,

(b) a pure capacitor,

(c) a pure resistor.

Explain the meaning of the terms *lag, lead.* Illustrate with a diagram.

20 A 0.5 μF capacitor, a 20 ohm resistor and a 0.050 H inductor are connected in series across a 25 V, 1000 Hz power source. Calculate the voltage readings across the capacitance and inductance.

21 What does 0.20 represent if a wave is expressed by the equation:

$$y = 0.20 \sin 0.4\pi(x - 60t)$$

The alternating current in a circuit is represented by the equation $I = 80 \sin 200\pi t$, where I is in amperes. What is the root mean square value of the current?

22 (a) Define (i) ***Reactance*** (ii) ***Impedance***

(b) Explain

(i) resonance frequency of an RLC circuit.

(ii) Explain the statement - ***The power supply voltage of a source is 230v.***

(c) A source of e.m.f 240V and frequency 50 Hz is connected to a series arrangement of a resistor and inductor and a capacitor. When the current in the capacitor is 10A, the potential difference across the resistor is 140V and that across the inductor is 50V, calculate the

(i) potential difference across the capacitor;

(ii) capacitance of the capacitor;

(iii) inductance of inductor.

(d) Draw and label one Vector diagram for the potential difference across the inductor, the capacitor and resistor in (c) above.

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23 A 40 μF capacitor in series with a 40Ω resistor is connected to a 100 v, 50 Hz a.c. supply.

(i) Draw a circuit diagram of the arrangement.

(ii) Calculate the impedance, I, in the circuit, the current in the circuit, the potential difference across the capacitor.

(SSCE)

24 A coil of inductance L = 0.007 H, a resistor of resistance 8Ω and a capacitor of capacitance 0.001 F are connected in series to an a.c. source of frequency $\frac{500}{\pi}$ Hz.

If the r.m.s. voltages across the coil, the resistor and the capacitor are 30V, 30V and 70V respectively,

(i) Draw a vector diagram to illustrate the voltage across the components in the circuit.

(ii) Calculate

(a) the r.m.s. voltage of the circuit

(b) the r.m.s. current in the circuit

(c) the power dissipated in the circuit

(iii) Write down the sinusoidal equation for the r.m.s. voltage, V, in terms of the time, t.

(SSCE)

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9 MODELS OF THE ATOM

The first hint that matter was electrical in nature came in 1834 when Michael Faraday found that chemical changes would occur when an electric current was sent through certain solutions — a process known as electrolysis. Atoms or ions in solution carried the electric current through the solution. Also experiments with gas discharge tubes led to the observation that a gas under low pressure and high electric potential conducts electric current in some way. To answer some basic questions of chemical behaviour, we must look deeply at the structure of the atom. We begin by briefly examining how our knowledge of the basic atomic structure came about.

There have been several models put forward by scientists to explain the structure of the atom. Some of these models – the Thomson, the Rutherford, the Bohr, and the Electron Cloud models, will be reviewed in this chapter.

At the end of this chapter the student should be able to:

1 state and discuss what chemical evidence there is for the existence of atoms.
2 state and discuss what experimental evidence there is for believing that matter is electrical in nature.
3 describe the Bohr-Rutherford model of the atom.
4 explain the existence of line spectra in terms of electron transitions between atomic energy levels.
5 explain the similarity in the behaviour of the elements in terms of their electronic configuration.

9.1 Various Models of the Atom

Modern atomic theory provides a reasonably satisfactory explanation of the properties of matter, the mechanisms of chemical change and the interaction of matter and energy. Such a theory emerged from the synthesis of the work of several scientists only a few of which will be discussed here.

John Dalton (1766-1844) is generally credited as the father of the atomic theory but the early Greek philosophers were the originators of the concept of atoms. They taught that matter is composed of atoms and is therefore finitely divisible. John Dalton considered the atom as constituting the simplest component of matter. He viewed the atom as 'indestructible', tiny hard spheres. The discovery of *radioactivity* by Henry Becquerel (1891) showed that atoms are complex rather than 'indivisible'. Radioactivity showed that atoms are not 'indivisible' or 'indestructible' but can disintegrate forming atoms of different elements. The discovery of *cathode rays* in electric discharge tubes (1895) by William Crookes revealed that negatively charged *electrons* were components of the atom.

By 1900, it was already established that matter consists of atoms but nothing was known about the structure of the atom. It was only known that the atoms contain electrons but that on the whole the atom was electrically neutral. This neutrality means that there must exist within the atom enough positively charged components to balance the negatively charged electrons. This led Sir *J.J. Thomson* the English Physicist, to propose an atomic model which visualised the atom as a homogeneous sphere of positive charge inside of which are embedded negatively charged electrons as shown in fig 9.1.

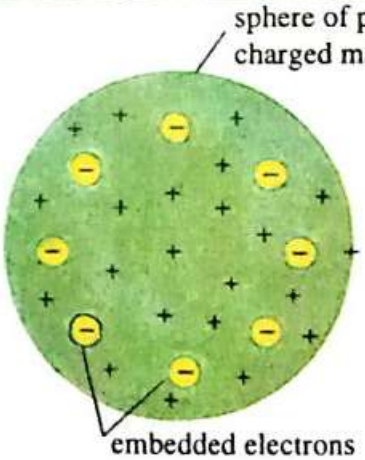


Fig. 9.1 Thomson's model of the atom

J.J. Thomson also determined the ratio of the charge to mass, $\frac{e}{m}$, of electrons, and found $\frac{e}{m}$ to be identical for all cathode-ray particles, irrespective of the kind of gas in the tube or the metal the electrodes are made of.

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Rutherford Model – the Nuclear Idea and the Planetary Model

Ernest Rutherford (1911) and his co-workers performed experiments whose results contradicted Sir J.J. Thomson's model. In these experiments a beam of positively charged alpha (α) particles was directed at a thin sheet of a metal foil. It was found that most of the alpha-particles passed through the foil without deflection as if the foil were mostly empty space. Only a few of them were diverted from their paths. Some of these few actually rebounded backwards.

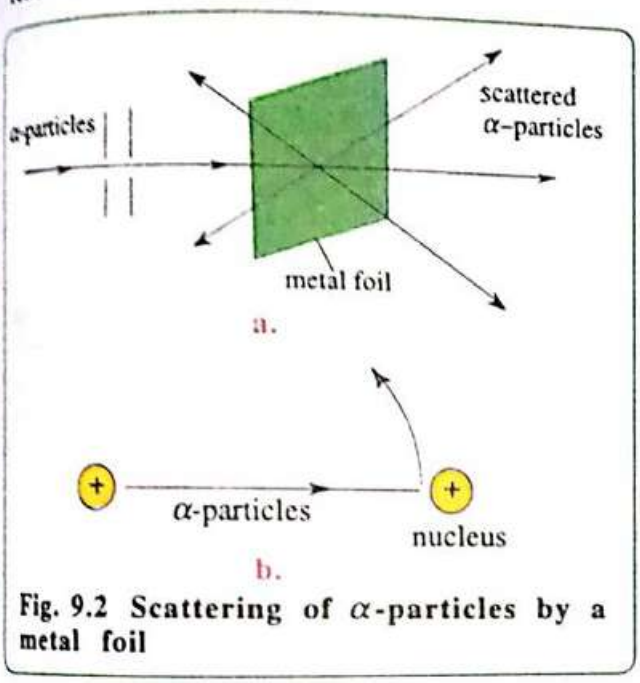


Fig. 9.2 Scattering of α-particles by a metal foil

This scattering of alpha-particles by the metal foil was explained as a repulsion from a heavy positively charged nucleus present at the centre of the atom of the metal foil. This follows because an abrupt change in path (as noted for a few α-particles) of a relatively heavy and positively charged α-particle can result only from its hitting or from its close approach to another particle (the nucleus) with a highly concentrated, positive charge. All these contradicted Thomson's atomic model which supposed that the distribution of the charges was diffuse. Hence Rutherford proposed his model of the atom.

Rutherford proposed a planetary model of the atom which suggested that the atom consists of a positively charged heavy core called the *nucleus* where most of the mass of the atom was concentrated. Around this nucleus negatively charged electrons circle in orbits much as planets move around the sun. Each nucleus must be surrounded by a number of electrons necessary to produce an electrically neutral atom.

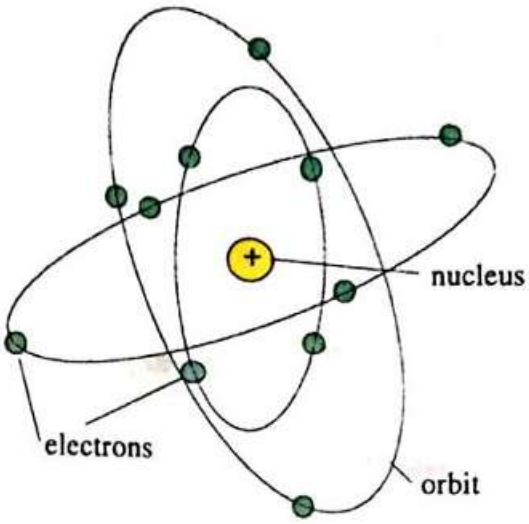


Fig. 9.3 Rutherford's planetary model

This model was a major step toward how we view the atom today. It however had two problems. According to Newtonian Physics, such an atom as Rutherford's would collapse by spiralling into the nucleus, since there is an attractive force between the oppositely charged nucleus and electrons. Further experiments indicated that charged particles moving in a field of opposite charge lose energy by emitting radiation. But atoms in their normal state neither collapsed nor emitted radiation.

Thus the two main difficulties of the Rutherford model are these:

1 it predicts that light of a continuous range of frequencies will be emitted, whereas experiment shows line spectra instead of continuous spectra.

2 it predicts that atoms are unstable – electrons quickly spiral into the nucleus – but we know that atoms in general are stable, since the matter around us is stable.

Clearly Rutherford's model was not sufficient to explain experimental observations. Some sort of modification was needed and this was provided by Neils Bohr.

The Bohr Model – Idea of Quantized Energy States

In 1913 Neils Bohr, a Danish scientists who had studied in Rutherford's laboratory, suggested a model of the hydrogen atom in which

(i) the electron moves around the nucleus in certain specific circular orbits (sometimes called *energy levels*) and that the centrifugal force due to this motion counterbalances the electrostatic

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attraction between the electron and the nucleus. In such orbits the electrons can move without radiating or losing energy. He called the possible orbits *stationary states*. Only orbits of particular radii were possible. In general the higher the energy of the electron, the farther its orbit from the nucleus.

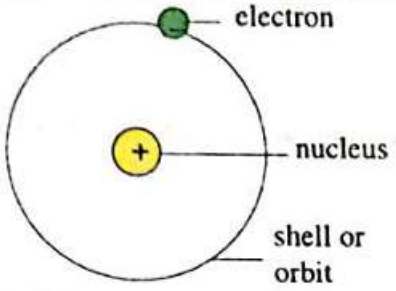
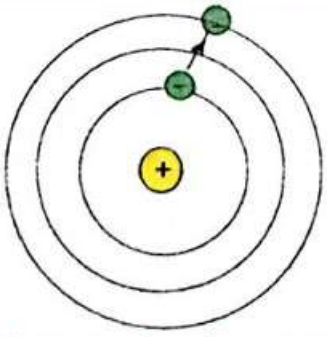


Fig. 9.4 Bohr model of the hydrogen atom

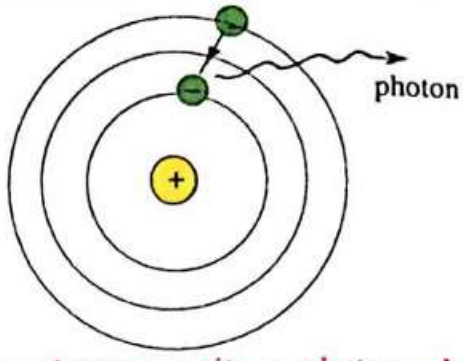
(ii) the energy of an electron in an atom cannot vary continuously but is restricted to a limited number of *discrete* or *individual values*. The energy of the electron is said to be *quantized* (i.e. can have only discrete values). Bohr thus postulated that the electrons in an atom cannot lose energy continuously but must do so in quantum 'jumps'. Bohr postulated that light is emitted only when an electron jumps from one stationary state to another of lower energy. When such a jump occurs, a single photon of light would be emitted whose energy is given by

$$hf = E_u - E_l \qquad 9.1$$

where E_u and E_l refer to energy of upper and lower states respectively, f is the frequency of the emitted light and h is a constant known as Planck's constant. Thus Bohr was able to account for the appearance of *line spectrum* rather than a *continuous spectrum*.



a. An electron absorbs energy when it transfers to a higher energy level (excitation)



b. An electron emits a photon when it moves to a lower energy level

Fig. 9.5 Line spectrum from an atom

(iii) The third postulate of Bohr's model was that the angular momentum values of the electron in an atom are *quantized*, that is they are restricted to a limited number of *discrete* values that are integral (n) multiples of a constant, the Planck's constant h ($= 6.626 \times 10^{-34}$ Js) divided by 2π. That is, angular momentum $L = n(\frac{h}{2\pi})$, where n = 1, 2, 3, 4. The integer n is called a *quantum number*.

The *quantization of angular momentum* first proposed by Neils Bohr is a key point in modern atomic theory.

The Bohr model is also known as the Bohr-Rutherford model since it was an extension of Rutherford's planetary model. The model was both a success and a failure.

The great success of Bohr's theory is that

1 it gives a model for why atoms emit line spectra and accurately predicts, for hydrogen, the wavelengths of emitted light or the frequencies of the lines in the hydrogen spectrum.

2 It offers an explanation for absorption spectra: photons of just the right wavelength can knock an electron from one energy level to a higher one. To conserve energy, the photon must have just the right energy. This explains why a continuous spectrum passing through a gas will have dark (absorption) lines at the same frequencies as the emission lines.

3 It ensures the stability of atoms by stating that the ground state is the lowest state for an electron and there is no lower energy level to which it can go and emit more energy.

4 It accurately predicts the ionization energy of 13.6 eV for hydrogen.

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Though the theory met with only limited success, the introduction of the ideas of quantum numbers and fixed energy levels were important steps forward. Though the idea of discrete energy levels applies to all atoms, many other aspects of the Bohr model explain only the spectra of hydrogen-like atoms or ions. Attempts to explain the spectra and other physical properties of more complicated atoms require the use of a more complex model.

The Electron-Cloud Model

This model visualises the atom as consisting of a tiny nucleus of radius of the order of 10^{-15}m. The electron is visualised as being in rapid motion within a relatively large region around the nucleus, but spending most of its time in certain *high-probability* regions. Thus the electron is not considered as a ball revolving around the nucleus but as a particle or wave with a specified energy having only a *certain probability* of being in a given region in the space outside the nucleus. The electron is visualised as spread out around the nucleus in a sort of *electron-cloud*. Some chemists prefer to consider the electron in terms of a cloud of negative charges (*electron cloud*), with a cloud being dense in regions of high electron probability and more diffuse in regions of low probability.

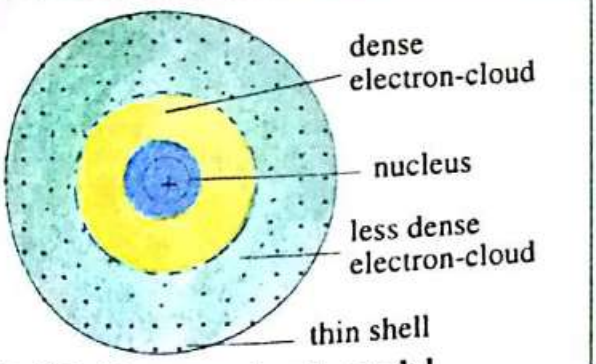


Fig. 9.6 Electron-cloud model

This model is therefore known as the *electron-cloud model*. Fig. 9.6 illustrates the electron density distribution of the electron cloud nearest the positive nucleus. Most of the time, but not always, the electron will be located inside the spherical outline. In other words, the probability of finding the electron inside the spherical boundary is high. The probability then decreases rapidly as the distance of the thin shell from the nucleus increases.

9.2 Atomic structure and chemical behaviour

Today we consider the atom as made up of a tiny but massive nucleus at the centre and outside the nucleus is a cloud of *electrons* which move in wave-like orbits or shells around the massive nucleus. The *nucleus* consists of *protons* which carry positive charges and *neutrons* which carry no charge. The proton and the neutron together constitute the *nucleon*. Practically all the mass of an atom is concentrated in the central nucleus. The protons, neutrons, and electrons are the fundamental subatomic particles of the atom.

The *electron* is the lightest particle of an atom, with a mass (m_e) of 9.1×10^{-31} kg and an electronic charge $e^- = 1.6\times10^{-19}$ C.

The *Proton* has a mass of 1.67×10^{-27} kg which is over 1836 times heavier than the mass of an electron. It carries a positive charge, $e^+ = 1.6\times10^{-19}$ C (i.e. $e^+ = e^- = 1.6\times10^{-19}$ C). There are the same number of protons in the atoms of a given element but different number of protons in the atoms of different elements. For example, there is only one proton in a hydrogen atom but eleven protons in a sodium atom. In a neutral atom, the number of protons equals the number of electrons.

The *Neutron* has the same mass as the proton but it carries no charge.

We denote the atom of an element X by ${}^{A}_{Z}X$ where A is the *mass number* and Z is its *atomic number*.

The *Atomic Number* or proton number (Z) is the number of protons in the nucleus of an element. The *Mass number* or *Nucleon number* (A) is the total number of protons and neutrons in an atom of an element. Thus the number of neutrons in the atom of an element equals the difference between the mass number and the atomic number (i.e. A-Z). The carbon atom denoted by ${}^{12}_{6}C$ has 6 protons and 6 neutrons and a mass number of 12; the Nitrogen atom denoted by ${}^{14}_{7}N$ has 7 protons, and 7 neutrons and a mass number of 14. Since a neutral atom has the same number of electrons as the number of protons, ${}^{12}_{6}C$ has 6 protons and 6 electrons.

An important question to consider is if all atoms are composed of these same components, why do different atoms have different chemical properties?' The answer lies in the number and

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arrangement of the electrons. The electrons comprise most of the atomic volume and thus are the parts that 'intermingle' when atoms combine to form molecules. Therefore the number of electrons possessed by a given atom greatly affects its ability to interact with other atoms.

The properties of the elements are periodic functions of their atomic number. When arranged in order of increasing atomic numbers, the elements with similar chemical properties recur at definite intervals, i.e. *periodically*. In regard to the *electronic structure* of *electronic configuration* (i.e. the arrangement of electrons in shells around the nucleus), this suggests a periodicity in the number of electrons in the outer shells of the atoms of the elements. The electrons in the outermost shell or shells of an atom are called *Valence electrons. These valence electrons are largely responsible for the chemical behaviour of the atoms of an element.* If elements having the same number of *valence electrons* are grouped together, the elements falling within each group have similar chemical properties.

Thus we note that atomic mass, atomic number, valence and periodicity, all indicate particularity of matter.

Isotopes are atoms of the same element which have the same atomic number (Z) but different mass number. Isotopes are thus atoms with the same number of protons but different number of neutrons. Isotopes have similar chemical properties because they have the same number of electrons round the nucleus. *Chemical combination is due to an exchange of outer or valence electrons between elements.*

In nature, elements are usually found as a mixture of isotopes. For example, two naturally occurring isotopes of chlorine are

(i) $^{35}_{17}Cl$ (17 protons, 17 electrons, 18 neutrons)

(ii) $^{37}_{17}Cl$ (17 protons, 17 electrons, 20 neutrons)

For Carbon we have:

(i) $^{12}_{6}C$ (6 protons, 6 electrons, 6 neutrons)

(ii) $^{13}_{6}C$ (6 protons, 6 electrons, 7 neutrons)

For Oxygen we have:

(i) $^{16}_{8}O$ (8 protons, 8 electrons, 8 neutrons)

(ii) $^{17}_{8}O$ (8 protons, 8 electrons, 9 neutrons)

(iii) $^{18}_{8}O$(8 protons, 8 electrons, 10 neutrons)

For Uranium we have:

(i) $^{238}_{92}U$ (92 protons, 92 electrons, 146 neutrons)

(ii) $^{235}_{92}U$ (92 protons, 92 electrons, 143 neutrons)

(iii) $^{234}_{92}U$ (92 protons, 92 electrons, 142 neutrons)

Summary

1 The first hint that matter was electrical in nature came through Faraday's experiments on electrolysis. Other evidence came from the production of cathode rays by electric discharge through gases under high electric potential and low pressure.

2 Some early models of the atom include the Thomson, the Rutherford, the Bohr and the Electron-Cloud models.

Thomson's model visualised the atom as a homogeneous sphere of positive charge inside of which are embedded negatively charged electrons.

Rutherford's planetary or nuclear model visualised an atom which consists of a positively charged heavy core called the nucleus around which negatively charged electrons circle in orbits much as planets move round the sun. The model was created to explain the back-scattering of alpha particles from thin metal foils. The model could neither explain the stability of atoms or line spectra.

To explain the line spectra emitted by atoms as well as their stability, Bohr's model proposed an atomic model in which

(i) the electrons found in an atom can only occupy orbits for which the angular momentum is *quantized* which results in *discrete* values for the radius and energy

(ii) the electrons in such *stationary state* emits no radiation, but

(iii) if an electron jumps to a lower state, it emits a photon whose energy equals the difference in energy between the two states ($hf = E_u - E_l$), f is the frequency of the light emitted

(iv) the angular momentum L, of the atomic electron is quantized by the rule $L = \frac{nh}{2\pi}$ where n can take only discrete values 1, 2, 3, etc.

The model, also known as the *Bohr-Rutherford model* successfully predicted the frequencies of the lines in the hydrogen spectrum. Also its introduction of the ideas of

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were important steps forward. The model could not explain spectra from multielectron atoms.

The *Electron-Cloud Model* visualises a central nucleus with a cloud of negative charges or *electron cloud* spread out around the nucleus.

The density of the cloud indicates the probability of electrons being in a given region around the nucleus.

3 The fundamental constituents of the atom are the proton and neutron located at the nucleus and the electrons moving in wave-like orbits around this nucleus.

4 The *atomic number* is the number of protons in the nucleus of an element. The mass number is the total number of protons and neutrons in the atom of an element.

5 *Valence electrons* are largely responsible for the chemical behaviour of an element. *Valence* is the chemical combining capacity of an atom.

6 *Isotopes* are atoms of the same element which have the same atomic number but different mass number. They have similar chemical properties because they have the same number of electrons round the nucleus.

Exercise 9

1 Which of the following statements is *not* correct? Isotopes of an element have
A the same number of electric charges on the nucleus
B the same chemical properties
C different nucleon numbers
D different proton numbers
E different atomic masses
(SSCE)

2 Which of the following representations is correct for an atom *X* with 28 electrons and 30 neutrons?
A $^{30}_{28}X$ B $^{28}_{30}X$ C $^{58}_{30}X$
D $^{58}_{28}X$ E $^{30}_{2}X$
(SSCE)

3 Which of the following statements are correct?
I The neutron has no charge
II The electron is lighter than the proton
III The proton is positively charged
IV The algebraic sum of the number of charges on the protons and on the electrons in a neutral atom is zero
A I, II, III, IV
B I, II and III only
C I, III and IV only
D II, III and IV only
E II and IV only
(SC/GCE)

4 An element and its isotopes only differ in the number of
A protons B electrons C ions
D X-particles E neutrons
(SC/GCE)

5 Which of the following particles determine the mass of an atom?
A Protons and neutrons
B Neutrons only
C Protons and electrons
D Neutrons and electrons
E Protons only

6 Which of the following names is *not* associated with models of the atom
A Isaac Newton B Neils Bohr
C J.J. Thomson D Ernest Rutherford
E John Dalton

7 a. In what ways are the isotopes of a given element:
(i) different from each other
(ii) similar to each other

b.

Atom	Number of protons	Number of neutrons
X	4	3
Y	3	4
Z	4	4

(i) Which of the atoms in the table above are isotopes? Give reasons.
(ii) How are protons, neutrons and electrons distributed in the atom of an element with atomic number 3 and mass number 7?

8 Define the terms: *Atomic Number*, *Mass Number*, *Nucleon*, *Isotopes* and *Valence electron*.

Which of the above terms is/are associated with the chemical properties of atoms.

9 Describe the essential features of the Bohr-Rutherford Model of the atom. What are its successes and its failures? How does it account for line spectra.

10 What are the essential features of the *Electron-Cloud Model* of the atom. Illustrate with a diagram.

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NUCLEUS, RADIOACTIVITY, NUCLEAR REACTIONS

10

This chapter will consider only the activities of the nucleus. At the end of the chapter the student should be able to:

1 Identify the radiations from radioactive substances using their characteristics.
2 Solve simple problems involving half-life of radioactive substances.
3 State some uses of radioactive substances.
4 Determine the products of a radioactive decay.

10.1 Radioactivity – Natural and Artificial

The phenomenon of radioactivity was discovered in 1896 by a French physicist Henri Becquerel. He noted while performing some experiment that uranium gave out some kind of rays that would penetrate through thick black paper and affect a photographic plate placed on the other side. This phenomenon is called *radioactivity* and substances that give rise to radioactivity are called *radioactive substances* or *elements*. The observation of this phenomenon was confirmed in 1898 by Marie Curie and her husband Pierre Curie. They discovered that a uranium ore, pitchblende, was a more intense source of radioactivity than uranium. They isolated the first concentrated radioactive substances or elements called *Polonium* and *radium*. Other radioactive substances such as *thorium*, *radon* and *ionium* were later isolated by the Curies and other workers.

Types of radiation from radioactive elements

The experimental genius of the British Physicist Ernest Rutherford unravelled the nature of the radiations emitted by radioactive elements. He and his co-workers discovered that the emissions were penetrating and are of three different types.

In their experiment, they placed a small sample of radium at the bottom of a small hole drilled in a block of lead. The radiation emitted from this radium emerged from the hole in a narrow beam. The rays were subjected to a strong magnetic field placed at the side of the beam. A photographic plate situated at appropriate sides to receive the rays showed that the paths of some rays were bent to the right, some to the left and some went straight on, unbent.

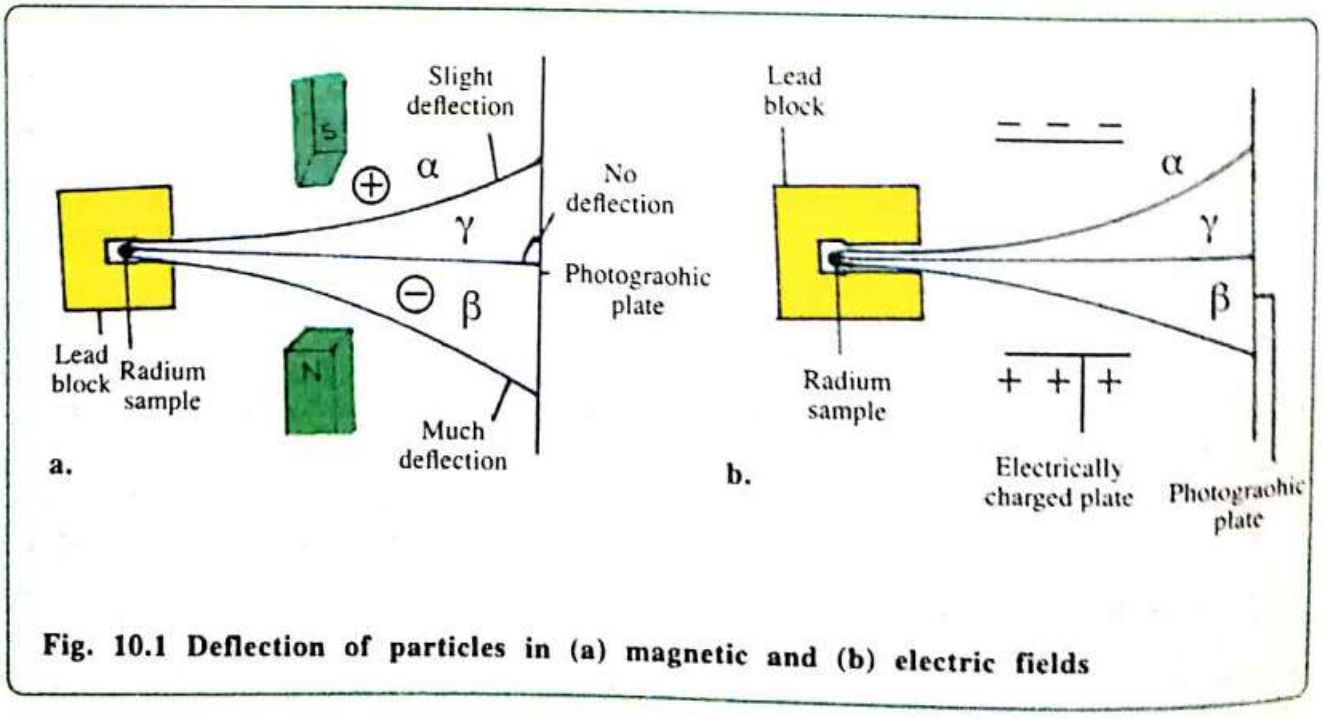


Fig. 10.1 Deflection of particles in (a) magnetic and (b) electric fields

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Table 10.1: Properties of the emitted Particles

Radiation	Alpha (α)-particles	Beta (β)-particles	Gamma (γ) rays
Nature	Helium nuclei ^4_2H	High energy electrons	Electromagnetic waves of very short wavelength
Velocity	5–7% speed of light	Travel at approx. speed of light	travel at speed of light
Charge	$+2e$ ($+3.2\times10^{-19}$ C)	$-e$ (-1.6×10^{-19} C)	Electrically neutral
Mass	Relatively massive	Relatively light	Negligible
Effect of magnetic field	Slightly deflected in a magnetic field in a direction expected for a +ve charge	Strongly deflected in a magnetic field in a direction expected of a –ve charge	Small or no effect
Ionizing power	Large, cause heavy ionization	Medium. About 0.1% of that of α-particles	Small
Penetrating power	Little penetrating power e.g. by thin sheets of paper	Good penetrating power in air, several mm of aluminium	High penetrating power in air and in solid e.g. many cm of lead
Fluorescence	Cause fluorescence in ZnS	No fluorescence in ZnS	

Electrically charged plates placed at the side of the beam as shown gave the same effect. Thus one of the three types of radiation was deflected in a direction showing that it consists of positively charged particles. They were bent towards the negative electric plates, or towards the south pole of the magnetic field. These were called *Alpha-particles* (α-particles).

Another type of radiation was deflected towards the positive electric plate, or the North pole of the magnetic field. This shows them to be negatively charged. They were called *Beta particles* (β-particles)

The third type of radiation that was neither affected by the electric or magnetic field indicated they were not charged. They travelled straight on unbent. They were actually an electromagnetic radiation and are called *gamma rays* (γ-rays).

The nature and properties of these three types of radiations from a radioactive element are summarized in Table 10.1. We are now in a position to define radioactivity thus:

Radioactivity is the spontaneous decay or disintegration of the nucleus of the atom of an element during which it emits α-, β- particles or γ-rays or a combination of any or all the three, and energy (or heat).

Radioactive elements are those elements that spontaneously emit radiation from their nucleus e.g. Radium, thorium, radon, ionium, polonium etc.

Radioactive Decay; Half-Life; Decay Constant

Radioactivity is a spontaneous process. It goes on independent of external control. It is not affected by temperature or pressure or by chemical treatment. It is also a random process as no one can predict which atom will disintegrate at a given time.

Experiments have shown that each radioactive element has a definite rate of decay which can be characterised by its *Half-Life*.

Half-life of a radioactive element is the time taken for half of the atoms initially present in the element to decay.

If the half-life of an element is T years, it means that after T years, 1 gm of the element will have a mass of $\frac{1}{2}$ gm, after $2T$ years, the mass of the element will be $\frac{1}{4}$ gm (or $\frac{1}{2}$ of $\frac{1}{2}$ gm) and so on. Thus if we have 1000 atoms of a radioactive element initially, whose half life is 10 years, then after 10 years, 500 atoms will remain; after 20 years 250 atoms will be left and after 30 years, 125 atoms will be left undecayed and so on.

Decay Constant, λ

The rate of decay of radioactive elements is found to be proportional to the number of atoms of the material present. Suppose there are N atoms of a radioactive element present at a time, t, then the probable number of disintegration per unit time or

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activity can be expressed by $-\frac{dN}{dt}$ (The minus sign arises from the fact that N is decreasing with time). Since the rate of disintegration is proportional to the number of atoms present at a given time, we have

$$-\frac{dN}{dt} \propto N$$

or $$\frac{dN}{dt} = -\lambda N \qquad 10.1$$

where λ is a constant of proportionality called the *Decay Constant* of the element.
From 10.1 we have that

$$\lambda = -\frac{1}{N}\left(\frac{dN}{dt}\right) \qquad 10.2$$

Hence Decay Constant is defined as the instantaneous rate of decay per unit atom of a substance
OR

$$\frac{\text{no. of atoms disintegrating per second}}{\text{no. of atoms in the source at that time}} = \lambda$$

Also by integrating equation 10.1 we have that

$$N = N_0 e^{-\lambda t} \qquad 10.3$$

where N_0 is the number of atoms present at time $t = 0$ (i.e. at the time when observations of decay were begun) and N is the number of atoms present at time t.

We obtain the time required for half of the atoms to disintegrate (half-life) by substituting $N = \frac{1}{2}N_0$ into equation (10.3) and eliminating N_0 we have

$$\frac{N_0}{2} = N_0 e^{-\lambda t}$$

$$\frac{1}{2} = e^{-\lambda t}$$

Taking the natural or Napierian logarithm of both sides we have

$$\log_e \frac{1}{2} = -\lambda t$$

But $\log_e \frac{1}{2} = \log_e 1 - \log_e 2 = 0 - \log_e 2$

$= -0.693$

Hence, $-0.693 = -\lambda t$

$$t = \frac{0.693}{\lambda} \qquad 10.4$$

The relation of N and t is displayed in a graph of N against t known as a decay curve (fig. 10.2)

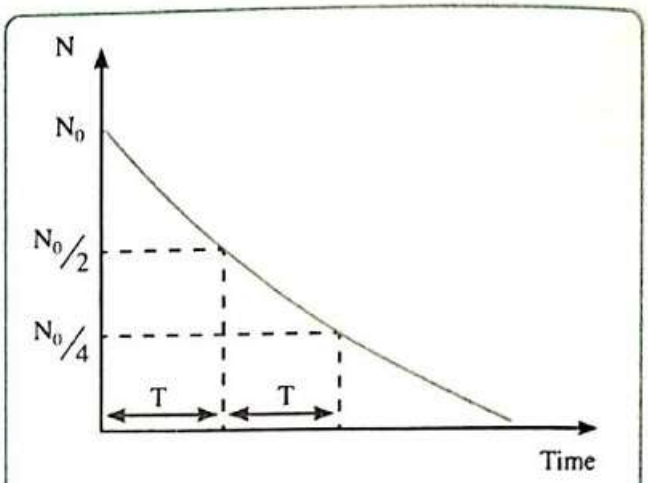


Fig. 10.2 Decay curve and half-life

Example 10.1

A certain radioactive element has a half-life of 10 years.

1 How long will it take to loose $\frac{7}{8}$ of its atoms originally present.

2 How long will it take until only $\frac{1}{4}$ of the atoms originally present remain unchanged.

Solution

If $\frac{7}{8}$ of its atoms has been lost, $\frac{1}{8}$ remains
Half-life = 10 years
If there were N-atoms originally present
$\frac{N}{2}$ atoms remain after 10 years
$\frac{N}{4}$ atoms remain after 20 years
$\frac{N}{8}$ atoms remain after 30 years

$\therefore$ It takes 30 years to lose $\frac{7}{8}$ of its atoms.

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2 $\frac{N}{2}$ atoms remain unchanged after 10 years

$\frac{N}{4}$ atoms remain unchanged after 20 years

Ans = 20 years

Transformation of Elements

There are two types of radioactivity – *natural* and *artificial* radioactivity.

Natural radioactivity is the **spontaneous disintegration** of the nucleus of an atom during which α-particle or β-particle, or gamma rays or a combination of any or all the three and heat (or energy) are released.

When a radioactive element undergoes radioactive decay, it may emit either α-particle, β-particle or γ-rays. This changes the atomic number of the element, hence a new element is formed.

For example, Radium-226 decays by emitting an α-particles to turn into a new element Radon. Radium-226 has a mass number 226 and an atomic number 88 and hence it is denoted by $^{226}_{88}Ra$. The α-particle it emits is a Helium nucleus denoted by $^{4}_{2}He$. So when Radium 226 emits an α-particle, we can write a nuclear equation:

$$^{226}_{88}Ra \longrightarrow {}^{4}_{2}He + {}^{222}_{86}Rn + \text{energy}$$

(Radon - 222)

Radon - 222 decays to Radium - 222 by emitting 2 β-particles. When the nucleus of an atom emits a β-particle (i.e. an electron), the atomic number of the atom increases by one unit, but its mass number remains unaltered. Hence since two β-particles are emitted from Radon 222 we can write the equation

$$^{222}_{86}Rn \longrightarrow 2\,{}^{0}_{-1}e + {}^{222}_{88}Ra + \text{energy}$$

Uranium-238 decays by emitting two α-particles and two β-particles to thorium-230. Hence we can write the nuclear equation thus:

$$^{238}_{92}U \longrightarrow 2\,{}^{4}_{2}H + 2\,{}^{0}_{-1}e + {}^{230}_{90}Th + \text{energy}$$

2α-particles 2β-particles

Thorium-234 decays by emitting a β-particle to the element Protactinium-234 thus:

$$^{234}_{90}Th \longrightarrow {}^{0}_{-1}e + {}^{234}_{91}Pa + \text{energy}$$

Generally we represent alpha (α) decay by

$$^{A}_{Z}X \longrightarrow {}^{4}_{2}He + {}^{(A-4)}_{(Z-2)}Y$$

α-particle

and Beta (β)-decay by

$$^{A}_{Z}X \longrightarrow {}^{0}_{-1}e + {}^{A}_{(Z+1)}Y$$

β-particle

Gamma radiation (γ) is a form of light, emitted as photons of energy hf, and has zero mass number and zero charge ($A = 0, Z = 0$).

To balance a nuclear equation we ensure that the sum of the atomic numbers, Z (subscripts) must be the same on the two sides of the equation. Also the sum of the mass numbers A (superscripts) must be the same on the two sides of the equation.

Artificial Radioactivity

If the radioactivity is induced in an element by *irradiation* with, for example, neutrons, the process is known as *artificial radioactivity*. By irradiation we mean exposure to radiation either by accident or by intent.

Artificial disintegration was first achieved by Rutherford when he disrupted nitrogen nucleus with energetic α-particles, to produce first of all an isotope of fluorine. The fluorine nucleus being unstable disintegrates immediately into an oxygen nucleus and a proton thus:

$$^{4}_{2}He + {}^{14}_{7}N \longrightarrow {}^{18}_{9}F \longrightarrow {}^{17}_{8}O + {}^{1}_{1}H + \text{Energy}$$

In **artificial radioactivity** an ordinary material, not normally radioactive is made radioactive by bombarding it with radioactive particles

$$^{4}_{2}He + {}^{27}_{13}Al \longrightarrow {}^{30}_{15}P + {}^{1}_{0}n \longrightarrow {}^{30}_{14}Si + {}^{0}_{1}e + \text{Energy}$$

Thus phosphorus nuclei which are not stable but radioactive can be produced by bombarding non-radioactive Aluminium with α-particles. The radioactive phosphorus nuclei then disintegrates spontaneously into stable Silicon atoms.

The neutron, the proton and the α-particles have been found very effective bombarding projectiles for disintegrating the nuclei of elements. Examples of such bombardments are:

$$^{1}_{0}n + {}^{6}_{3}Li \longrightarrow {}^{3}_{1}H + {}^{4}_{2}He + \text{Energy}$$

$$^{1}_{0}n + {}^{24}_{12}Mg \longrightarrow {}^{24}_{11}Na + {}^{1}_{1}P + \text{Energy}$$

$$^{4}_{2}He + {}^{9}_{4}Be \longrightarrow {}^{12}_{6}C + {}^{1}_{0}n + \text{Energy}$$

$$^{1}_{0}n + {}^{59}_{27}Co \longrightarrow {}^{60}_{27}Co + \text{Energy}$$

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Isotopes can be made artificially by bombarding neutrons or protons or deuterons at elements; e.g.

$$^{34}_{10}S + ^{1}_{0}n \longrightarrow ^{35}_{10}S + \text{Energy}$$

$$^{79}_{35}Br + ^{1}_{0}n \longrightarrow ^{80}_{35}Br + \text{Energy}$$

Such artificially produced isotopes are unstable and decay with the emission of α-particles, β-particles and γ-rays. They are therefore called *radioisotopes*.

Radioisotopes or radioactive isotopes are isotopes that are made artificially by bombarding neutrons or protons or deuterons at elements

In the above examples, the isotope sulphur-35 was made by bombarding the normal sulphur-34 by neutrons, isotope cobalt-60 was made by bombarding normal cobalt-59 by neutrons, and carbon-12 was produced by the reaction between α-particles and beryllium nuclei.

Radioisotopes are used in (1) medicine as radiotherapy, (2) industry to study defects in metals and welded joints, (3) agriculture as radioactive tracers and preservatives and (4) in geological research for radio-dating.

Health hazards from Radioactive Substances

When the body is exposed to radioactive substances for a long time, the radiations penetrate the body and can destroy cells in tissues and upset natural chemical reactions. Such radiations can bring about genetic changes or *mutations*, causing cancer and undesirable hereditary effects. The ionising properties of α-particles can damage the skin. Radiations from radioactive substances can kill a person, cause sterility, and cause blood abnormalities.

Because of these effects, workers in radiological laboratories must monitor their radiation exposure. They should be shielded with thick blocks of lead, iron or concrete.

Applications of Radioactivity

Radioactive substances find very much application in

(i) Agricultural and scientific research,
(ii) Medical field and
(iii) Industrial field.

(i) Agricultural and Scientific research

Radioactive elements are used in agriculture as *radioactive tracers* and to induce mutations in plants and animals to obtain new and improved varieties. Biologists used them as tracers or markers to help trace the paths of metabolic processes in plants and animals. Geologists and archaeologists use the measurement of half-life to estimate the age of rocks, and carbon-14 (radiocarbon) to date recent organic remains. This is called *radioactive dating*.

(ii) Medical Field

Gamma-rays from radioactive substances are used to treat cancer patients, to sterilize surgical equipment, foods etc.

(iii) In Industry

Radioactive elements are used in industry (a) to study the defects in metals and welded joints, and to check metal fatigue. (b) As radioisotopes, they are used to trace underground pipe leakages.

10.2 Nuclear Energy

The protons and neutrons (nucleons) in the nucleus of each atom are held together by very powerful nuclear forces. An enormous amount of energy is therefore required to tear the nucleon apart. This energy is over 10^6 times more than that required to remove the electrons from an atom.

Nuclear Fission

As was first shown in 1934 by Enrico Fermi, the heavy nucleus of Uranium-235 can be split into two other elements, Krypton and Barium, by bombarding it with a slow neutron

$$^{1}_{0}n + ^{235}_{92}U \longrightarrow ^{141}_{56}Ba + ^{92}_{36}Kr + 3\,^{1}_{0}n + \text{Energy}$$

It was found that the total mass of the component products is less than the mass of the original Uranium. The difference in mass (*mass defect*) is a measure of the nuclear energy released. According to Albert Einstein

$$E = \Delta mc^2$$

where E is the energy released, Δm is the difference in mass and c is the velocity of light. The amount of energy released when 1 g of Uranium-235 undergoes fission is about 7.4×10^{10} J, a prodigious amount of energy. This energy is released in the form of heat.

Nuclear Fission is the splitting up of the nucleus of a heavy element into two approximate equal parts with the release of a huge amount of energy and neutrons.

Fission can occur with most of the very massive nuclei (e.g. Plutonium and Uranium) and has been produced by slow neutrons, high-energy alpha particles, protons, X-rays and gamma-rays.

In the bombardment of Uranium-235 by slow neutron, several neutrons are produced as by-products. These neutrons may cause the splitting of other Uranium nuclei which, in turn yield more neutrons which may further split other Uranium nuclei and so on. Thus a *chain reaction* is set in motion. A chain reaction is a multiplying and self-

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maintaining reaction. When the size of Uranium exceeds a certain *critical mass*, there is a rapid production of neutrons accompanied by a release of a tremendous amount of energy in a *nuclear explosion*. This is the principle of the atomic and nuclear fission bombs.

Fission is also the process used in the present day nuclear power stations.

Nuclear Fusion

Fusion is a nuclear process in which two or more light nuclei combine or fuse to form a heavier nucleus with the release of a large amount of energy.

A typical example of nuclear fusion is the fusion of two hydrogen nuclei to form Helium, He.

$$^{2}_{1}H + ^{3}_{1}H \longrightarrow ^{4}_{2}He + ^{1}_{0}n + \text{Energy}$$

↑ ↑ ↑ ↑

deuterium tritium Helium neutron

In the above fusion process the two isotopes of hydrogen, deuterium and tritium, combine to form the heavier nucleus of Helium. To bring the two light nuclei together in a fusion process, very high temperature of the order of 10^6 - 10^8 degrees Celsius are required to over-come the coulomb repulsive forces between the two nuclei. This poses severe technological problem due to the fact that materials to withstand this high temperature are difficult to come by. It is for this reason that fusion reactions have not been harnessed in nuclear power stations on Earth to produce power, even though they theoretically offer the prospect of enormous quantities of cheap power.

The sun and stars produce an enormous output of energy through nuclear fusion. This is because of the presence of hydrogen isotope in the sun and the conditions of extremely high temperature and pressure are found in the interior of the sun and stars.

Advantages of Fusion over Fission

1 Fusion is more easily achieved with lightest elements such as hydrogen; nuclear repulsion is easily overcome as nuclei approach each other.
2 The raw materials required for fusion are more cheaply and readily available. For example, hydrogen can be obtained by the electrolysis of sea water which is cheap and in plentiful supply.
3 Fusion process produces less dangerous (i.e. non radioactive) by-products.
4 There is no upper limit to the mass of hydrogen that can be exploded in a nuclear fusion process, so very large energies can be obtained.

Nuclear fusion has been used in the hydrogen bomb.

Peaceful Uses of Nuclear Energy

Nuclear energy released during nuclear reactions find very many uses today.

1 Many nuclear power plants are now being used to generate cheap electricity. For countries which lack oil or coal, nuclear power stations have become the ultimate solution to the problem of energy.
2 In the medical field, several fission products obtained in nuclear reactors are used for radiotherapy. For example such radioisotopes as ^{131}I, ^{60}Co and ^{192}Ir have been found very useful in radiotherapy.
3 Radioisotopes from nuclear plants are used in agriculture as radioactive tracers and preservatives.
4 Some spacecrafts, ships and submarines are powered by nuclear energy. Such vehicles have built-in reactors.

Nigeria Nuclear Energy Programme

Nigeria has been showing interest in the development of a nuclear energy programme to help in solving the perennial problem of electricity supply in the country. It is estimated that Nigeria will suffer from an electricity shortage of some 11,000 to 17,000MW by 2017 when it is believed that the total demand of electricity will be between 28,000 and 32,000MW. The hope is that nuclear energy will contribute to the bridging of this gap in electricity supply.

The Nigeria National Atomic Energy Commission (NAEC) has proposed a nuclear road map, which consists of three phase technical frame work to generate at least 1000MW of electricity through power by 2017 and to increase this to 4000MW by 2027. Two technical committees were inaugurated the country's first power plant.

The challenges facing the development of nuclear energy in Nigeria include the handling of the radioactive material. There is concern about the safety and security of the radioactive sources. There is lack of dedicated storage facilities and detection capacity at point of entry, inadequately trained staff and inadequate tracking of sources.

Meanwhile, the International Atomic Energy Agency (IAEA) has certified Nigeria as having met the first requirement set by the agency for countries wishing to generate power from nuclear energy. Nigeria appears ready to undertake the significant work necessary work to prepare for ordering its first nuclear power plant.

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Summary

1 Radioactivity is the spontaneous decay of the nucleus of the atoms of an element during which α-particles, β-particles, and γ-rays and energy are emitted.

2 Radioactive elements are those elements that spontaneously emit radiation from their nucleus e.g. Radium, thorium, polonium etc.

3 Alpha (α-) particles (^4_2He) are positively charged nuclei of Helium. They ionise air considerably but have a low penetrating power. Due to their heavy mass, they are only slightly deflected by electric and magnetic fields.

4 Beta (β-)particles ($^0_{-1}e$) are fast-moving electrons, have very small masses and hence are deflected by electric and magnetic fields. They ionise air less than α-particles.

5 Gamma (γ-) rays are electromagnetic waves, have the most penetrating power, are electrically neutral and are not affected by magnetic or electric field. They ionise air slightly.

6 Half life (T) is the time required for one-half of the radioactive atoms of an element present to decay.

7 Decay constant (λ) is the instantaneous rate of decay per unit atom of a substance. It is related to the half-life by the equation

$$T = \frac{0.693}{\lambda}$$

8 *Radioisotopes* are isotopes that are made artificially by bombarding neutrons or protons or deuterons at elements. They are used in medicine, industry and agriculture.

9 Nuclear fission is the splitting up of the nuclei of heavy elements into approximately equal parts with the release of a huge amount of energy and neutrons. Atomic and Nuclear bombs are made by this process.

10 Nuclear fusion is the combination or fusing of two light nuclei to form a heavier nucleus with the release of a large amount of energy. The sun produces its heat through nuclear fusion.

11 Nuclear energy is used for the production of destructive hydrogen or nuclear bombs, but more peacefully, in the medical field for radiotherapy, in industry for production of cheap electricity from nuclear reactors, and for powering spacecrafts, ships and submarines.

Exercise 10

1 If a nucleus ^3_1H decays, a nucleus ^3_2He is formed accompanied with the emission of a
A neutron B proton
C beta particle D gamma particle
E alpha particle
(SSCE)

2 The phenomenon of radioactivity was first discovered by
A Marie Curie B Sir J. J. Thomson
C Henri Becquerel D Neils Bohr
E Enrico Fermi
(SSCE)

3 What are Beta particles?
A Protons B neutrons
C electrons D Helium nuclei

4 Alpha particles are
A not charged
B highly penetrating
C Helium nuclei
D electromagnetic radiation.

5 Two radioactive elements P and Q have half lives 10 and 20 minutes respectively. If they started off with the same number of atoms, what is the ratio of the number of remaining atoms of P to that of Q.
A 8:1 B 1:4 C 4:1
D 1:8 E 1:2

6 When the nucleus of a uranium atom is split into two fragments of nearly equal masses, the sum of the masses of the fragments is less than the mass of the original nucleus. This difference is a measure of the
A atomic mass
B change of momentum of each fragment
C potential energy lost
D kinetic energy lost
E nuclear energy released.
(SC)

7 A substance has a half life of 3 mins. After 6 mins the count rate was observed to be 400. What was its count rate at zero time.
A 200 B 1200
C 1600 D 2400
(JAMB)

8 The number of neutrons contained in the nucleus of $^{238}_{92}U$ is
A 92 B 146
C 238 D 330
(JAMB)

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9 Two radioactive elements *A* and *B* have half lives of 100 and 50 years respectively. Samples of *A* and *B* contain equal number of atoms. What is the ratio of the number of the remaining atoms of *A* to that of *B* after 200 years?

A 4:1 B 2:1 C 1:1
D 1:2 E 1:4

(SSCE)

10 Distinguish between *natural* and *artificial radioactivity*. What are alpha, beta and gamma radiations? Use a table to display their properties.

11 What are *radioisotopes*? Describe the uses of radioisotopes.

12 a. Explain the terms *nuclear fission*, and *nuclear fusion*.

b. State two advantages of fusion over fission and explain briefly why, in spite of these advantages, fusion is not normally used for the generation of power.

c. $^{238}_{92}U$ is a long *half-life* alpha emitter and decays to Thorium Th which, in turn decays by beta emission with a small *decay constant* to an *isotope* of protactinium Pa. The protactinium isotope finally decays by beta emission to an element *Y*. Explain the italicized terms and write down the decay scheme $^{238}_{92}U$ as stated above.

d. State three uses of radioisotopes.

(SSCE)

13 a. What do you understand by the statement: 'the half life of a certain radioactive element is 60 days'?

b. What is the effect of the emission of a β-particle from an atom of radioactive phosphorus of atomic number 15 and mass number 32?

14 Distinguish between natural and artificial radioactivity. Mention two ways in which artificial radioactivity can be induced in a nucleus. (SC/GCE)

15 Write down four peaceful uses of nuclear energy.

16 a. Define *radioactivity*: *half life*; *Decay constant*.

b. Write down the relation between half life and Decay constant.

c. In 180 minutes, the activity of a certain radioactive substance falls to one-eight of its original value. Calculate its half life.

17 The half life of a radioactive iodine is measured to be 8.0 days. A solution contained 1.5 mg of this iodine on a certain day. Write down on a table the mass of iodine remaining after 8, 16, 24, 32, days. Plot a graph of mass against time and from it deduce the mass of iodine that would remain at the end of 30 days.

18 State two facts about nuclear fission and about nuclear fusion. Which of these processes is believed to occur in the sun? What changes take place in the atomic structure of an element *X* with atomic number 88 and mass number 226 if it emits an alpha particle?

19 (a) State two
(i) differences between **nuclear fusion** and **nuclear fission**.
(ii) peaceful uses of atomic energy.
(b) (i) Explain chain reaction.
(ii) State (1) one conditions necessary for **chain reaction** to occur.
(iii) Two components in a nuclear reactor used to control **chain reaction**.
(c) (i) A nuclear reaction is given by: $^{2}_{1}H + ^{3}_{1}H = ^{4}_{2}He + ^{1}_{0}n$ + energy
what type of nuclear reaction is it?
(ii) The isotopes of a nuclide has a half life of 5.4×10^{3}S. Calculate its decay constant.

(SSCE)

20 (a) Explain (i) Nuclear fission, (ii) Nuclear fusion.
(b) (i) State three applications of atomic energy.
(ii) Define ***half-life***.
(iii) Give the expression that relates the half-life, T, and the decay constant, λ, of a radioactive material.
(c) A radioactive element X with atomic number 88 and mass number 226 emits in succession
(i) an alpha particle
(ii) a beta particle and
(iii) gamma radiation.
Explain, using equations where necessary, the changes that take place in the atomic structure of the element at each stage.

(SSCE)

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ENERGY QUANTIZATION

11

Some of the important postulates of the Bohr's model of the atom was that the electrons moved around the nucleus in specified orbits. In these orbits they can move without radiating energy. They can acquire or lose energy only in discrete units called *quanta*. Thus Bohr suggests that *electrons in the atom exist in discrete* (or quantized) *energy states*.

In 1902, Max Planck was able to show that experimental observations in the radiation emitted by substances could be explained on the basis that the energy from such bodies is emitted in *separate* or *discrete* packets of energy known as *energy quanta* of value hf, where f is the frequency of radiation and h is a constant known as *Planck's Constant*. Thus the energy E, of one quantum of radiation or photon is given by

$$E = hf$$

This is known as *Planck's theory of radiation*. The term *quantum* means *fixed amount, or discrete or separate amount* as distinguished from a continuous quantity.

Planck's quantum hypothesis thus suggests (and this is accepted today) that the energy of radiation can be $E = hf$, or $2hf$ or $3hf$, etc. but there cannot be vibrations or radiations whose energy lies between these values. That is energy radiated is not a continuous quantity but is rather quantized - i.e., it exists only in *discrete* amounts. This is the meaning of the concept of *energy quantization*.

At the end of this chapter, the student should be able to:

1. Explain the concept of energy quantization.
2. Use the photon concept to explain the ejection of electrons in the photo-electric effect.
3. Describe X-ray production and state it's characteristic properties, uses, and safety precautions.
4. Solve simple problems involving threshold frequency, work function and Planck's constant.

11.1 Atomic Energy Levels

Electrons in atoms are believed to be arranged around their nuclei in positions known as *energy levels* or *electron orbits* or *electron shells*. The electron is held in its orbit by the force of attraction between it and the nucleus. Each electron in a given atom possesses a specified amount of energy that is related to the distance between the electron and the nucleus. Since the electrons in an atom are restricted to individual orbits they are said to occupy *energy levels*. Electrons in the orbit closest to the nucleus have the highest energy, i.e., it takes more energy to remove such electrons from the first energy level than to remove electrons from any of the other higher levels. Such electrons in the closest orbit to the nucleus are said to be in the *ground state* or *lowest energy level*. The energy of an electron is given by the relation

$$E = -\frac{1}{n^2}R \qquad 11.1$$

where n is the electron's quantum number and R is

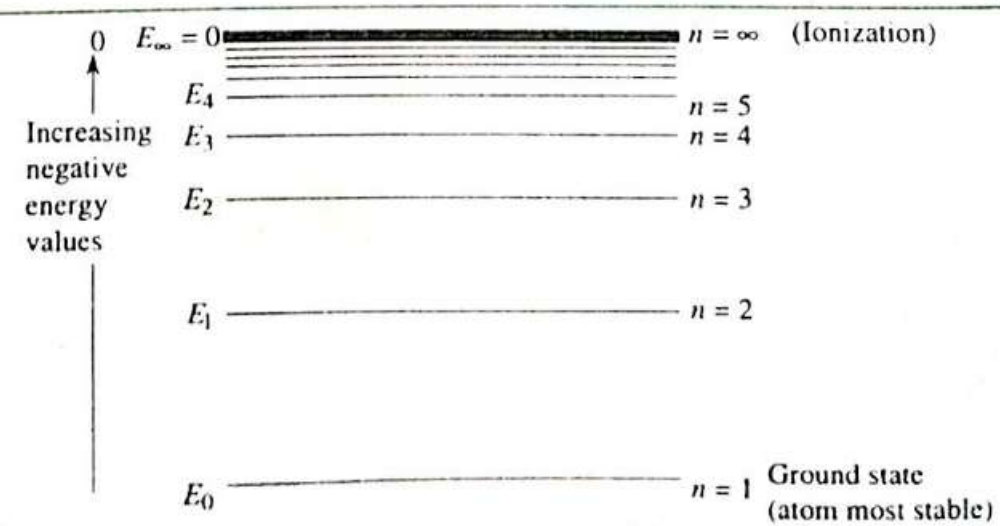


Fig. 11.1 Energy levels in the atom (not to scale)

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a constant. The minus sign signifies that one must do work on the electron to remove it from the atom.

It is customary to draw horizontal lines to represent energy levels as shown in fig. 11.1 for hydrogen. This is called an energy level diagram. Energy states or levels are numbered $n = 1, 2, 3 \ldots$ starting from the *ground state* (E_0) as $n = 1$. When the electron is infinitely far from the atom (or is completely removed from the atom) the atom is said to be *ionized*. The energy of this state with $n = \infty$ is defined as zero.

When an atom is heated or bombarded with an energetic particle, its energy is increased and the atom is said to be *excited*. For the atom to be excited, its energy must be increased by a *definite amount*. No intermediate energy change is possible. During excitation (i.e. the process of increasing the energy of the atom), one or more of the atomic electrons may move to a higher energy level. The electron possesses this higher energy as long as it remains in the *excited state*. Because excited states are unstable, the electron soon jumps back to its original orbit or energy level and emits a *photon* or *quantum* of light with a characteristic frequency, f, (or wavelength, λ) in the process, according to the relation

$$E_n - E_0 = hf_n = \frac{hc}{\lambda_n} \qquad 11.2$$

where E_n is the energy in the excited state and E_0 is the energy in the ground state ($n = 1$).

Not all jumps are to the ground state or ground level. Electrons can also jump from one allowed higher energy level to a lower allowed energy level. The energy of each radiated photon is therefore proportional to the length of its corresponding arrow as shown in fig. 11.4

Example 11.1

An electron jumps from one energy level to another in an atom radiating 4.5×10^{-19} Joules. If Planck's constant is 6.6×10^{-34} Js, what is the wavelength of the radiation? Take velocity of light = $(3\times10^{8}\ \text{ms}^{-1})$

Solution

$\Delta E = E_n - E_m = hf$

$\Delta E = 4.5\times10^{-19} = 6.6\times10^{-34} f$

But $f = \frac{c}{\lambda}$

Hence

$\Delta E = 4.5\times10^{-19}$

$= 6.6\times10^{-34} \times \frac{3\times10^{8}}{\lambda}$

$\lambda = \frac{6.6\times10^{-34}\times3\times10^{8}}{4.5\times10^{-19}}$

$= \underline{4.4\times10^{-7}\ \text{m}}$

Example 11.2

An atom excited to an energy level E_2 ($=-2.42\times10^{-19}$ J) falls to the ground state E_0 ($=-21.8\times10^{-19}$ J). Calculate the frequency and the wavelength of the emitted photon. ($h = 6.6\times10^{-34}$ Js).

Solution

$\Delta E = E_2 - E_0 = hf$

$-2.42\times10^{-19}\ \text{J} - (-21.8\times10^{-19}\ \text{J}) = hf$

$19.38\times10^{-19} = 6.6\times10^{-34} f$

$f = \frac{19.38\times10^{-19}}{6.6\times10^{-34}}$

$= \underline{2.94\times10^{15}\ \text{Hz}}$

$\lambda = \frac{c}{f} = \frac{3\times10^{8}}{2.94\times10^{15}}$

$= \underline{1.02\times10^{-7}\ \text{m}}$

11.2 Franck-Hertz Experiment

Experiments performed by Franck and Hertz gave one of the most direct proofs of the existence of *energy levels* or *electron shells* within an atom.

In these experiments it was possible to measure the energy necessary to raise an electron from the ground state in an atom to an outer orbit or higher state (excitation) or to remove it from the atom entirely (*ionization*).

They bombarded atoms in form of mercury or sodium vapour with electrons emitted from a hot filament, F, and accelerated to higher energy (several electron volts).

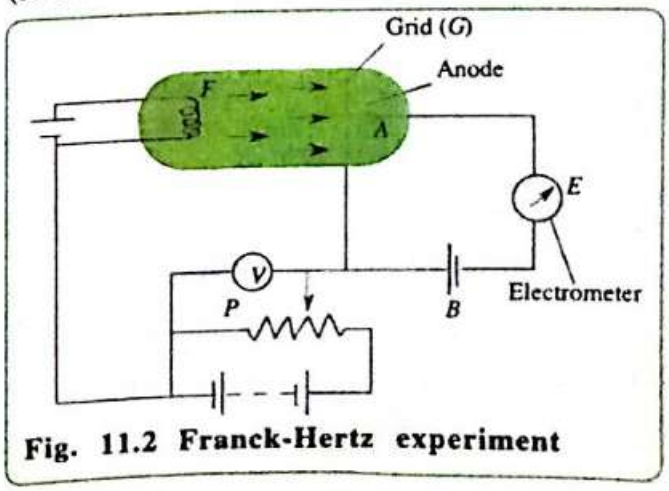


Fig. 11.2 Franck-Hertz experiment

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The electrons were accelerated towards a grid plate (G). The distance between F and G was arranged to be much greater than the mean distance between atomic collisions (the mean free path). This was to ensure that the electrons would make collisions before reaching G. The potentiometer P was used to vary the p.d. (V) between F and G. This is the electron accelerating p.d. The anode (A) close to G, was made negative in potential relative to G by applying a small p.d. between A and G.

The electrons accelerated by the high p.d. acquire high kinetic energy before reaching G. They are however subjected to a retarding field before passing through G to the anode A. The number per second reaching A was measured by an electrometer (E) As V was slowly increased from zero, the speed and number of electrons reaching G increased and the anode plate current increased until the p.d. reached a value V_1. As V was further increased, the current dropped to a minimum, rises again to another p.d, V_2. This pattern of drop and rise in the current was repeated as V was further increased. A graph of current against voltage is as shown in fig. 11.3. It was found that the potential interval between successive peaks was constant and equal to 2.1 volts. (for sodium vapour). This constant potential interval between the successive peaks is known as the excitation potentials (V_c) of the atom.

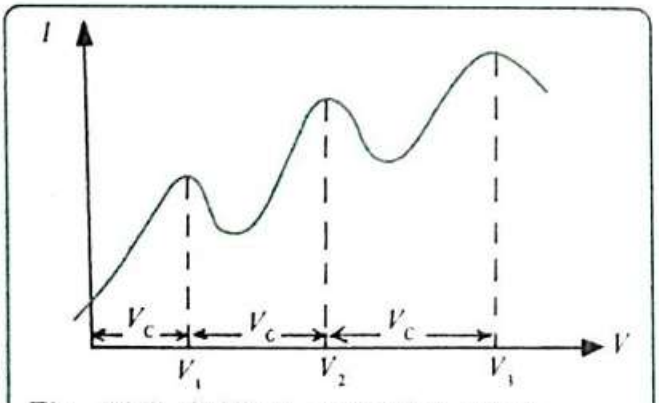


Fig. 11.3 Critical potential curve

At each of the critical potentials (V_1, V_2, or V_3 etc.) the electrons have just sufficient kinetic energy to raise the internal energy of the sodium atom by collision.

11.3 Atomic Spectra; Colour and Light Frequency

An ordinary tube of hydrogen or any other gas does not give off light. The gas can give off light only if energy is first given to the atoms of the gas. A gas atom given energy is said to be *excited*. A gas atom can be excited or energised either by heating or by sending an electrical discharge through the gas. Electrical discharge through the gas can be made by placing a high voltage across a tube containing the gas at low pressure.

When the gas atoms are excited, they give off light which when analyzed is seen to consist of a large number of spectral lines. Each line consists of light of one wavelength or colour. This type of spectrum is called a *line spectrum* or the *atomic spectrum* of the element.

A Line Spectrum is a spectrum that consists of a number of well defined lines each having a particular frequency or wavelength or colour

A *continuous spectrum* on the other hand, consists of light of all colours or wavelengths.

In a *line spectrum* only certain wavelengths of light are emitted, and these are different for different elements and compounds. The emission from a material is known as its *emission spectrum*. The emission spectrum is characteristic of the material and can serve as a type of finger print' for identification of the gas. If a continuous spectrum (a spectrum containing all frequencies or colours or wavelengths) is passed through a gas, dark lines are observed in the spectrum. These dark lines correspond to lines normally emitted by the gas. This is called an *absorption spectrum*. Gases absorb light at the same frequencies at which they emit.

Because the frequencies of light given off by an excited atom of an element are unique or peculiar to that element, each element has its own characteristic *atomic spectrum*. The fact that excited atoms emit light of only certain characteristic frequencies tells us that only characteristic energy changes take place within the atom. *Atomic spectra*, then tell us that when an excited atom loses energy, not just any arbitrary amount is lost. Only certain specific energy changes can occur, which is why only certain specific frequencies of light are emitted. This is the only way atomic spectra can be explained.

The electron in an atom can only have energies corresponding to the set of electron energy levels in the atom. When the atom is supplied with energy (or excited), an electron is raised from a low energy level(E_1) to a higher one (E_2). When the electron drops back, energy equal to the difference, $E_2 - E_1$, between the two levels is released and emitted as a photon of light of frequency f, according to the equations

$$E_2 - E_1 = hf$$

where h is *Planck's constant*. Because only certain energy jumps can occur, only certain frequencies

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can appear in the spectrum.

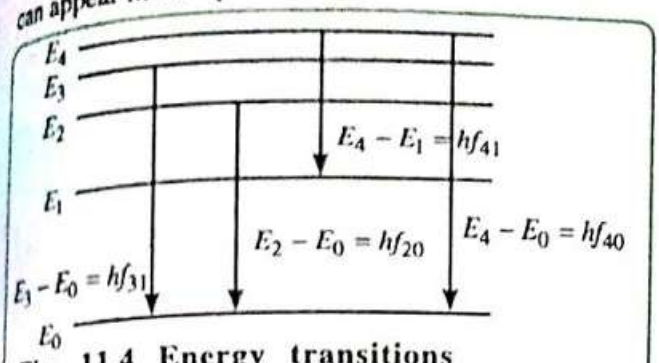
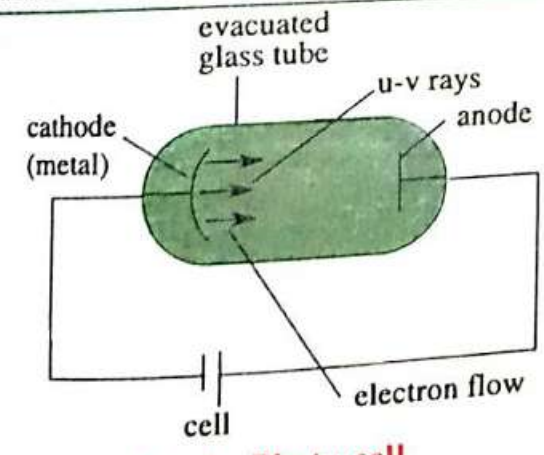


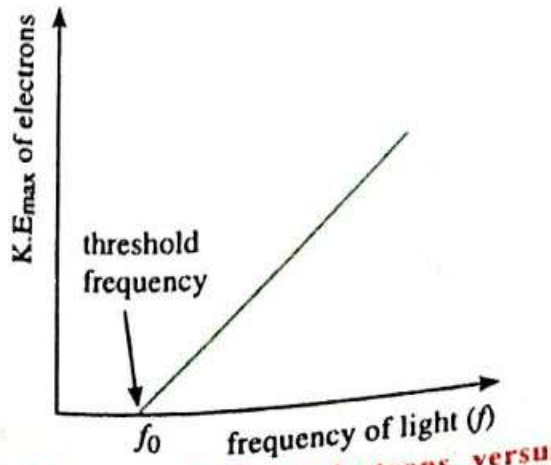
Fig. 11.4 Energy transitions

11.4 The Photo-electric Effect

When light falls on metal surfaces, electrons are emitted. This is the *photo-electric effect*. The emitted electrons are known as photo-electrons. A puzzling feature of this phenomenon as was first discovered by Lenard in 1902, was that the maximum kinetic energy (or maximum velocity) of the emitted electrons was independent of the intensity of the incident light but was proportional to its frequency (or wavelength). When the maximum energy of the photo-electrons are plotted against the frequency of the incident light for a given metal, a graph as shown in fig. 11.5b was obtained.



a. A Photo-cell



b. Maximum K.E of electrons versus frequency of incident light

Fig. 11.5

It shows a straight line graph cutting the frequency axis at f_0 which is known as the *threshold frequency*, because no electrons will be emitted at any lower frequency (or correspondingly at any higher wavelength).

Increasing the intensity of the light increases the number of photo-electrons, but not their energy or velocity. For light with frequency less than threshold frequency complete absorption of incident light energy takes place. The absorbed energy is used to raise the energy level of the surface electron or to overcome the potential barrier of the photoelectron.

Photocells are used in (a) Burglar alarm (b) Television cameras (c) automatic doors (d) automatic street lighting (e) automatic digital counters.

Einstein's equation and explanation

In 1909, *Einstein* proposed an explanation of the photoelectric effect using a concept first put forward by Max Planck that light-waves consist of tiny bundles or packets of energy known as *photons* or *quanta*. Each photon of light (or electromagnetic wave) has an energy, hf, where h is Planck's constant ($= 6.62 \times 10^{-34}$ Js) and f is the frequency of the light.

Einstein's idea can be expressed in an equation.

$$hf = W + \frac{1}{2}mv^2 \qquad 11.3$$

The term hf represents the total energy content of a single photon of incident light. Part of this energy, W, is used to get the electron free from the atom and away from the metal surface. This energy W is known as the *Work function*.

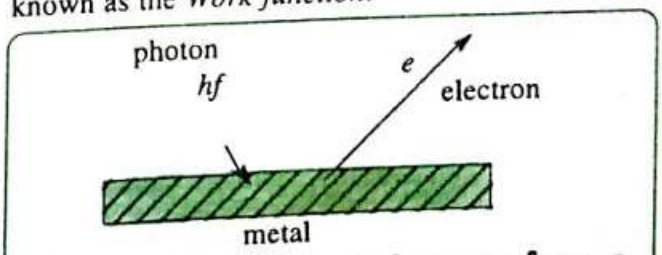


Fig. 11.6 Liberation of electrons from a metal surface

The remainder of the energy is used to give the liberated electron a kinetic energy, $\frac{1}{2}mv^2$ where v is the velocity of the photoelectron of mass, m. W is different for different metals. The work function W is given by

$$W = hf_0 \qquad 11.4$$

where f_0 is the threshold frequency.

The **threshold frequency** is the frequency of light which, falling on a surface, is just sufficient to liberate electronics without giving them any additional kinetic energy.

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The **Work function** (W) of a metal is the minimum energy required to liberate an electron from a metal surface.

Equations 11.3 and 11.4 can be combined to obtain

$$hf = hf_0 + \frac{1}{2}mv^2 \qquad 11.5$$

or $$hf = hf_0 + E \qquad 11.6$$

where $E = \frac{1}{2}mv^2$ is the kinetic energy of the photoelectrons.
From (11.6) we have that

$$E = hf - hf_0 \qquad 11.7$$

This is *Einstein's photoelectric equation.*

The energy of the ejected electron may be found by determining what potential difference V must be applied to stop its motion.
Then

$$E = \frac{1}{2}mv^2 = eV \qquad 11.8$$

The product eV is the electron-Volt.

$$1\,eV = 1.6\times10^{-19}\text{ Joules}$$

Hence we can also write equation 11.7 as

$$E = eV = hf - hf_0 \qquad 11.9$$

For any surface the incident light must be of short enough wavelength (or high enough frequency) so that the photon has enough energy (hf) to eject the electron.

Applications of photocells are in (i) Photometers, (ii) Burglar alarms, (iii) Automatic doors, (iv) Television cameras, (v) Sound production from film tracks.

Example 11.2

Calculate the energy in joules of ultraviolet light of wavelength 3×10^{-7} m. Take the velocity of light as $3\times10^{8}\,ms^{-1}$ and Planck's constant as 6.6×10^{-34} Js.

Solution

$$E = hf = \frac{hc}{\lambda}$$

$$= 6.6\times10^{-34}\times\frac{3\times10^{8}}{3\times10^{-7}}\text{ Joules}$$

$$= \underline{6.6\times10^{-19}\text{ J}}$$

Example 11.3

Compute the frequency of the photon whose energy is required to eject a surface electron with a kinetic energy of 3.5×10^{-16} eV if the work function of the metal is 3.0×10^{-16} eV. Take $h = 6.6\times10^{-34}$ Js, 1 eV = 1.6×10^{-19} J.

Solution

$$E = hf - W$$

$$3.5\times10^{-16} = hf - 3\times10^{-16}\text{ eV}$$

$$hf = 6.5\times10^{-16}\text{ eV}$$

$$= 6.5\times10^{-16}\times1.6\times10^{-19}\text{ J}$$

$$f = \frac{6.5\times10^{-16}\times1.6\times10^{-19}}{6.6\times10^{-34}}$$

$$= \underline{0.157\text{ Hz}}$$

Example 11.4

A certain metal has a work function of 2.5 eV. Calculate (i) the threshold frequency (ii) the maximum energy of the liberated electrons when the metal is illuminated by light of wavelength 3.5×10^{-5} cm. Take $h = 6.6\times10^{-34}$ Js, $c = 3.0\times10^{8}\,ms^{-1}$, 1 eV = 1.6×10^{-19} J.

Solution

$$W = hf_0$$

where W is work function, h is Planck's constant and f_0 is threshold frequency

$$2.5\times1.6\times10^{-19}\text{ J} = 6.6\times10^{-34}\times f_0$$

$$f_0 = \frac{2.5\times1.6\times10^{-19}}{6.6\times10^{-34}}\text{ Hz}$$

$$= \underline{6.06\times10^{-34}\text{ Hz}}$$

Maximum energy of electron E is given by

$$E = hf - W$$

$$\lambda = 3.5\times10^{-5}\text{ cm} = 3.5\times10^{-7}\text{ m}$$

$$f = \frac{c}{\lambda}$$

$$E = \frac{hc}{\lambda} - W$$

$$= \frac{6.6\times10^{-34}\times3\times10^{8}}{3.5\times10^{-7}} - 2.5\times1.6\times10^{-19}$$

$$= 5.66\times10^{-19} - 4\times10^{-19}$$

$$= \underline{1.66\times10^{-19}\text{ J}}$$

Example 11.5

Calculate the energy in Joules of a light photon of wavelength 7000 Angstrom(Å). Take 1 Å = 10^{-10} m, velocity of light, $c = 3\times10^{8}\,ms^{-1}$, $h = 6.6\times10^{-34}$ Js.

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Solution

$$\text{Energy of photon} = hf = \frac{hc}{\lambda}$$

$$= \frac{6.6\times10^{-34}\times3\times10^{8}}{7000\times10^{-10}}\ \text{J}$$

$$= 2.83\times10^{-19}\ \text{Joules}$$

11.5 X-rays

X-ray was discovered accidentally in 1895 by William Roentgen. In principle the production of X-rays is the inverse of the photo-electric effect and can therefore be interpreted in a similar manner.

Production of X-rays

The apparatus for the production of X-rays is as shown in fig. 11.7. The X-rays are produced when thermally generated electrons from a hot filament are accelerated through a high voltage and focused onto a tungsten target.

The electrons are produced by thermionic emission from the hot cathode which is a heated tungsten filament. This filament lies inside a curved metal cylinder which acts to focus the emitted electrons on to a tungsten target embedded in a copper block which also acts as the anode. Tungsten is used as the target because it is a metal of high melting point.

The metal cylinder is kept at a high negative potential relative to the anode. This serves to accelerate the electrons through the X-ray tube which is also evacuated in order that the accelerated electrons can get to their target without losing much of their energy. A large fraction of the kinetic energy of the accelerated electrons is transformed into heat when these electrons are suddenly stopped on hitting the tungsten target. The target must therefore be made of a metal of high melting point. The rest of the energy of the electrons is converted into X-rays which are radiated from the tungsten target through a window.

The heat so generated is conducted through the copper anode of high conductivity and removed by cooling fins.

The following energy conversions take place during the process of producing X-rays. The electrical energy is converted into the thermal energy of the thermionic electrons. This thermal energy is converted into the mechanical kinetic energy of the accelerated electrons. This is in turn converted into the electromagnetic energy of the X-rays and thermal energy.

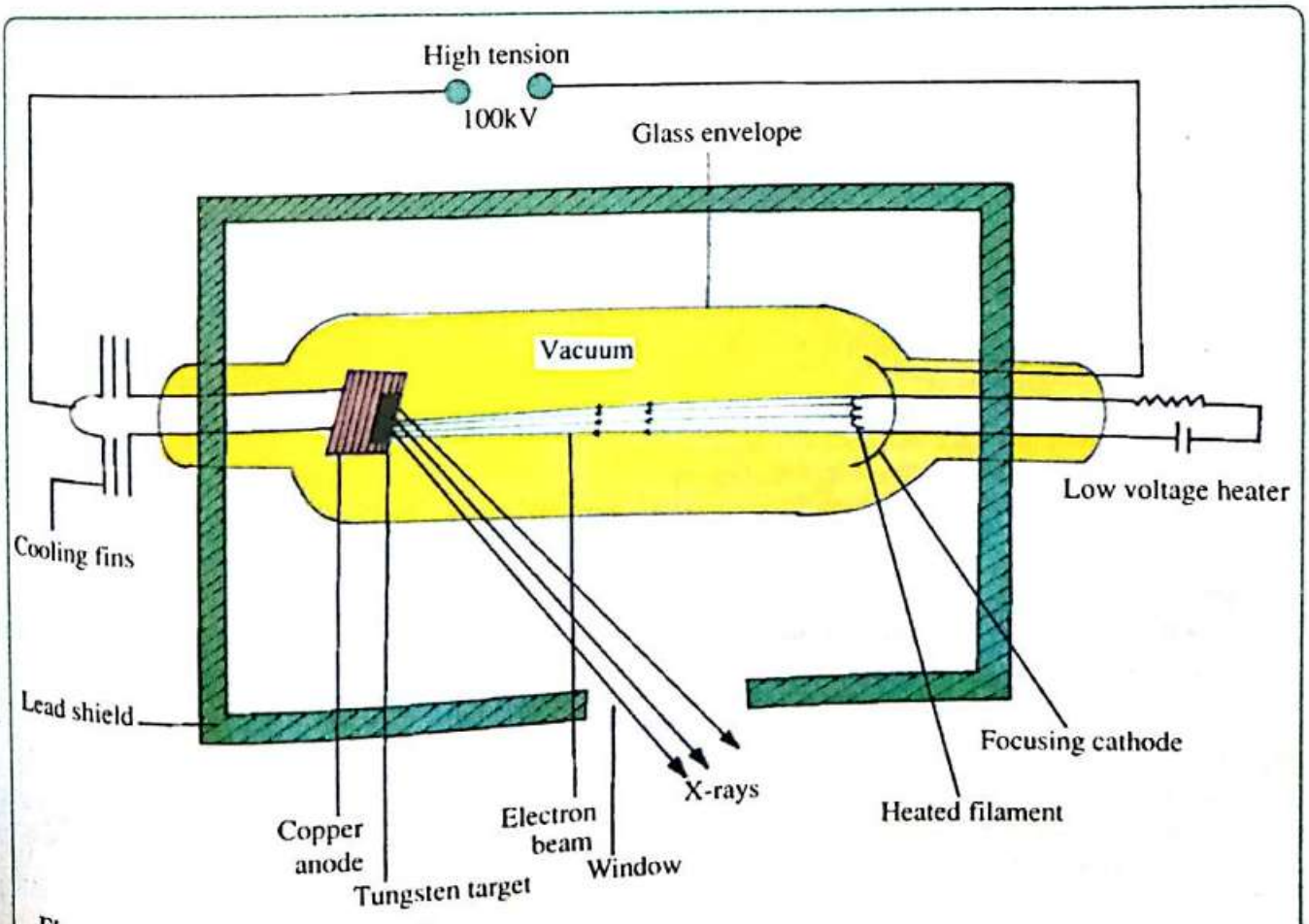


Fig. 11.7 A Modern x-ray tube

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Types of X-rays

There are two types of X-rays - the *Hard X-rays* and the *Soft X-rays*. The Hard X-rays have a high penetrating power and a short wavelength. They can be produced by increasing the p.d. between the cathode and anode of the X-ray tube. The higher this p.d, the faster and more energetic the bombarding electrons.

Soft X-rays have a lower penetrating power and longer wavelengths.

Hardness is a measure of the strength or penetrating ability of the X-rays.

Intensity is the energy radiated per unit time per unit area by the X-rays. The intensity of the X-rays is governed by the current of the hot filament.

Characteristics and Properties of X-rays

1 X-rays are electromagnetic waves of very high frequency. Like light, they have a velocity of 3×10^{8} ms^{-1} in free space.
2 They have very short wavelength (2×10^{-10} m), much shorter than that of light waves.
3 Their major characteristic is that they have a high penetrating power. They can easily pass through most solid substances which are opaque to ordinary light, e.g. flesh, wood, paper, aluminium foil, etc. Soft X-rays do not penetrate dense metals, and bones.
4 Like light, X-rays travel in straight lines.
5 They are not deflected by electric or magnetic fields.
6 They are diffracted by crystals.
7 They cause fluorescence in zinc substances e.g. zinc sulphide.
8 They cause the liberation of electrons when they fall on certain substances (Photoelectric effect).
9 They ionise gases.

Comparison of X-ray and Optical Spectra

X-ray spectra from different metals such as iron, tungsten, copper or zinc are all similar in appearance. Each spectrum indicates energy changes of electrons in the interior of the atom, close to the nucleus. On the other hand, optical spectra of such metals are related to the energy changes of electrons in the outermost shells or orbits of the atoms, which are different for different metals. Hence the optical spectra are different for different metals.

Practical Application of X-rays

1 Because of their high penetrating power, X-rays are used for examining the body to locate broken bones or hidden metallic objects like bullet in a patient's leg or a pin swallowed by a child. X-rays penetrate the flesh but not the bone or the metal object. They thus disclose the positions of the bone or the metal objects.
2 They are used in airports to detect metals and contraband in a baggage.
3 In industry they are used to detect cracks and flaws in metal castings and welded joints.
4 They are used as analytical tool for investigating crystal structure.
5 They are used in the treatment of tumours and some skin diseases. Cancer cells can be destroyed by X-rays.
6 They can be used to detect alterations which have been made on works of art.
7 They are used in radio-therapy; radiography; and in agriculture to kill germs;

Hazards of X-rays and Safety Precautions

Exposure to X-rays can burn the skin (skinburns), can also cause cancer. Deeper within the body, the X-rays can also destroy body cells. X-rays cause genetic mutations with undesirable hereditary consequences. They can damage body tissues and cause Leukaemia.

Special precautions are therefore necessary for people who work with X-rays. They should wear lead coats, and have regular health checks.

Summary

1 Electrons in atoms exist in discrete or quantized energy states.
2 *Planck's theory of radiation* suggests that energy is emitted from heated bodies in discrete or separate packets known as energy quanta. The energy of one quantum of radiation is given by
$$E = hf$$
3 The Franck-Hertz experiment gave the most direct proof of the existence of energy levels in atoms.
4 *Line spectrum* is a spectrum that consists of a number of well defined lines each having a particular frequency or wave-length. They arise when electrons in excited atoms jump from one level to a lower level.
5 In the *Photoelectric effect*, electrons are emitted from the surface of metals when an electromagnetic wave e.g. light, falls on the metal surface. The maximum Kinetic energy of the photoelectrons is independent of the intensity of the incident light. It depends on the frequency or wavelength of the incident light.

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6 The *threshold frequency*, f_0, is the lowest frequency that can cause photoemission from a metallic surface.

7 *Work function* ($W = hf_0$) is the minimum energy required to liberate an electron from a metallic surface.

8 *Einstein's photoelectric equation* states that the maximum energy (E) of the emitted electrons is given by

$E = hf - W$ or $E = eV = hf - hf_0$

9 X-rays are produced when thermally generated electrons from a hot filament are accelerated through a high p.d. and focused onto a tungsten target of high melting point where the electrons are suddenly stopped. The energy of the impinging energetic electrons is converted mostly into heat. A small percentage of the energy is converted into X-rays. The heat generated is removed by cooling fins.

X-rays are electromagnetic waves of short wavelength ($\sim 10^{-8}$ cm). They have high penetrating power. They are used in the medical field and in industry.

Exercise 11

Take $h = 6.6 \times 10^{-34}$ Js

1 Which of the following gives rise to the line spectra observed in atoms?
- A Kinetic energy of moving atoms
- B Potential energy of an electron inside an atom
- C Change of an electron from a higher to a lower energy level in the atom
- D Disturbed proton in the nucleus
- E Excitation of an electron in the atom.

(SSCE)

2 Which of the following phenomenon is called *photoelectric effect*?
- A High energy electrons impinge on a metallic anode which then emits photons
- B A high energy photon emits photons as it is slowed down.
- C A metal absorbs quanta of light and then emits electrons.
- D Two electrons are created from a quantum of light.

(JAMB)

3 Which of the following statements about X-rays is/are correct?

I. X-rays do not consist of electrically charged particles.

II. X-rays are not deflected in an electric or magnetic field.

III. The wavelength of X-rays is longer than that of light.

- A I only
- B I and II only
- C II only
- D II and III only
- E I and III only.

(SC/GCE)

4 When a radiation strikes a metal surface, electrons may be ejected from the metal. The maximum kinetic energy which may be acquired by an ejected electron depends on the
- A Intensity of the radiation
- B Source of the radiation
- C Wavelength of the radiation
- D Detection device for the electron.

(JAMB)

5 If a small child swallowed a safety pin, why would an X-ray photograph clearly show the location of the pin?

6 State the nature and properties of X-rays. What are the uses of X-rays in Medicine and in Industry.

7 Define *Photoelectric effect*, *Work function*, *threshold frequency*. State and explain Einstein's photoelectric equation.

8 a. (i) Explain the terms: *photoelectric emission* and *threshold frequency*

(ii) Einstein's photoelectric equation can be written as

$E = hf - hf_0$

What does each of the symbols used in the equation above represent?

b. Calculate the frequency of the photon whose energy is required to eject a surface electron with a kinetic energy of 1.97×10^{16} eV if the work function of the metal is 1.33×10^{-16} 6eV

[1 eV = 1.6×10^{-19} J, $h = 6.6 \times 10^{-34}$ Js]

c. In a photoelectric cell, no electrons are emitted until the threshold frequency of light is reached. Explain what happens to the energy of the light before emission of electrons begins. State one factor that may affect the number of emitted electrons.

(SSCE)

9 Calculate the energy of a photon of red light of wavelength 6000 Å. (Take 1 Å = 10^{-10}m)

10 Ultraviolet light of wavelength 2000 Å is incident on a potassium surface. What is the maximum kinetic energy of the photoelectrons ejected? What retarding potential difference will be required to stop the emission of electrons? Assume that the photoelectric threshold wavelength of potassium is 4400 Å and 1 Å = 10^{-10} m.

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11 Calculate the energy in Joules of ultraviolet light of wavelength 3×10^{-7} m.

12 With the aid of a diagram describe the production of X-rays. State five properties of X-rays and three uses to which X-rays are put in practical life.

13 a. (i) By means of a labelled diagram, describe the mode of operation of a modern X-ray tube.
(ii) State the energy transformations which take place during the operation.
b. Explain the terms *hardness* and *intensity* as applied to X-rays
c. (i) State *three* uses of X-rays.
(ii) State one hazard of over-exposure to X-rays in a radiological laboratory, indicating two safety precaution. (SSCE)

14 a. What is the energy of a photon of light of wavelength 4500 Å, in Joules and in electron-volts? (1 Å = 10^{-10} m)
b. What is the work function of a metal whose photoelectric threshold wavelength is 6.8×10^{-3} Å.

15 Define *photoelectric effect, threshold-frequency, Work function.*
The threshold frequency for a metal is 5.5×10^{14} Hz. What is the work function for the metal? If light of wavelength 4.5×10^{-7} m shines on the metal, calculate the maximum kinetic energy of the emitted photoelectrons in Joules and in electron-volts. (1 eV = 1.6×10^{-19} J)

16 Describe Franck-Hertz experiment and the conclusions that can be drawn from the result of the experiment.

17 a. Explain what is meant by *photoelectric emission.*
b. Draw a labelled diagram showing the structure of a simple type of a photocell and explain its mode of operation.
c. State *four* applications of photoelectric emission. (SSCE)

18 (a) Explain the term **Photoelectric Effect.**
(b) Draw a photocell with its associated electric circuit and label its parts.
(c) What factors determine the
(i) Current produced by the photocell?
(ii) the maximum kinetic energy of the photoelectrons?
(d) State one similarity and one difference between photoemission and evaporation.
(e) Name two methods by which a beam of free electrons may be produced other than photoemission.
(f) State two applications of photoelectric effect.
(g) A light wavelength 5.0 x 10^{-7}m is incident on metal resulting in photoemission of electrons. If the work function of the metal is 3.04 x 10^{-19}J calculate the
(i) frequency of the light
(ii) energy of the incident photo
(iii) maximum kinetic energy of the photoelectrons (speed of light = 3.00 x 10^{8}ms^{-1}; Planck's constant = 6.6 x 10^{-34}JS. (SSCE)

19 (a) State two
(i) properties of x-rays
(ii) reasons to show that x-rays are waves,
(iii) uses of x-rays other than those in medicine
(iv) hazards of x-rays.
(c) The potential difference between the cathode and target of an x-ray tube is 5.00 x 10^{4}V and the current inn the tube is 2.00 x 10^{-2}A. Given that only one percent of the total energy supplied is emitted as x-radiation, determine the
(i) maximum frequency of the emitted radiation;
(ii) rate at which heat is removed from the target in order to keep it at a steady temperature. (Planck's constant, h = 6.63 x 10^{-34}JS, electronic charge, e = 1.6 x 10^{-19}C)

20 (a) With the aid of a labeled diagram describe the mode of operation of a modern x-ray tube.
(b) State the energy transformations that take place during the operation of the x-ray tube.
(c) Define as applied to x-ray, the following terms
(i) hardness
(ii) intensity.
(d) State (i) four uses of x-rays (ii) one hazard of over-exposure to x-rays in a radiological laboratory. (SSCE)

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21 (a) Define Ionization Potential

(b) (i) State the three types of emission spectr

(ii) Name one source each which produces each of the spectra stated in 21(b)(i)

(c) In an X-ray tube, electrons are accelerated to the target by potential difference of 80 KV, calculate

(i) speed of the electrons

(ii) threshold wavelength of the electrons ($h = 6.6 \times 10^{-34}$JS, $e = 1.6 \times 10^{-19}$C. $Me = 9.1 \times 10^{31}$Kg)

(d) An X-ray photon of frequency 4.5×10^{-18}Hz strikes an electron, assumed to be at rest. If the electron absorbs all the photon energy, calculate the speed acquired by the electron. ($h = 6.6 \times 10^{-34}$JS, $Me = 9.1 \times 10^{31}$Kg)

22 (a) State two conditions under which photo-electrons can be emitted from the surface of a metal.

(b) List two particle characteristics of electro-magnetic waves.

23 In a photoelectric emission experiment, when the incident photon is of frequency F_1, the maximum kinetic energy of the photoelectrons is E_1, where the photon of frequency F_2 is incident on the same surface the photoelectrons is E_2. Deduce an expression for Planck's constant.

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12 WAVE-PARTICLE PARADOX

Matter sometimes behaves as a wave and at other times as particles. There are therefore two theories of matter — the wave theory and the particle theory. This chapter discusses this paradox in the behaviour of matter.

At the end of this chapter the student should be able to:

1 Identify phenomena which are only satisfactorily explained by assuming that matter behaves like (a) waves and (b) particles.
2 Explain the uncertainty principle in very general terms and give examples.

12.1 Wave Nature of Matter

Electron Diffraction

The wave nature of X-rays was established by X-ray diffraction experiments. In the same way the Davisson and Germer experiment established the wave nature of electrons.

In the Davisson and Germer experiment, a beam of electrons emitted from a heated filament was made to impinge on a layer of a thin metal film or crystal at *C*. The electrons were diffracted and the diffraction rings were produced on a photographic plate placed behind the thin metal film as shown in fig. 12.1.

If the voltage *V* on the anode was increased, the velocity, *v*, of the electrons was increased. The rings were then seen to become narrower. Hence the wavelength, λ, of the electron waves decreases with increasing electron velocity.

The wavelength of a material object is given by

$$\lambda = \frac{h}{mv} \qquad 12.1$$

where mv is the momentum of the object and h is Planck's constant.

Later experiments showed that protons, neutrons and other particles also have the wave property of diffraction.

12.2 Particle Nature of Matter

a. Photoelectric Effect

In the photoelectric effect, it was shown that when light falls on a metal surface, electrons are emitted from the surface. Similarly when X-ray is allowed to fall on the surface of a thin sheet of metal like gold, the X-ray not only produces diffraction pattern but also acting like particles they may collide with the atoms of the metal and eject electrons as in the photoelectric effect.

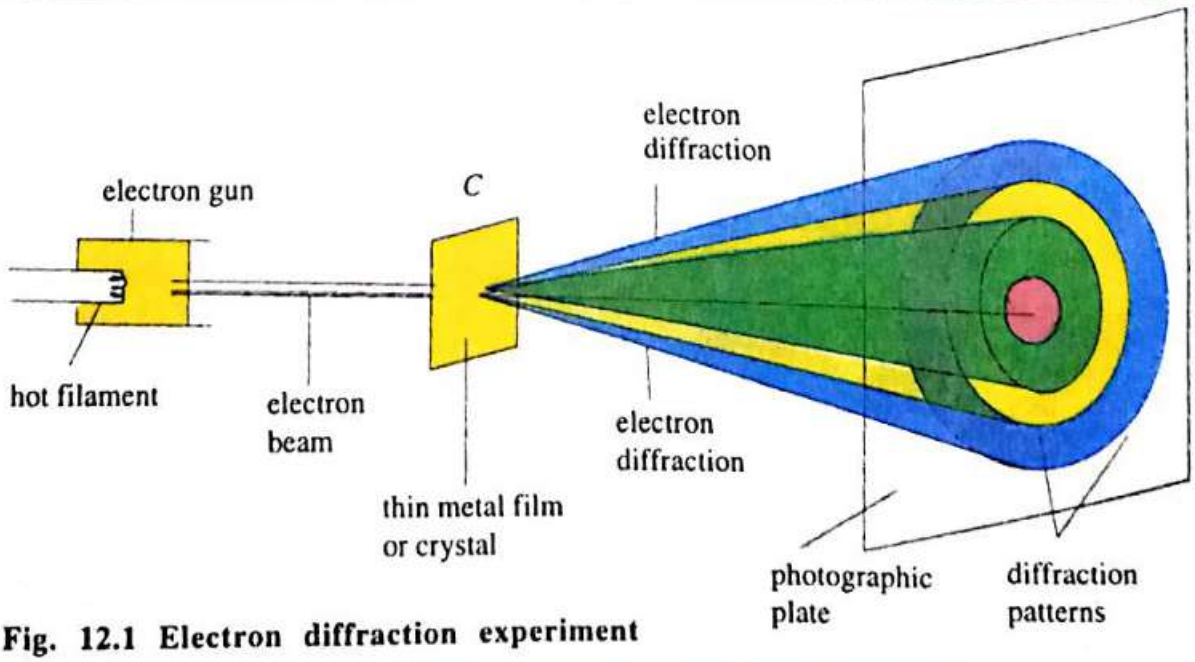


Fig. 12.1 Electron diffraction experiment

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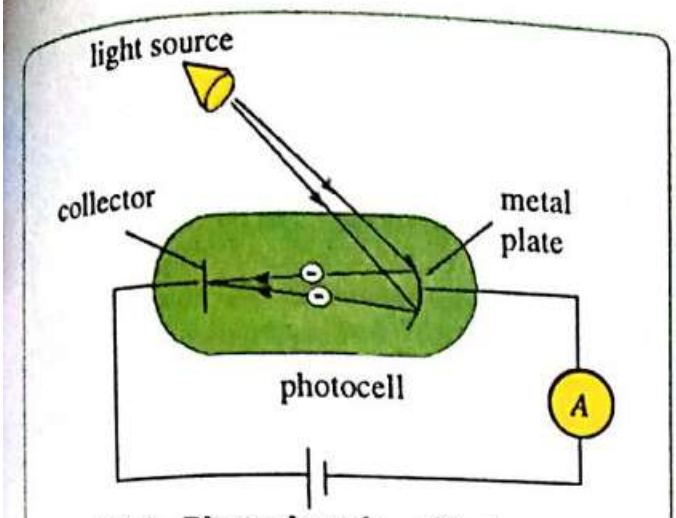


Fig. 12.2 Photoelectric effect

b. The Compton Effect

When a single X-ray photon collides with a free electron, the electron recoils off as though it were a perfectly elastic sphere.

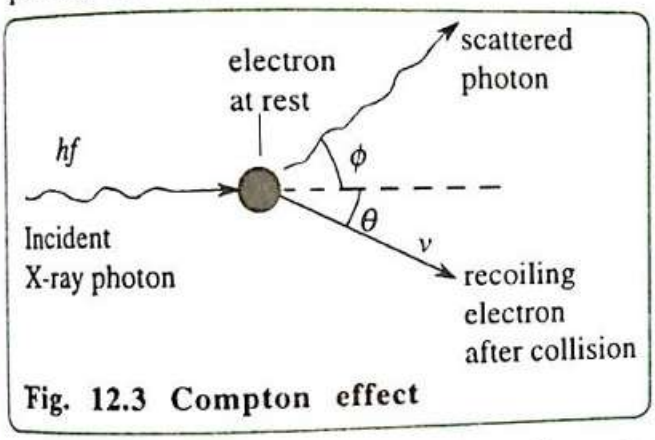


Fig. 12.3 Compton effect

This is the *compton effect*. In this effect, the scattered photon has a slightly lower frequency than the incident X-ray photon. In this phenomenon, matter in form of X-rays is shown to behave as a particle. The recoiling photon and electron are able to conserve energy and momentum.

12.3 Wave-Particle Duality

From the foregoing discussions, we seem to be in a dilemma as to the nature of matter. Some experiments e.g. electron diffraction indicate that matter behaves like a wave, but other phenomena such as photoelectric effect and compton effect experiments indicate that matter behaves like a stream of particles or photons.

These two theories seem to be incompatible but both have been shown to have validity. Thus matter appears to have a dual nature. This is referred to as the *wave-particle duality* or the *wave-particle paradox*.

As in matter, so is it with light. Some observable phenomenon in the nature of light, such as *reflection, refraction, diffraction, interference* and *polarization* can be interpreted or explained by assuming that light (or matter) behaves like waves. But other observable phenomena such as *emission* and *absorption of light, photoelectricity, radiation of energy* from heated bodies, *thermionic emission* can only be understood by assuming the particle nature of matter.

This dual nature of matter is known as the *wave-particle duality* or the *wave-particle paradox*.

The wave-particle duality refers to the idea that light and matter (such as electrons) have both wave and particle properties, that is, light behaves either as a wave or as a particle but not as both simultaneously.

12.4 The Uncertainty Principle

In general, the process of making a measurement tends to alter the quantity being measured. This is very much pronounced in very small scales such as the atomic and the nuclear scales. As was first pointed out by Heisenberg, it is impossible in principle to make precise measurement of both the position (x) and momentum (p) of a particle simultaneously. In fact the more precisely the location of the particle has to be specified, the more uncertainty is introduced into the determination of its momentum and conversely too. Any such measurements have inbuilt uncertainties Δx in the position and Δp in the momentum. Such measurements can only be expressed as probabilities. Heisenberg showed that:

$$\Delta x \,.\, \Delta p \geq h$$
$$\Delta x \,.\, \Delta v \geq h \qquad 12.2$$
$$\Delta E \cdot \Delta t \geq h$$

here ΔE, Δt, Δp and Δx are the uncertainties in the energy, time, momentum and position measurements.

1 *Momentum* and *position*,
2 *Energy* and *time*,
3 *Position* and *velocity*, are known as *complementary variables*.

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Heisenberg Uncertainty Principle states that it is impossible to know accurately the exact position and momentum of a particle simultaneously. The uncertainty in the momentum multiplied by the uncertainty in the position approximately equals the Planck's constant, *h*.

Because of the extremely small value of *h* ($= 6.63 \times 10^{-34}$ Js) the uncertainty principle is of no consequence for objects above atomic sizes.

Summary

1 The phenomena that can be rightly explained by assuming that matter behaves like a wave are: *electron diffraction or the diffraction of light, reflection, refraction, interference and polarization of light.*

2 The phenomena that can be explained by assuming that matter behaves like a Particle are: *emission and absorption of light, photoelectric effect, thermionic emission, compton effect.*

3 The *wave-particle duality* refers to the idea that light and matter have both wave and particle properties.

4 According to the uncertainty principle, if Δx is the uncertainty in the measurement of x (position) and Δp is the uncertainty in the measurement of p (momentum) then $\Delta p \cdot \Delta x \geq h$, where h is Planck constant. Momentum and position, energy and time, velocity and position, are known as *complementary variables.*

The greater the accuracy in measuring one variable (e.g. energy) the smaller the accuracy in measuring the other complementary variable (e.g. time).

Exercise 12

1 Which of the following is/are not complementary variables.
- A Energy and Time
- B Energy and position
- C Energy and Mass
- D Velocity and position.

2 In which of the following is/are the *Uncertainty Principle* most applicable.
- A The momentum of an electron at a particular speed.
- B The momentum of an electron at a particular time.
- C The energy of a satellite at a particular time.
- D The energy of an electron in a cathode-ray tube at a particular time.
- E The momentum of a fast car at a particular position.

3 Which of the following phenomena is called photoelectric effect?
- A High energy electrons impinge on a metallic anode which then emits photons.
- B A high energy photon emits photons as it is slowed down.
- C A metal absorbs quanta of light and then emits electrons.
- D Two electrons are created from a quantum of light.

(JAMB)

4 Which of the following statements is a correct consequence of the *Uncertainty Principle*?
- A The uncertainty in our knowledge of energy and the duration taken to measure it, are each less than Planck's constant.
- B The complete knowledge of the position of a particle implies the complete ignorance of its energy.
- C It is possible to measure exactly both the position and momentum of a particle at the same time.
- D Both momentum and energy of a particle can be known with absolute certainty.
- E A particle's kinetic energy cannot be measured accurately at any time.

(SSCE)

5 Explain what is meant by the *duality of matter*, illustrating your answer with observable phenomena.

(SSCE)

6 Explain briefly in general terms
(i) the *Uncertainty Principle*
(ii) *Wave-Particle Paradox.*

7 Mention three observable phenomena in which matter behaves like a wave and three other phenomena in which matter behaves like a particle.

8 (a) State Einstein's equation of photoelectric effect. What conservation principle does the equation represent?
(b) List three application of photocells.
(c) A photo emissive surface has a threshold frequency of $4.02 \times 10^{14} Hz$. Calculate the (i) threshold wavelength. (ii) work funtion. (iii) Kinetic energy of the emitted photoelectrons. ($c = 3.0 \times 10^{8} ms^{-1}$; $h = 6.63 \times 10^{-34} Js$).

(SSCE)

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BASIC ELECTRONICS 13

In this chapter we shall introduce some basic elements of electronics. We will describe semiconductors and semiconductor diode. We shall discuss briefly and qualitatively the *p*-type and *n*-type of semiconductors, the *p-n* junction diode and its current/voltage characteristics, and the use of a diode as a rectifier.

At the end of the chapter the student should be able to:

1 Distinguish between conductors, semiconductors and insulators in terms of their conductivity and modes of conduction.
2 Explain the two types of carriers of electricity in a semiconductor.
3 Distinguish between the *n*-type and *p*-type semiconductors.
4 Describe the *p-n* junction diode and its current/voltage characteristics.
5 Discuss the use of a diode as a rectifier.

13.1 Semiconductors

Substances which allow electricity to flow through them are called *conductors*. Examples of conductors are metals (such as copper, silver, iron, etc.) salts and inorganic acid solutions. Substances which do not allow electricity to flow are called *insulators*. Examples are plastic materials, glass, wood, air, rubber, etc. Insulators therefore have high electric resistances while conductors have low electric resistances. In other words, electric *conductivity* is high in conductors and low in insulators.

Certain materials have an electric resistance between the high value of insulators and the low value of conductors. In other words the electric conductivity of such materials is intermediate in value between those of conductors and insulators. Such materials are therefore called *semiconductors*. Examples of such semiconductors are silicon and germanium.

Semiconductors are materials which have an electric conductivity intermediate in value between that of a pure metal (good conductor) and that of good insulator

Semiconductors are widely used in the electronic industry, e.g., in computers and communications.

Semiconductor materials possess a crystalline structure, i.e. the atoms are arranged in an orderly manner. In both germanium and silicon each atom has four electrons orbiting in the outermost shell, and therefore said to have four valence electrons.

13.2 Modes of Conduction in Semiconductors — electrons and hole carriers

Another distinguishing characteristic between conductors and semiconductors is their mode of conduction. The carriers of electricity in a metal such as copper are only free electrons. Current is carried through an electrolyte by two carriers, the positive and negative ions. The current is carried through a semiconductor also by two types of carriers:

(i) *free electrons* which have negative charges and
(ii) *holes*, which have positive charges

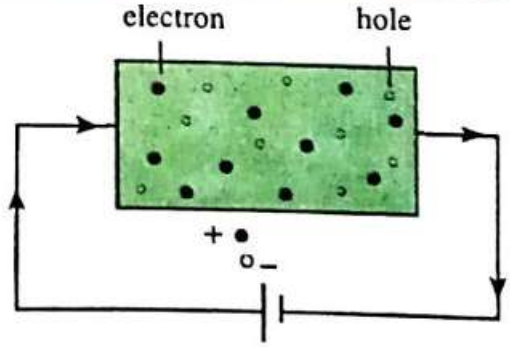


Fig. 13.1 Electron and Hole carriers in a semiconductor

How do free electrons and holes arise in a semiconductor? In an atom of a semiconductor like germanium or silicon, there are four *valence electrons* in the outermost shell. In its purest form at the lowest possible temperature (0 K), all the valence electrons are firmly bound to the nucleus of each atom. The crystal has no free electrons and is practically a non-conductor of electricity. At ordinary room temperatures, however, thermal energy of a valence electron may become greater than the energy binding it to its nucleus. Such an

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electron then becomes liberated from the bonds, and becomes 'free electron' by moving out of its position in the atom. This movement leaves a vacancy at the electron's former position. This vacant place (at A) is called a *hole*. It is equivalent to a positive charge, $+e$, where e is the charge on an electron.

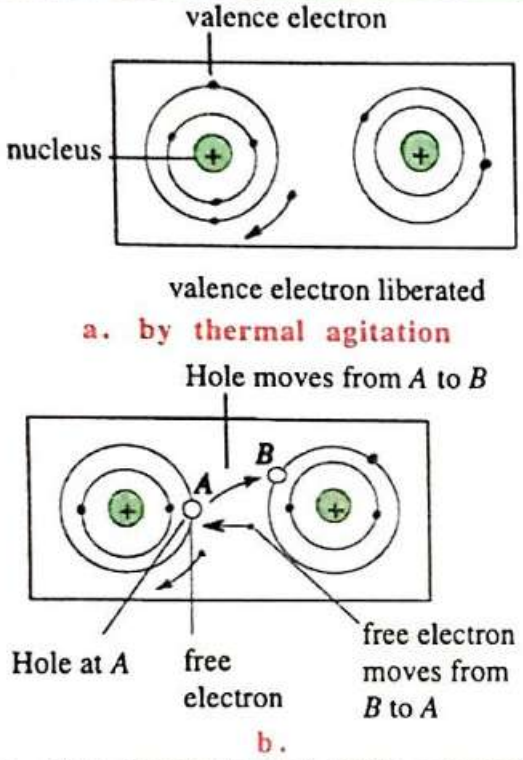


Fig. 13.2 Electron and Hole movement

An electron in a neighbouring atom is then attracted and moves to fill up this vacancy at A and in turn creates another vacancy or hole in its former position, B (fig. 13.2). Thus the hole appears to move from A to B. Thus while the free electron moves from B to A, the hole moves from A to B in the opposite direction.

If the semiconductor is not being subjected to an external electric field, the position of the hole moves at random from one point to another as the free electron moves from one point to another. When a battery is connected as in fig. 13.1, the valence electrons are forced to move in one particular direction opposite to the external field while the holes then drift in the direction of the field. Thus the current in a semiconductor is carried by the free electrons and holes.

In a pure semiconductor there are equal number of electrons and holes. It is the movement of valence electrons which creates electron-hole pairs in a semiconductor.

Effects of temperature on semiconductors

The resistance of a pure conductor (e.g. a pure metal) increased with rise in temperature. Thus increase in temperature decreases the conductivity of a good conductor. In contrast, the resistance of a semiconductor decreases with rise in temperature. Hence the conductivity of a semiconductor increases with temperature rise.

Thus we can distinguish between a pure metal and a semiconductor through the effect of temperature on their conductivities or electrical resistances.

13.3 Types of Semiconductors

There are two types of semiconductors – the *n-type* and the *p-type* semiconductors.

The *n*-type semiconductor is produced by introducing a tiny amount of a suitable foreign element or impurity to a pure semiconductor. For example when we add an impurity such as arsenic or antimony to germanium crystal we produce an *n*-type semiconductor. We are said to 'dope' the pure germanium crystal in the process to produce an impure semiconductor. The effect of this 'doping' is to alter the structure of the germanium crystal and produce more free electrons or negative charges in the semiconductor. Therefore there are more free electrons or negative charges in an *n*-type semiconductor than there are positive charges (*n* means negative).

When a battery is connected to an n-type semiconductor as shown in fig. 13.3a, the excess free electrons will move towards the positive terminal of the voltage source and cause a current to flow in the external circuit.

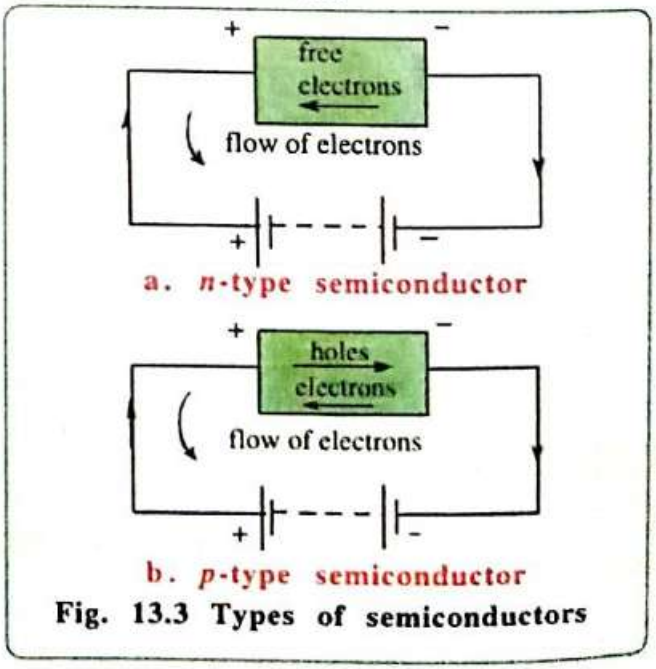


Fig. 13.3 Types of semiconductors

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In an *n*-type semiconductor, the majority carriers are electrons, and the *minority carriers* are the holes or positive charges.

The *p*-type semiconductor is produced by 'doping' pure germanium or silicon crystal with an impurity such as indium or boron that have three valence electrons. The result of such doping is to produce an impure semiconductor with more holes or positive charges than electrons. Such a semiconductor with more holes than electrons is called a *p*-type semiconductor. (*p* means positive). The holes behave like positively charged particles when a potential difference is applied to the crystal. They move towards the negative terminal of the voltage source. In a *p*-semiconductor, conduction is due mainly to positive charges or holes.

The holes are said to be the *majority carriers* and the electrons are the *minority carriers*.

The impurity atoms (such as arsenic) added to pure germanium or silicon to produce the *n*-semiconductor are called *donors* because they donate electrons as carriers. The impurity atoms such as boron or indium added to pure germanium or silicon crystal to produce the *p*-semiconductor are called *acceptors* because each accepts an electron when the atom is introduced to the crystal.

13.4 The *p-n* junction diode

A *p*-type semiconductor crystal joined to an *n*-type semiconductor crystal to act as a single semiconductor crystal is called a *p-n junction*. Such an alloy with a very thin junction between the two semiconductor types can be produced by a special industrial process. The *p-n* junction can function like a thermionic diode. It is therefore called a *p-n junction diode*.

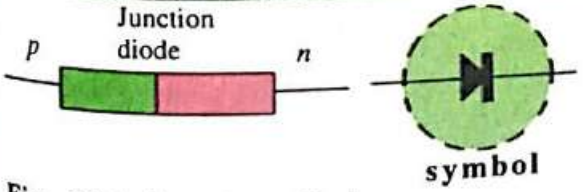


Fig. 13.4 Junction diode

If we apply a p.d. to a *p-n* junction as shown in fig. 13.5a with the positive terminal of the battery connected to the *p*-type and the negative terminal connected to the *n*-type semiconductor, the milliameter connected in series with the *p-n* junction diode shows a large flow of current. Thus the *p-n* junction diode in this arrangement has a fairly low resistance and conducts electricity well.

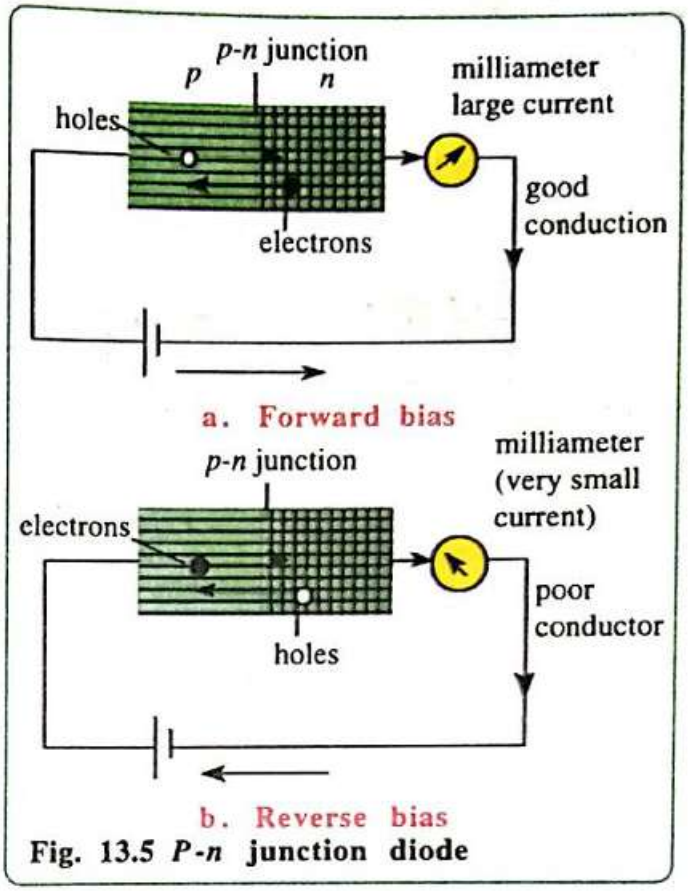


Fig. 13.5 *P-n* junction diode

Under this arrangement as shown in fig. 13.5a, the diode is said to be *forward-biased*. In this arrangement the positive terminal of the battery repels the numerous holes or positive charges across the junction and the negative terminal repels the electrons across the junction to constitute a current flow in the crystal and in the external circuit. When the applied p.d. is increased, the current increases.

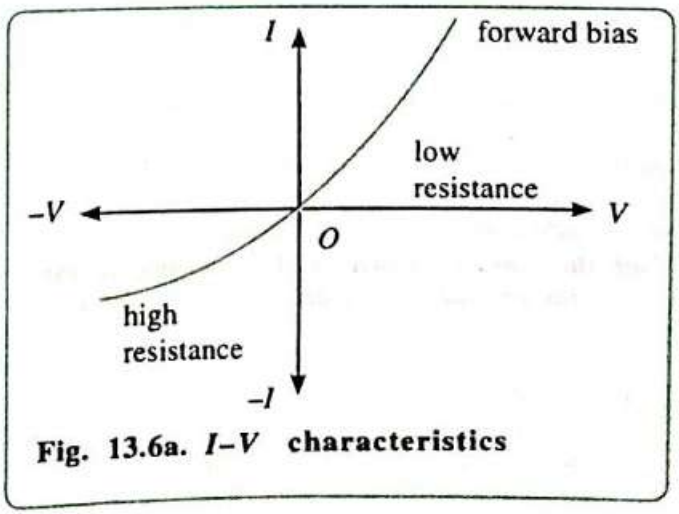


Fig. 13.6a. *I–V* characteristics

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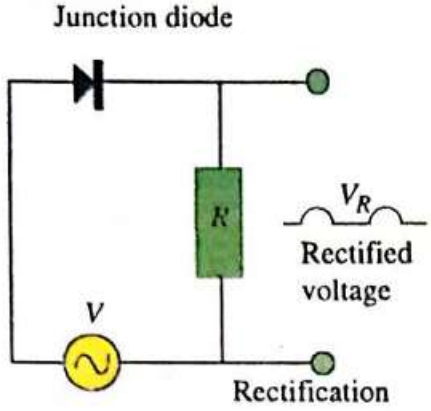


Fig. 13.6 Diode junction *I-V* characteristics and rectification

When the battery polarity is reversed so that the negative terminal is connected to the *p*-type and the positive terminal is connected to the *n*-type semiconductor, only a very small current flows in the external circuit. This is because only few electrons in the *p*-type semiconductor and only few holes in the *n*-type semiconductor are repelled across the junction by the negative and positive terminals respectively. The *p-n* junction diode arranged in this way is said to be *reverse-biased.*

We can thus see from the above that the *p-n* junction diode has a low resistance in one direction (forward-bias) and a high resistance in the reverse direction (reverse-bias). The *p-n* junction diode can hence be used as a *rectifier* and can therefore change a.c. to d.c. voltages.

The current versus voltage graph (*I-V* characteristic) of the junction diode is as shown in fig. 13.6. The graph does not follow the Ohm's law graph. The junction diode is therefore a 'non-ohmic' conductor just as the thermionic diode.

The *p-n* junction diode has several advantages over the thermionic diode. For example

(i) it is much smaller,
(ii) it needs only a small p.d.to operate in a radio receiver,
(iii) it requires no time to warm up to produce the current carriers since these are already available in the semiconductor,
(iv) it is cheaper to manufacture in large numbers and less liable to break.

Summary

1 A semiconductor is a material which has an electrical conductivity intermediate in value between that of a pure metal and that of a good insulator. Examples are silicon and germanium.

A *Hole* is the space that is left in the atomic arrangement of a crystal when an electron is removed. This space is mobile and behaves as if it carries positive charge.

2 Current is carried through a semiconductor by electrons and holes. Hole movement is equivalent to movement of a positive charge numerically equal to the charge on an electron.

3 In a pure semiconductor at room temperature there are an equal number of electrons and holes.

4 The conductivity of a semiconductor increases with temperature. That is, the resistance of a semiconductor decreases with rise in temperature.

5 There are two types of semiconductors – the *p*-type and the *n*-type. The *p*-type semiconductor has more holes than electron. Thus holes are the majority carriers of current. The *n*-type semiconductor has more electrons than holes. Thus electrons are the majority carriers.

6 A *p-n* junction diode consists of a p-type and n-type semiconductors in contact. When in forward bias (i.e. the p-semiconductor is connected to the positive terminal of a battery and the n-semiconductor is connected to the negative terminal of a battery), it has a low resistance to current flow. When in reverse bias (i.e. the *p*-semiconductor is connected to the negative terminal of the battery and the *n*-semiconductor is connected to the positive terminal), it has a high resistance to current flow.

7 The *p-n* junction diode acts as a rectifier like the thermionic diode. Its *I-V* characteristic does not obey Ohm's law.

Exercise 13

1 Which of the following are semiconductors?
A Copper B Silicon
C Germanium D Plastic E Iron

2 Which of the following statements is/are not correct?
A The carriers of electricity in a metal are electrons
B The carriers of electricity in an electrolyte are positive and negative ions.
C The carriers of electricity in a semiconductor are electrons and ions.
D The carriers of electricity in a semiconductor are electrons and holes.

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E The resistance of a semiconductor is more than that of an insulator at the same temperature.

3 Which of the following statements is/are not correct?

A The resistance of a pure conductor increases with temperature.

B The resistance of a semiconductor increases with rise in temperature.

C The resistance of a semiconductor decreases with rise in temperature.

D The conductivity of a semiconductor increases with rise in temperature.

E The conductivity of a semiconductor is not affected by temperature.

4 Explain what is meant by a *semiconductor* and state how an *n*-type semiconductor is produced.

Using a circuit diagram, explain what is meant by *forward-bias* as applied to a *p-n* junction diode.

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5 Explain what is meant by (i) an *n*-semiconductor (ii) a *p*-semiconductor (iii) a *p-n* junction

Explain in terms of electrons and holes how current is conducted in (a) a *p*-type semiconductor (b) an *n*-type semiconductor.

6 Describe what happens when a *p-n* junction diode is (i) forward-biased (ii) reverse-biased.
Draw the *I-V* characteristic of a junction diode.

7 (i) Describe briefly how a *p*-type semiconductor is produced.

(ii) State three advantages and one disadvantage of a junction diode over a thermionic valve.

(iii) Sketch a graph to show how the current through a junction diode varies with the potential difference across it.

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14 ROCKETS AND SATELLITES

14.1 Introduction

Rockets and satellites are parts of the very important and exciting technological advances of the modern world. Both instruments are mostly often linked to one another as the rockets are always used to lift or transport the satellites to space.

The Chinese are believed to have invented the first rockets which they used as devastating weapons of war. They were described as "arrows of flying fire" by historians. They later found popular use in firework displays. They have many other uses which we shall discuss in later sections.

In this chapter, we shall explore these two spectacular instruments. We will discuss the components parts of each instruments, their functions and uses.

At the end of the chapter the student should be able to:

1. Identity the components parts of rockets and satellites.
2. Describe the functions of rockets and satellites.
3. State the uses of rockets and satellites.

14.2 Component Parts of Rockets

A rocket is an engine or intrument that produces a great power, more power for its size than any other kind of engine.

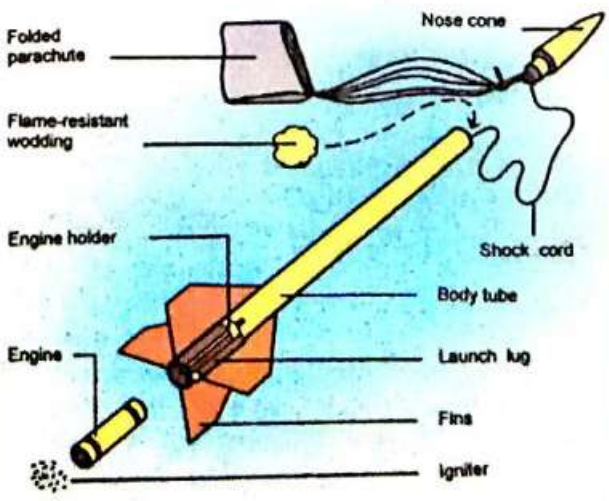


Fig. 14.1a Basic parts of a model rocket

A model rocket consists of seven basic component:

(a) The body tube
(b) The launch lug
(c) The fins
(d) The engine holder
(e) The rocket engine
(f) The nose cone
(g) The recovery device
(h) A lunch system or pad to get the rocket into the air

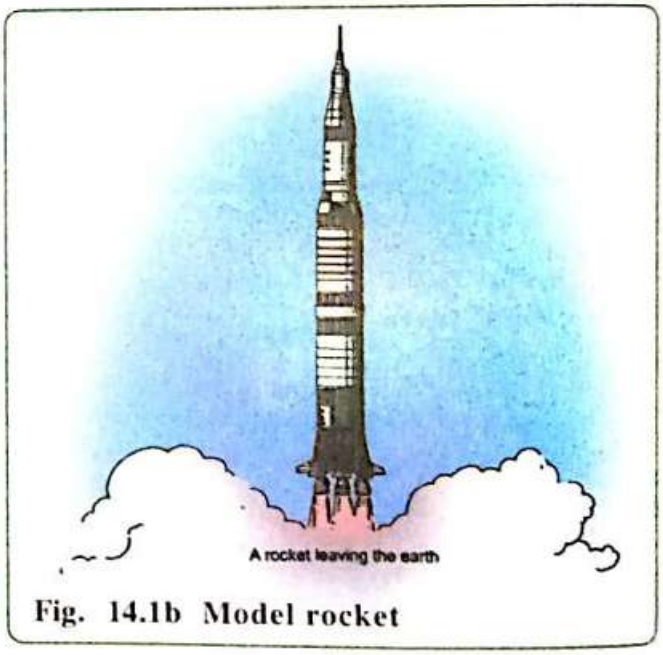


Fig. 14.1b Model rocket

Fig. 14.1 is a sketch diagram of a model rocket. It is made up of:

(a) *The body tube:* this is a hallow cylinder to which all the other parts are attached.

(b) *The launch lug:* this is a narrow tube fastened to the side of the body tube. The launch lug guides the rocket and keeps it vertical during lift off.

(c) *The fins;* these are attached to the bottom part of the body tube just as fins are attached to a fish. They help the rockets to travel straight during flight.

(d) *The engine holder* or engine mount is a ring cemented inside the bottom of the body tube of the rocket. It is constructed in such a way that the rocket engine fits into the engine holder.

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(e) *The rocket engine* is usually in the form of a tube and contains the solid fuel or propellant.

(f) *The nose cone* forms the top of a model rocket. Its tip has a rounded point that reduces the air resistance.

(g) *The recovery device* is wired to return the rocket slowly to the ground after it has reached its highest point or maximum altitude called apogee. One such device is a parachute behind the nose cone.

(h) *The launch system*. This consists of a launch pad and an engine ignition system with a battery.

Component parts of a Multistage Rocket

A multistage rocket consists of two or more sections called stages. Each stage has a rocket engine and propellant. The entire rocket itself consists mainly of rocket engine and nose-cone.

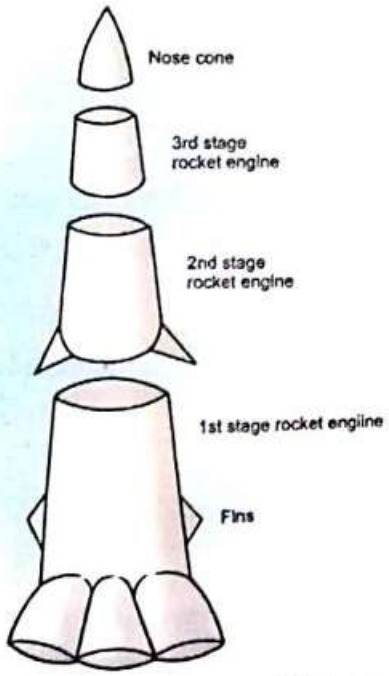


Fig. 14.2a Major parts of a multistage rocket

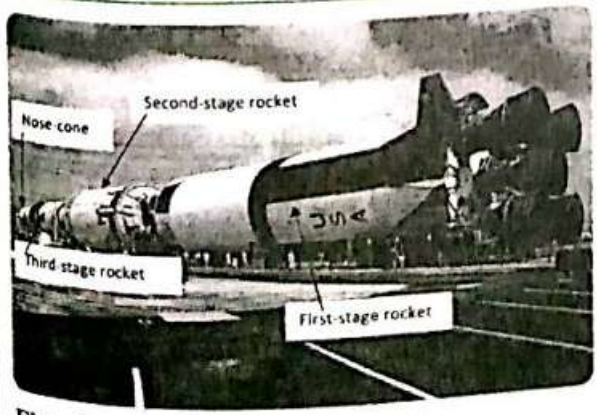


Fig. 14.2b Multistage rocket

A rocket travelling to space requires more than one engine. The reason is because it will require much large amount of fuel or propellant to gain the speed required to escape the Earth's gravity; and this is not economically favourable. A multistage rocket can reach higher speed because it lightens its weight by dropping Stages as it used up fuel or propellant. A system of two or more rocket engines called a multistage rocket is therefore employed. (see fig.14.2)

The first rocket engine called "first stage rocket" or booster, launches the rocket. After the propellant of this stage is used up, burnt or exhausted, the vehicle drops that section and uses the second stage, and the process repeats itself, after which the third stage rocket comes into operation.

Another component part of the rocket is the *Nose Cone*. This is located at the pointed end of the entire structure as shown in fig 14.2. It is in the Nose Cone that the astronauts and pay load stay during space travel. Pay load is the equipment carried by a spacecraft or rocket. It can be satellite or other research equipment.

14.3 Launching a Multistage Rocket

The rocket is launched or sent off into space from a launch pad located at a special site called *launch sites*.

The rockets are prepared for launching in a step by step process called the countdown. Each step is scheduled for a specific time during the countdown and the rocket is launched when the countdown reaches "zero". A hold may however occur at any stage if the weather conditions are not conducive for launching the rocket.

The first stage fires to push the rocket to a considerable height as illustrated in fig. 14.3. After which the engine is disconnected and let down by means of parachutes as illustrated in fig. 14.3b or allowed to fall into the ocean at predetermined position and is recovered for reuse. This makes the rocket lighter. The second stage starts firing when the stage is disconnected (or jettisoned). This increases the speed of the rocket. When the fuel in the second stage finishes, it is jettisoned and may not be recovered like the first stage as illustrated in fig. 14.3c. This is because, by now the rocket must have been very close to target orbit. At this point, the third stage fires the rocket into the target orbit. This is illustrated in fig. 14.3d.

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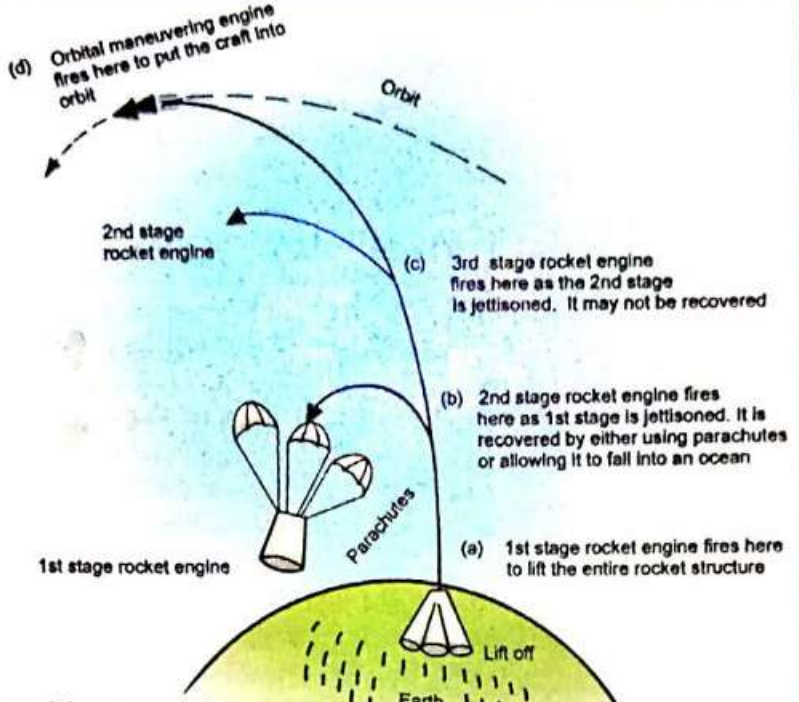


Fig. 14.3 Launching a multistage rocket

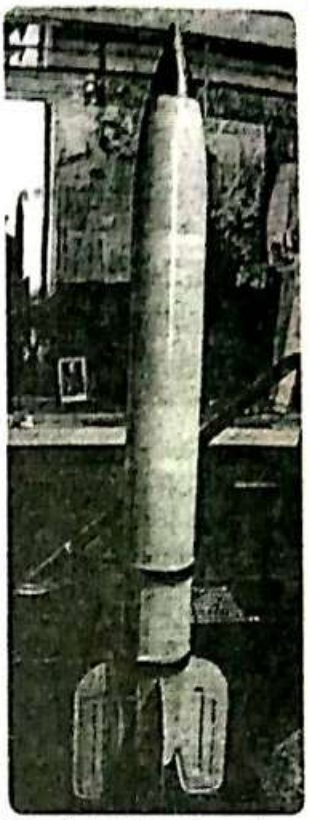


Fig. 14.4(b) A rocket-driven bomb

14.4 Functions of Rockets

Some of the functions of rockets are as follow:

1. Rocket carries astronauts to space to study the environment and the Moon.
2. Rocket carries robotic probes (i.e. specialized equipment) to study other planets.
3. Rocket carries artificial satellites such as weather satellites, Earth satellites, communication satellites, scientific and astronomical satellites.
4. With rockets, portable laboratories can be carried to space to carry out some scientific researches which are impossible on earth's surface.
5. It is hoped that in future, man will have colonies or cities in space; and rocket powered vehicles will be used to shuttle between these cities.
6. With appropriate design, rocket engine can serve as a vehicle for carrying a bomb to explode in an enemy's territory. Such a rocket-bomb structure moves in the form of a missile as shown in fig. 14.4.



Fig. 14.4(a) A missile fired from a warship it uses rocket engine similar to those used by spacecrafts

14.5 Uses of Rockets

i. *High speed transportation:* Rockets are chiefly used for providing high-speed transportation both within the earth's atmosphere and in space.

ii. *Military use:* Rocket are a very destructive military weapon. They were first used by the Chinese in this fashion.
The bazooka is a small rocket launcher carried by soldiers for use against armoured vehicles. They are used as *guided missiles* by the Navy against enemy aircrafts. They are used to propel a type of long-range missiles called an *intercontinental ballistic missile*. Such missiles travel for thousand of kilometres to hit enemy target with nuclear explosives and bombs. (fig. 14.4)

iii. *Scientific Research:* Rockets are used by scientist to explore the earth's atmosphere. The rockets carry meteorological equipment such as barometers, cameras, thermometer, etc. up to the atmosphere and up there, they are used to measure weather conditions and make weather forecasts. These weather equipment are carried in weather satellite and transported into the atmosphere by a rocket.
Some astronomical telescopes like *Hubble Space telescope* were launched by rockets. With this type of telescope, astronomers can see distant

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stars and galaxies to study them and propound theories.

iv. Some experiments which are not possible on the earth due to gravity and inadequate vacuum are now possible in space. This is because rockets can carry laboratories to space where there is no gravity and appropriate vacuum can be created. Rockets lift artificial satellites into orbit around the earth. These orbiting satellites gather information for scientific research and relay them to earth stations. Others relay telephone conversations, radio and television broadcasts across the oceans and lands.

The Rocket has made it possible for Information and Communication Technology (ICT) to make the world a **'global village'**. In a village when anything happens everybody hears about it quickly or the same time; ICT has made the world like that. Events in any part of the world are instantly conveyed to all over the world and can impact the rest of the world. Without rockets, communication satellites that make this possible would not have been there.

v. *Space travel – Carrier Rockets:*
Rockets provide the power to launch space craft into orbits around the earth and also to the moon and other planets. Such rockets are called *carrier rockets*.

vi. *Distress signals*
Rockets are used as distress signals from ships and air plane and from the ground.

vii. *Fireworks*
Rockets are used in fireworks displays.

14.6 Rocket Propulsion

The force which moves a rocket comes from burning fuels. When the rocket fuel or propellant (e.g. liquid hydrogen) enters the engine, it mixes with liquid oxygen in the right proportion and burns. During burning, ensuing gases expand and are forced out through the openings at the rear of the rocket. When this happens according to Newton's 3rd law of motion (Action and Reaction are equal and opposite), the rocket is pushed in the opposite direction as shown below fig. 14.5.

To illustrate the above action in the laboratory, inflate a balloon, use a pin to open up a small hole on one portion of the balloon. When you let the balloon go, you will notice that as air rushes out of the balloon through the pin-hole, the balloon is thrust forward in the reverse direction as shown in fig. 14.6. (see also Book 2, chapter 6 on Jet and Rocket propulsion)

We can summarise by saying that rockets are used for scientific and atmospheric research, for military use, for space travel, for launching probes and satellite.

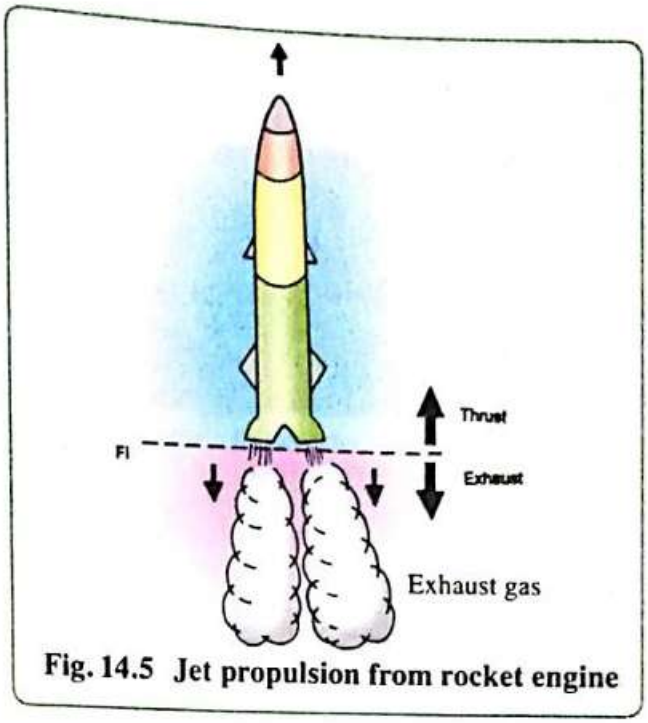


Fig. 14.5 Jet propulsion from rocket engine

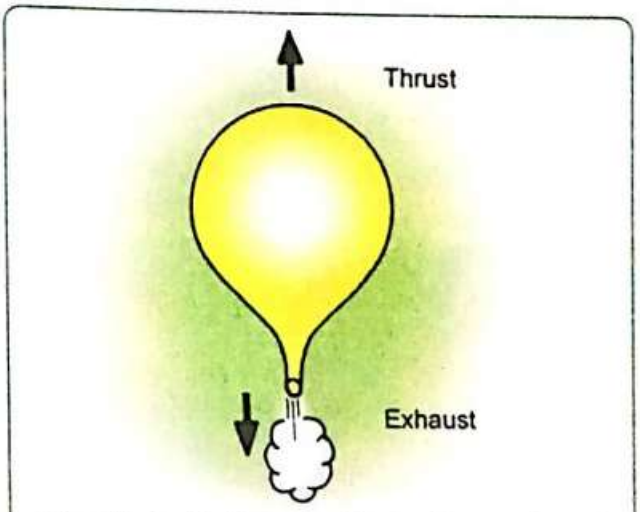


Fig. 14.6 Balloon is thrug forward as it lets air out. This is in accordance with the Newton's 3rd law of motion

14.7 Satellites

Satellites have been fully discussed in chapter 2 of this book 3 and the student is advised to refer to that chapter to refresh his/her memory.

Meaning and types of satellites

Generally, a satellite is any object that orbits or moves round a planet. The two types of satellites are Natural satellite and artificial satellite. Natural satellite refers to the Moon. Artificial simply means man-made satellite. Some of them are shown in fig. 14.7. There are many artificial satellites in the orbits; some of them are weather satellite, earth's satellite, communication satellites, scientific and

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Astronomical satellites. All these satellites are very important to man.

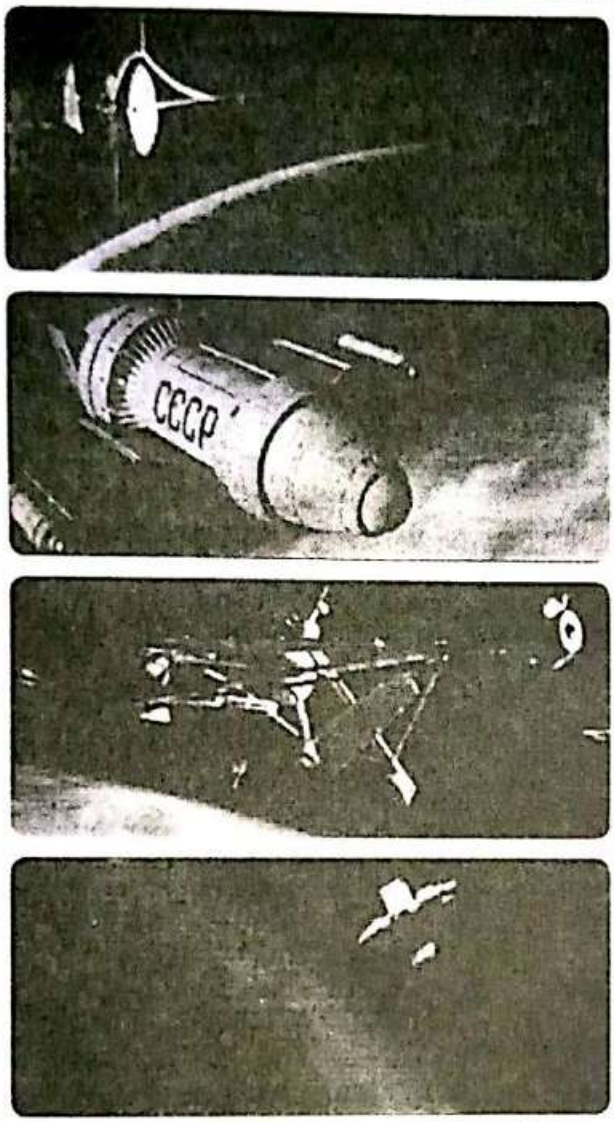


Fig. 14.7 Different artificial satellite orbiting the Earth in space; in each case the Earth is seen in the background

14.8 Component Parts of a Satellite

We can summarize the basic parts of a satellite and their functions.

i. **Power**: Electricity is needed for the instruments and to enable the spacecraft directional control. In most cases, power is provided by solar arrays. Sometimes, in cases like the recently launched, Saturn-bound satellite, called **Cassini**, solar arrays do not supply enough power for planetary probes; instead, nuclear generators are used.

ii. **Communication**: The satellite has to communicate with ground control so directions and commands can be sent up and down. To provide radio contract, satellites have radio antennae.

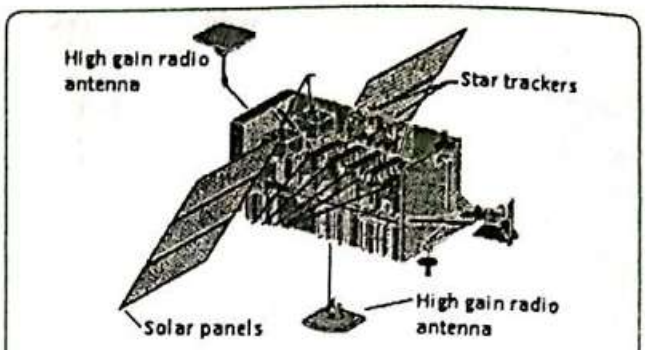


Fig. 14.8 Main component parts of a satellite

iii. **Guidance**: Star trackers and gyroscopes help satellites align themselves in the right direction. This is common with astronomical satellites.

iv. **Scientific Instruments**: Satellites have scientific instruments on them. Such scientific instruments are cameras with diverse functions (as in the Earth observation satellites), infrared sensors (as in weather satellite), radio antennae (as in communication satellites), etc.

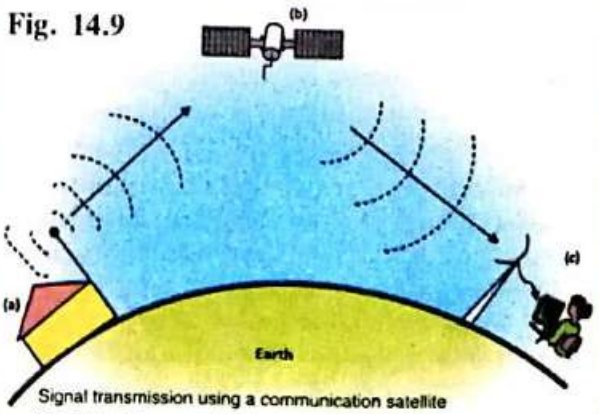
14.9 Uses of Satellite

Satellites have found many uses in the following areas:

i. Communications

Artificial Satellites called Communication Satellites are used to relay or transfer telephone messages, radio and television signals across the whole world. Such world events like the Olympic games, World Cup Football matches are transmitted simultaneously from the field to all over the world via satellite television. People now stay in their homes and watch World Cup football matches. The whole world watched the funeral ceremony of the late pop star, Micheal Jackson. A 'satellite dish' connected to your television makes these possible.

Fig. 14.9



Signal transmission using a communication satellite

(a) Transmitting ground station transmits signal

(b) Communication satellite receives the signal from the transmitting station and boasts/amplifies it and beams the signal downwards.

(c) Someone whose computer system is connected to server accesses the signal and downloads the information.

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Fig. 14.10 Photographic picture taken by a satellite in space; inside the photograph is a satellite dish, snow covering the ground and tracks made by moving vehicles.

ii. Photography

Satellite has made it possible to take pictures of places on the Earth. Figure 14.10 is a picture taken by a satellite in space. For instance, pictures of an accident scene, like the scene of an aeroplane crash can be taken by an appropriate satellite in space. A piece of land can be watched or monitored for possible invasion by enemies, terrorists, etc, without actually sending security agents to the place.

iii. Mapping

Satellite has made it possible to map pieces of land in order to check erosion menace, potholes on the main roads, topography, boundaries, etc. This helps the Government to determine what to use a particular piece of land for.

iv. Geographic Information System (GIS)

Satellite has made it possible that some information concerning the geography of an area can be obtained.

v. Geographical Position System (GPS)

This is generally used in navigating aeroplanes, ships, etc. Satellite has made it possible that one cannot get lost, say in a city. This is because, through GPS, he can easily download and see all the roads and major buildings in that city on his GSM handset; and hence, find his way. Also through (GPS), properties like automobiles may be difficult to be stolen. This is because, with appropriate registration, the vehicles can easily be located.

vi. Weather Forecast

The weather forecasts we receive every time are made possible by appropriate satellites in the space. Cloud maps used on TV weather forecasts are created from images relayed by satellite about 36000km above the earth.

vii. Astronomical Satellites

An astronomical satellite, like Hubble Space Telescope (HST) in space is used by scientists to look deep in space and view distant galaxies and make their theories concerning the universe.

viii. Defence Industry

In this modern world, some weapons of war can be carried into space by satellites. Such weapons can be fired into the space from the ground station; this is illustrated in figure 14.11.

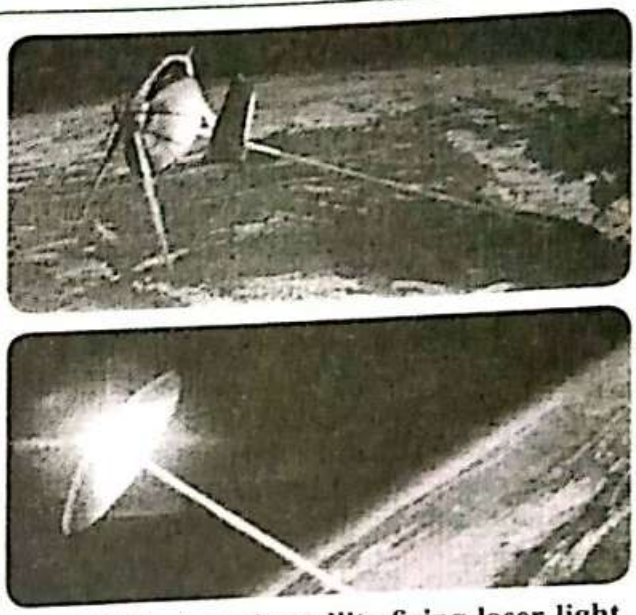


Fig. 14.11 Armed satellite firing laser light to enemy camp on the Earth

In this figure, the two pictures show two armed satellites firing laser light to enemy camps or territories on the Earth. This type of light sets anything it touches on fire; therefore, can be uses as a weapon during war.

Exercise 14

1. With the aid of a diagram, describe the component parts of a model rocket.
2. Sketch and identify the component parts of a multi-stage rocket.
3. State 4 uses of rockets.
4. Discuss the principle of rocket propulsion.
5. Distinguish between natural and artificial satellites. Give examples.
6. List the major component parts of a satellite.
7. a. Briefly describe 5 uses of satellites

   b. Which of the above uses benefit you most and how?

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15 NIGERIAN SATELLITE

15.1 Introduction

In April, 1999. The Government of Nigeria established the National Space Research and Development Agency (NASRDA) with a mandate to organise all space science and technology-related activities in order to make a greater impact on development effort in Nigeria. Subsequently, in June, 2001 the National Space Policy and programmes was approved, thereby creating an enabling environment for the achievement of the Nigeria space agenda. In pursuance of the objectives of this agenda, NASRDA embarked on a technical initiative by building knowhow amongst indigenous Nigerian engineers and scientists in space technology, which has crystallized in building and launch of the first Nigerian satellite called NigeriaSat-1.

Following this and to enhance Nigerian compliance with Information and Communication Technology (ICT), NigCom-Sat-1 was launched. NigCom-Sat-1 is Nigerian second satellite but first communication satellite and was launched on May 13, 2007.

In this chapter, we shall discuss features of NigeriaSat-1, operation, uses of NigeriaSat-1, features of NigComSat-1, its operation and uses.

At the end of the chapter the student should be able to:

1. State the features of NigeriaSat-1.
2. Describe the operation of NigeriaSat-1.
3. State the uses of NigeriaSat-1.
4. State the features of NigComSat-1.
5. Describe the operation of NigComSat-1.
6. State the uses of NigComSat-1.

15.2 NigeriaSat-1

NigeriaSat-1 is the Nigerian contribution to the international Disaster Monitoring Constellation (DMC) project. It was launched on 27 September 2003. It is shown in Fig. 9.1

The Disaster Monitoring Constellation (DMC) is an international co-operation in space, bringing together organizations from seven countries. The countries are Algeria, China, Nigeria, Thailand, Turkey, the United Kingdom, and Vietnam. The DMC Consortium (or group) is forming the first-ever mirosatellite constellation bringing remarkable Earth observation capabilities both nationally to the individual satellite owners and internationally to benefit world-wide humanitarian aid effort.

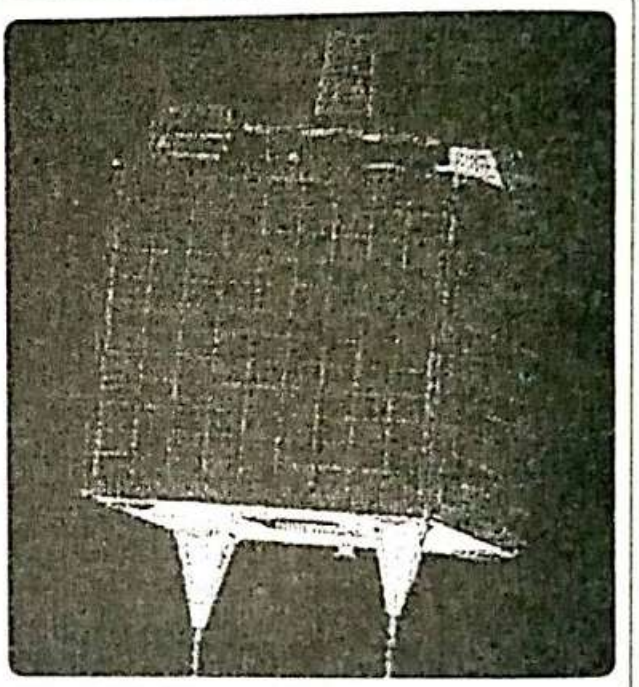


Fig. 15.1 Nigeria Sat-1

15.3 General Features of NigeriaSat-1

1. It is a product of advanced technology in micro-satellite, operating in low Earth orbit.
2. It weighs 100kg.
3. It is in constellation (company) with six other satellites; and this gives it added advantage of global coverage and daily revisit with provision of real-time (as it happens) data.
4. It provides opportunity to develop appropriate payloads (or facilities) to solve local problems.
5. It provides opportunity for capacity building in satellite technology – some Nigerian Engineers were trained in the process
6. It provides opportunity for bridging the digital divide within and across Africa.
7. Ground control station is located in Asokoro Abuja for the control and operation of the satellite.

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15.4 Operation of NigeriaSat-1

NigeriaSat-1 is for Earth's Observation. Earth's Observation here means that the satellite will take photos or images of Nigerian piece of land on which it is focused and beam them back to the ground station for use in various areas of socio-economic development.

15.5 Uses of NigeriaSat-1

The information obtained from NigeriaSat-1 can be used by Nigerians to take care of some needs.

Some of such needs are as follows:

a. Disaster Management

Some natural disasters that need control are:

i. Flooding: This includes lake and river flooding.

ii. Erosion: This includes coastal erosion, wind erosion and gully erosion.

iii. Water Pollution: This includes marine (i.e. sea) pollution and fresh water (i.e. rivers and streams) pollution.

iv. Desertification: This is the process by which an area of land turns into a desert. Desert is a large area of land that has very little water and very few plants growing on it.

v. Forest Fire or any Major fire Outbreak: This can occur during dry seasons; and can be man-made or natural.

vi. Documents the following:

   a. Pre, on-time and post disastrous events such as those already mentioned with possibility for early warning, planning, evacuation, refugee displacement location and relief management.

   b. Cultural habits of the affected people and human activities in the immediate and distant environment.

   c. Associated natural phenomenon.

   d. Total potential area of spread and damages.

b. Agriculture

This includes:

i. Investigating and monitoring pest infestation.

ii. Crop performance, yield forecasting.

iii. Planning afforestation (planting of trees) programmes.

iv. Mapping land use planning and management e.g. flood plains suitable for rice cultivation.

c. Water Resources Development and Management

This includes

i. Location of dams for hydroelectric power generation.

ii. Planning water impoundment for various uses such as irrigation, domestic, industrial, navigation, fishery, recreation, etc.

iii. Surface hydrology and watershed management including analysis of major drainage basis.

iv. Structural analysis for aquifers springs and bore.

v. Preparation of water resources master plan.

d. Solid Mineral Exploration/ Exploitation

This includes

i. General geological mapping and update/ revision.

ii. Planning solid mineral exploitation including mine reclamation.

iii. Structural, analysis and regional geological reconstruction and mineral exploration.

iv. Mapping/differentiation of host mineralized zones and rocks in oil/gas and solid mineral exploration.

v. Preparation of geomorphologic maps.

e. Ecosystems Management with Associated Goods and Services

This includes

i. Mapping/differentiation of friendly/economic floral or plant species and fauna or wild life habitat (e.g. forest and game reserves).

ii. Forest and rangeland monitoring.

iii. Pollution of marine ecosystem due to effluents from domestic and industrial waste.

iv. Planning protection and management of lakes, reefs and mangrove ecosystems.

v. Mapping and monitoring wild aquatic plants, e.g. water hyacinth.

vi. Local and regional planning for tourism and tourism potentials, as well as investigation of historical/archaeological sites.

f. Demography

This includes

i. Mapping/planning population survey and census enumeration areas.

ii. Rural and urban growth mapping and monitoring.

iii. Monitoring land/use/cover dynamics.

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g. Transportation and Utilities

i. Terrain evaluation for various uses and regional planning.

ii. Route location and road allocation planning with road maintenance strategy.

iii. Planning economic railway network development.

iv. Mapping and planning utilities location: oil pipeline, PHCN transmission line, NITEL cables and telecommunication network.

v. Planning waterways and general land and water transportation network.

h. Environmental Management

This includes

i. Mapping/planning sewage location, and domestic and industrial waste disposal sites.

ii. Mapping/monitoring soil and land degradation.

iii. In the determination of environmental impact assessment.

iv. Risk zone mapping and environmental inventory and monitoring.

i. Defense and Security

This includes

i. Mapping of state and international boundaries.

ii. In the mapping/planning terrain traffic ability for movement of ground troops and military wares.

iii. Provision of maps/geoinformation for vehicle tracking.

iv. Mapping/determination of camouflages.

v. Mapping of the coastal areas for surveillance purposes.

j. Health and Public Health Delivery

In malaria epidemiology, it will enhance study of the relationship between malaria vectors and environmental variables (larvae habitats) e.g.

i. Flooding pastures and transitional swamps and other breeding sites and plants.

ii. Combine with weather data for delineating 'harmattan' dust area with associated diseases.

iii. In remote sensing and Geographical Information System (GIS) for meningitis early warning.

iv. Prevention and isolation of river blindness environment.

k. Education and Capacity Building

i. Facilitate/enhance remote sensing education or capacity building for the use of Earth observation data from the satellite.

ii. Availability of readily accessible data for research and development.

iii. Instructional/teaching aid for secondary institutions.

15.6 NigComSat-1

Operation of NigComSat-1

NigComSat-1 is a Nigerian communication satellite. It was launched on 13 May 2007. Its purpose is for communication. This goes a long way to boost Information and Communication Technology (ICT) in Nigeria.

Uses of NigComSat-1

The satellite provides the following services and applications under communication as follows:

a. Telepresence

Telepresence means to be present by electronic means in order to render services. Telepresence includes tele-education (i.e. distance learning), tele-medicine (i.e. distance curing of illness and disease), e-Goverrnment (i.e. electronic Government) and e-commerce (electronic commerce). In all these the resource persons are not bodily present; instead, they render their services by electronic means.

b. Broadcasting

This involves mainly television broadcast.

c. Telecommunication

This involve the use of telephones, mobile phones, etc. This will also go a long way to reduce the cost of making phone calls within and outside the country thus bringing the much needed growth of the industrial and commercial sectors of the economy.

d. Internet and Multimedia

This involves communication and exchange of information through the internet.

Exercise 15

1. State the features of NigeriaSat-1.
2. Describe the operation of NigeriaSat-1.
3. State the uses of NigeriaSat-1.
4. State the features of NigComSat-1.
5. Describe the operation of NigComSat-1.
6. State the uses of NigComSat-1.

In questions 7-10 tick the appropriate answer.

7. One of the following options is the purpose of NigeriaSat-1
   - A defence
   - B information and communication
   - C Earth's observation
   - D weather forecast

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8. NigeriaSat-1 was launched on
   A 27 September 2003
   B 30 December 2002
   C 2 February 2005
   D 3 March 2003
9. NigeriaSat-1 can help Nigerian government solve one of the following needs:
   A build industries
   B Develop space technology
   C disaster management
   D to be ICT compliant
10. One of the following options summarizes the function of NigComSat-1:
   A Earth's observation
   B communication
   C disaster monitoring
   D electric power generation

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16 ENERGY AND SOCIETY

In Books 1 and 2 of this book, we discussed the concept of **Energy** as the ability to do work. We also discussed the various types and forms of energy. How energy can be transformed from one form to another, together with the law governing the conservation of energy were fully discussed with the various numerical examples to illustrate the concept of energy as understood by the physicist or natural scientist.

In the human society, the term **energy** is used as a synonym of **energy resources** and most often refers to substances like fuel, petroleum products and electricity in general.

In this chapter, we shall discuss energy as it is understood by the human society and how it interacts and affects the society.

At the end of the chapter, the student should be able to:

1. Identity the sources of energy.
2. Distinguish between renewable and non-renewable sources of energy.
3. Identify the various ways energy is used.
4. Explain the importance of energy in the development of society.
5. Explain the impact/effect of energy use on the environment.
6. Identify energy sources that are environmentally friendly and those that are hazardous to the environment.

16.1 Sources of Energy

There are many energy resources now available to the society. These are

1. Petroleum and Natural gas (Fossil Fuel)
2. Nuclear energy
3. Solar energy
4. Wind energy
5. Hydroelectric energy
6. Geothermal energy
7. Energy from coal and wood
8. Tidal and Ocean energy
9. Biomass energy

The world is still researching and developing other sources of energy as some of the above are depleting and will need to be replaced by other sources.

i. **Energy from Fossil fuels: Petroleum and Natural Gas (Hydrocarbons), Coals**
Petroleum occurs naturally in two forms – **liquid** and **gas**. Liquid petroleum as it occurs naturally in a reservoir or oil well is called **Crude oil**, and gaseous petroleum is often referred to as **Natural Gas**. Petroleum is a mixture of organic substances called **Hydrocarbons**. It is a valuable substance used as fuel and as raw material for the production of other products called **petrochemicals**.
Liquid petroleum or crude oil is further classified into **Light Crude** which means the crude oil that can flow easily and **Heavy Crude**, that implies the type of crude oil that does not flow easily. **Bonny Light** is a type of light crude oil produced in Nigeria.
Crude oil is refined in refineries to produce petrol, diesel, kerosene, lubricating oil, etc.
These refined products are used for every mode of transportation, for powering industrial machinery, electricity generation, and for various domestic uses for cooking, lighting of lamps, and heating and cooling of houses and offices.
About two-third of the world energy is provided by petroleum. It is such an important energy resource that it has caused wars between countries and civil unrests within countries.
Nigeria is a petroleum producing country whose crude oil is highly valued by the world. It is the country's main export commodity and the source of most of her wealth.

ii. **Nuclear Energy** – see chapter 10 of this book for discussion on energy released from nuclear reactions.
The use of nuclear power is for reducing world fuel shortages. It has the best prospect of providing the world enough of its energy needs, but fear of nuclear accidents always haunts the society for it can be very catastrophic, e.g. the 1986 Chernobyl nuclear disaster. Nuclear energy is environmentally friendly as it does not pollute the atmosphere.

iii. **Solar Energy** is the energy available from the Sun. See Chapter 12 of Book 1 for discussion on solar energy resource. Solar energy is a very useful form of clean energy. In sunny areas of

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the world, solar energy is collected by Solar panel and concentrated on water to produce steam to drive turbine and for domestic heating purposes. It can also be converted to electricity for lighting our home and streets. Solar energy is the energy available from the sun. The sun is the ultimate energy resources available to man. Its energy does not deplete, it cannot be used up. Solar energy is used by green plants in the process of photosynthesis. Solar energy can be converted to electrical energy using photocells. Solar radiation can be converted to chemical energy by photochemical process. Solar energy can be used for heating homes and boiling water.

v. **Wind Energy** results from uneven heating of the earth's surface. It is driven by the thermal energy of the Sun and can be used by sailing ships and for turning windmills. Wind energy resource is best generated along coastlines as well as on hills. Wind power can be used to produce large-scale electricity that could be tied to the national electric networks. It is a clean form of energy.
Hydroelectric energy or water energy provides electricity from dams, hydroelectric plant etc. (see Dams and Hydroelectricity Book 1 chapter 7)

vi. **Geothermal energy** is the enormous energy or heat within the earth itself. People living close to the thermal springs or volcanic fissures have the heat of the earth readily available for energy generation to heat their homes or to generate electricity. Geothermal energy is generated wherever water comes in contact with hot rocks below the earth's surface. The rocks give out heat that makes the water hot enough to turn into steam.

vii. **Energy from coal and wood** is released when they are heated. The released energy is used in powering electric plants and driving some locomotives, and for domestic uses. Wood is still a major energy resource for cooking in Nigeria. Exploitation of wood energy leads to deforestation. The use of coal and wood is not environmentally friendly as it pollutes the atmosphere. Coal was the source of electric power station in Oji River in Enugu state of Nigeria with installed capacity of over 30kw. The power station has since gone out of service.

viii. **Tidal and ocean energy** is produced from tides and oceans. Tides are due to the gravitational pull of the moon on water surrounding the earth. Tidal power is environmentally friendly.

ix. **Biomass energy** is the energy released from the degradation or fermentation of such organic materials like cow dung, poultry waste, rice husk, weeds, human and animal excrements, garbage, cornstalks, etc.
To date only hydroelectric power, and nuclear power have been significant alternatives to fossil fuels (Petroleum and Natural gas)

16.2 Renewable and Non-Renewable Energy

Renewable Energy Resources are those that can be replaced as they are used up. They can be continually replenished as they are exploited and utilized.

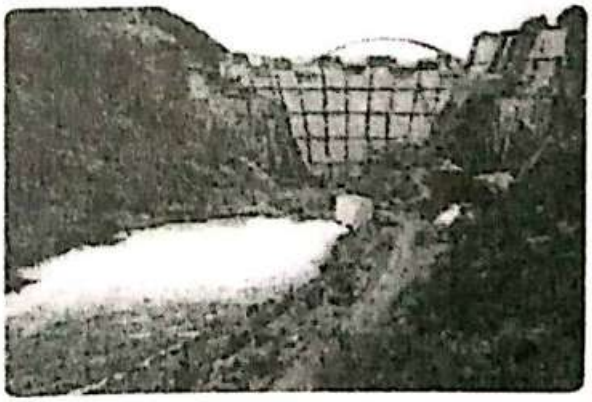
Examples are

(a) solar energy
(b) wind energy
(c) water energy or hydroelectric power
(d) Biomass
(e) Tidal power
(f) Biofuels

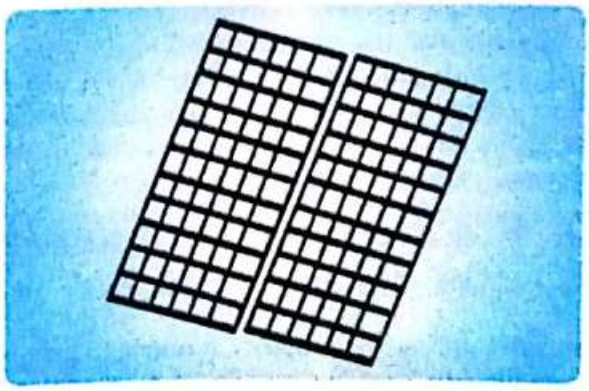
Non-Renewable Energy Resources are those that cannot be replenished as they are used up.

Examples are

(a) Nuclear Energy from radioactive materials
(b) Petroleum and Natural gas
(c) Coal and wood
(d) such other fuel that are classified as fossil fuel.

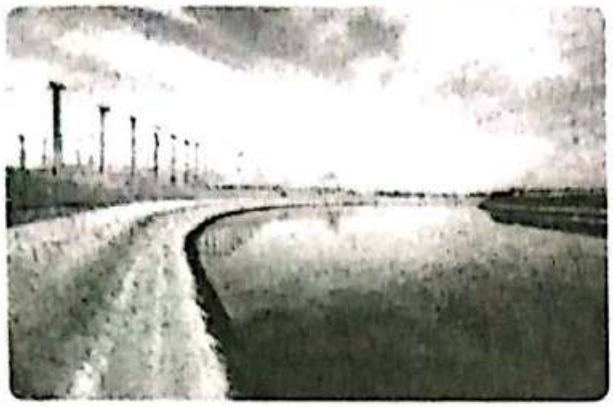


(a) Hydroelectric power source



(b) Solar plate

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(c) Windmills

Fig. 16.1 Renewable Energy

16.3 Uses of Energy

The use of energy has been a key in the development of the human society by helping it to control and adapt to the environment. Managing the use of energy is inevitable in any functional society. In most countries, the development of energy resources has become essential in these vital areas of use such as

(i) Agriculture – irrigation etc
(ii) all forms of transportation
(iii) waste collection and recycling
(iv) information technology
(v) communication
(vi) electricity supply to homes and running industrial machinery
(vii) for heating or cooling of homes
(viii) for operating domestic equipment such as refrigerators, freezers, grinding machines, lighting our homes and streets etc.

Energy is of extreme importance in the development of any society. It is the engine that makes the society to function. Developed countries are those whose energy resources are being put into efficient and maximum use in improving the social and economic life of their society.

The poor state of the electricity situation in Nigeria has been impacting very negatively on the life of the people, paralysing industrial and manufacturing machinery, and stultifying economic growth of the country.

16.4 Environmental Impact of Energy Use:

Utilization of energy resources in running factories, transportation, heating, cooling and lighting our homes and offices, etc, has a great effect on the environment. Burning of fossil fuels produces available and instantaneous supply of electricity for industry or power to move vehicles. It however, also, generates air pollutants including carbon dioxide, sulphur dioxide and nitrogen oxides etc. If you live near a factory belching out black smoke from chimneys or you are following closely behind a smoky vehicle, you will readily appreciate the amount of black soot that is readily poured into the surrounding environment, polluting the entire area and beyond. Burning fossil fuel for electricity generation also releases trace metals such as chromium, beryllium, cadmium, cooper, mercury, nickel, silver, etc into the environment. These act as pollutants.

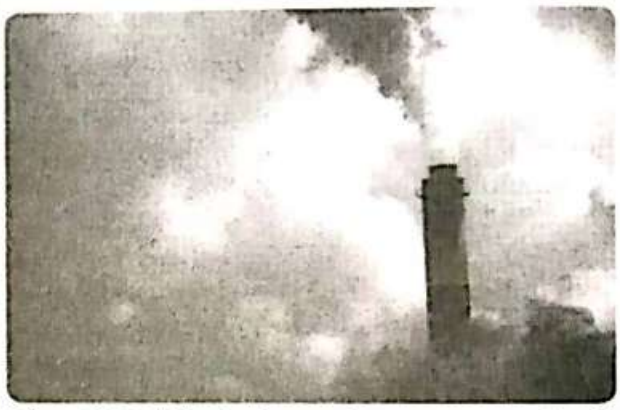


Fig. 16.2 Much dependence on crude oil causes air pollution

Carbon dioxide is an important **greenhouse gas**, which is thought to be responsible for some fraction of the rapid increase in global warming seen especially in increasing temperature records observed all over the world including Nigeria.

There are known global implications of the various forms of pollution being introduced into the atmosphere. These include:

a) **Acid Rain** – This results when gaseous emissions of sulphur oxide, nitrogen oxide etc come into contact with water vapour and sunlight and are chemically converted to strong acidic compounds such as sulphuric acid, nitric acid etc, and are carried to the earth by rain drops, snow flakes, fog or dew. Environmental consequences of acid rain include considerable damage to building and monuments in highly industrialised areas and their environs. Acid rain also affects crops adversely and cause skin problems.

b) **Greenhouse Effect** is the trapping of the sun's radiation and heat in the lower atmosphere of the earth or any other planet caused by an increase in carbon dioxide which is more transparent to solar radiation than to the reflected

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radiation from the earth. It is the process by which absorption and emission of infra-red radiation of the sun by such gases as carbon dioxide in the atmosphere, warm the earth's lower atmosphere and surface. The problem of the **greenhouse effect** has led to increase of the earth's average temperature and this has led to increase in the melting of polar ice caps. This in turn is leading to increase in sea levels resulting in catastrophic flooding of the coastal areas.

c) **Global Warming** Global warming is simply the increase in the temperature of the earth's atmosphere caused when certain gases such as carbon dioxide, smokes, etc trap the sun's heat. Global warming is currently causing havoc and disruption on the earth's weather and climate. This increase is as a result of the increasing concentration of **greenhouse** gases such as carbon dioxide methane, etc which result from burning of **fossil fuel** and **deforestation**. An increase in global temperature will cause sea level to rise and will change the amount of and pattern of precipitation or rainfall and expansion of subtropical deserts, retreat of glaciers etc; also it could cause changes in the frequency and intensity of extreme weather events, and changes in agricultural yields. These events are already being observed around the world including Nigeria.

d) **Oil Spillage** This is a release of liquid petroleum hydrocarbon into the environment due to human activity and is a form of pollution. Oil spills include release of crude oil from tankers, offshore platform drilling rigs and oil wells as well as spills of refined petroleum products, e.g. petrol, diesel and their by-products, and heavier fuels used by large ships.

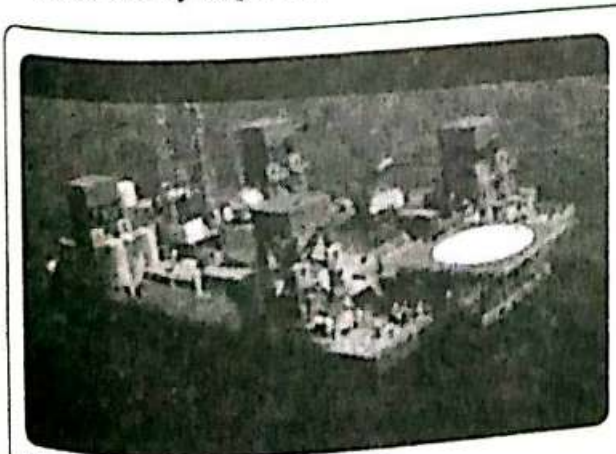


Fig. 16.3 Off-shore oil rig

Oil spillage has destroyed both human and aquatic lives and impairs birds flight abilities and kills off birds and fishes. Oil spillage also destroys farmland and pollutes rivers and streams, poisoning the drinking water of villages in the oil producing areas.

16.5 Energy Crisis

An energy crisis is any great bottlenecks (e.g. price rise, shortage/scarcity, decisive moments) in the supply of energy resources to an economy. We are all familiar with sudden petrol scarcity or price hike in Nigeria and the attendant upheavals with fight and struggles at the petrol station. Energy or energy resources are used to power or move vehicles, run industrial machinery, light up our cities, operate gargets, and equipment in the homes etc. With increasing population growth and industrialisation, there has been an enormous demand on energy supply. Supply of energy is therefore far less than the actual demand.

Energy crisis can occur due to shortage in supply or increasing demand, monopoly manipulation of market, industrial action by unions (e.g. strike by tanker drivers), bottlenecks at refineries and port facilities, vandalisation of pipelines, military coup, youth restiveness, political decisions, power outages, financial mismanagement of resources, political upheavals etc.

Summary

1. The world's energy resources include, fossil fuel (Petroleum, Natural gas), Solar energy, Nuclear energy, geothermal energy, wind energy, water energy, Biofuels, hydroelectricity.
2. Renewable energy is energy resource that is replenishable. Non-renewable energy is energy resource that cannot be replenished or renewed after use or depletion.
3. Environmentally friendly energy resources are those that do not pollute the environment. Examples are Solar Energy, Nuclear energy, Wind energy etc Hazardous energy resources include fossil fuels, petroleum, coal, wood etc.
4. Energy resources are used for transportation, electricity supply, communication, Agriculture, information technology running industrial machinery, domestic use etc.
5. The use of energy impacts on the environment through the greenhouse effect, global warming, depletion of the ozone layer, oil spillage, acid rain etc.

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Exercise 16

1. Discuss the various energy resources available (a) to the whole world (b) to your country
2. Which of the available energy resources are environmentally friendly and which are hazardous to the environment.
3. Distinguish between **renewable** and **non-renewable energy resources**. Give two examples of each type.
4. Explain the impact of the use of available resources on the environment.
5. Write short notes on
   (i) Greenhouse effect
   (ii) Global warming
   (iii) Acid rain
   (iv) Oil spillage
   (v) Energy crisis
6. Outline the various uses of the energy resources of the world and the effect of such uses on the environment.
7. What are energy crisis? How is it experienced in your country from time to time?

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ANSWERS

BOOK 1

Exercise 1

1 A, C
2 A 0.1 cm B 0.01 cm
C 0.001 cm D 0.1 sec
3 (1, 2, 3)
4 a. metre rule
b. vernier callipers or micrometer screw guage
c. Micrometer screw guage
d. Micrometer screw guage
5 a. Metre rule for length, micrometer screw guage for radius
b. radius
6 3.1536×10^7 sec; 8.64×10^4 sec; 2.6784×10^6 sec;
9 E
10 C
11 E
12 D
13 C
14 E
15 $kg^{-1}m^4s^{-1}$
16 B
17 C
18 D
19 $kgm^{-1}s^{-1}$

Exercise 2

2 106 $kmhr^{-1}$, 64 $kmhr^{-1}$, 94 $kmhr^{-1}$
5 66.7 $kmhr^{-1}$, 66.7 $kmhr^{-1}$ due east
6 8.3 hrs
8 49.38 $kmhr^{-1}$
9 2 hr
10 B
11 B
12 B
13 A
15 20 ms^{-1}; 2 ms^{-2}
19 A
20 37.5 ms^{-1}, 2.25 ms^{-1}, 750 m.
21 E
22 D
23 B
25 Examples of rotational motion:
(1) Motion of a windmill
(2) Motion of a turn table or disc.
(3) Rotation of the earth about its axis.
(4) Movement of the blade of an electric fan etc.
Examples of linear motion:
(1) An athlete running on a straight track.
(2) A car moving on a straight road.
(3) A ball rolling on a level ground etc.
(e) (i) net force = $F - F_1 = ma = 10$ N.
(ii) $F = ma + F_1 = 10 + 5 = 15.0$ N
(iii) $\mu = \frac{F}{R} = \frac{F}{mg} = \frac{15}{5 \times 10} = 0.3$
26 A
27 C

Exercise 3

1 E
2 No work
3 B
6 80 J
7 C
8 4800 J
9 A
10 C
11 225 J
12 75 W
13 20 J, PE converted to KE
15 3.33 m
16 12.5 J
17 2000 J
18 300 J in each case.
19 6.8×10^3 times

Exercise 4

1 A
2 A
3 C
4 C
5 D
6 C
7 D
8 E
9 D

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10 1.0 m
13 3.6×10^{-4} m
20 445.05 kg
21 a. 1.335×10^{4} kgm^{-3}
b. 1.3546×10^{4} kgm^{-3}
22 $A = 7.5 \times 10^{-5}$ m, $B = 60$°C,
$C = 20.8 \times 10^{-6}$ K, $D = 5 \times 10^{3}$ mm^{3},
$E = 20$ K
23 70.46 m
24 C
25 16.6°C

Exercise 5

2 B
3 A
4 D
5 C
6 A
7 C
8 D

Exercise 7

1 C
2 III
3 E
4 E
5 B
6 D
12 11 Ω, 0.493 Ω
13 (i) 8.3 Ω (ii) 9.28
(iii) 6.03 Ω (iv) 7.53 Ω
14 (i) 6.67×10^{-3} A, 0.027 W
(ii) 0.012 Ω, 0.3 W
(iii) 160 V, 6400 W
(iv) 0.25 A, 0.5 W
(v) 2.27 A, 96.8 Ω
(vi) 200 V, 400 Ω
15 0.454 A, 484 Ω
16 N8.33

Exercise 8

1 D
2 D
3 A

Exercise 9

1 B
2 E
3 B
4 A
5 C
6 66.67 N, 7.5 cm,
elastic limit is less than 100 N
7 44.27 ms^{-1}
8 1.71 J
10 $F = \frac{ALE}{e}$
13 2.84

Exercise 11

1 C
2 B
3 B
4 D
5 B
6 C
7 A

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BOOK 2

Exercise 1

1 A
2 B
3 E
4 E
5 E
6 A
8 a. 17 N b. 3N
c. 14.8 N; 24° to 10 N
d. 4.18; 35° to 10 N
e. 12.21; 35° to 10 N
9 a. 6.77, 12.8°
b. 5.0, 36.87°
c. 3.61, 46.03°
10 96.42 N, 114.91 N
11 4.12 N, N 14° W
12 130.54 N, 83.88 N
13 25 N, 43.3 N
14 $X = 6.97$, $Y = -9.29$, $R = 11.61$, $\theta = 36.88$ East of South
15 41.8° West of North; 1.12 hr
16 102.86 m, 129 s.
17 $K = 8000\ Nm^{-1}$, $F = 400$ N, $a = 200\ ms^{-2}$, $V = 15.7\ ms^{-1}$
18 Wtan 60°
19 (i) 3gsin 30 = 15 N
(ii) R = mgcos 30 = 30 cos θ° = 25.9 N

Exercise 2

1 B
2 C
3 D
4 B
5 A
6 B
7 D
8 B
9 E
10 C
11 $21.79\ ms^{-1}$
12 $2.828\ kmhr^{-1}$
13 812.2 km
15 $4.5\ ms^{-1}$
16 t = 10s, d = 200m
17 Draw a graph of s vs t^2, slope = $\frac{1}{2a}$;
19 $3.5\ kms^{-2}$, 128.5 km
20 4500 cm; 1500 cm
21 a. 5×10^4 m b. 100 s
c. $800\ ms^{-1}$ and $500\ ms^{-1}$
d. 50 s and 150 s. At t = 50 s shell is at height of 37.5 km but ascending; at t = 150 s shell is at the same height of 37.5 km but descending.
22 450 m
23 20 m
24 105 m, 7.47 sec.
25 b. (i) $340\ ms^{-1}$ (ii) $17\ ms^{-1}$
(iii) 9400 m (iv) $188\ ms^{-1}$
26 225 m, 5 secs, 11.71 s, 1013.22 m
27 1 sec; $-10\ ms^{-1}$

Exercise 3

1 B
2 B
3 B
4 A
5 C
6 $U_x = 1732\ ms^{-1}$, $U_y = 10^3\ ms^{-1}$, $H = 50$ km, $R = 346.4$ km
7 8.94 s, 63.43°, 200 m
8 26.5°
9 4.24 sec, 212 m, 42,42 ms^{-1}, $\theta = 40.3°$
10 9 sec, 701.5 m, 30°
11 0.222 m, 0.087 m, 0.196 m, 4.9 m
12 20 sec, 3464 m.
13 20 m
14 $\theta = 30°$
15 C
16 t = 10s, d = 2000m
17 $\sqrt{2}$ mv or 1.414 mv
19 7.5 s

Exercise 4

1 C
2 B
3 B
4 C
5 C
6 C
7 B
8 D
9 D
10 A
11 C
12 C
13 A
15 21.79 N, 23.43° N of E; 21.79 N, 23.42° S of W
16 4.04 N, 8.08 N
17 $P = 36.25$ N, $Q = 58.75$ N
18 $P = 160$ N, $P_x = 240$ N
19 $N = 80$ g; 33 cm from zero end of the rule
21 125 N, 95 N
23 1.5 m from the boy or at the centre of the pole
24 a. 8 N acting to the right at 30 cm above C
b. 100 N up at 50 cm
25 (i) $R_1 = 175$ N, $R_2 = 125$ N;
(ii) 0.79 m to the left
(iii) 100 N, 200 N
26 40 N, 70 N
27 (i) 150 Nm anticlockwise
(ii) By a couple of moment 150 Nm clockwise
(iii) 1.25 m

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(iv) +50 Nm (if we take anticlockwise moment as +ve and clockwise moment as –ve).

28 $T_1 = 84.07$ N, $T_2 = 44.16$ N

30 5000 N, 4.8×10^3 N

31 (i) 30 N, 26 N, 28.67 N, 26.8 N
(ii) 4.635×10^4 kg
(iii) 2240 N

32 7.1 cm

33 2.499 cm^3, 9.20 gcm^{-3}

34 a. 1.5 b. 0.92 c. 104 g

35 b. $\rho_s = \frac{a\rho_l}{b-c}$
c. 318 cm^3, 682 cm^3.

36 2.06 x $10^6 N^{2\,cm3}$

37 $0.04m^2$

Exercise 5

1 A
2 A
3 C
4 C
5 A
6 A
7 B

10 0.01 sec; $4\times10^4\pi^2$ ms^{-2}; 20π ms^{-1}

11

12 5 ms^{-1}

13 $K = 5\pi^2$, $E = 0.0225$ Joule

15 a. 2.25 m b. 1.33 sec

16 a. 9470 cms^{-2}, 377 cms^{-1}
b. 5680 cms^{-2}, 301 cms^{-1}

17 22.4 kg

18 6π $rads^{-1}$, 120π cms^{-1}

20 a. 19.6 Nm^{-1} b. 8.08 ms^{-1}
c. 0.1 m d. 0.808 ms^{-1}
e. 6.53 ms^{-2} f. 0.78 sec, 1.29 Hz

21 $A_1 = A_2\sqrt{8}$.

22 9.87ms^{-1}; 3s; 45m

Exercise 6

1 C
2 B
3 B
4 C
5 B

8 4 $kgms^{-1}$, 363.6 N

10 3.33 ms^{-1}

11 16.32 $kgms^{-1}$

12 96 kg, 0, 48 kg

13 666.67 ms^{-1}

14 7.35×10^{22} $kgms^{-1}$, 2.8×10^8 $kgms^{-1}$, 12.5×10^7 $kgms^{-1}$

15 3.33 ms^{-2}, up.

16 2ms^{-1}; 7.5 x 10^3J; 0Ns

Exercise 7

1 E
2 B
3 D
4 B
5 D
6 B
7 B
8 C
9 7500 J

10 (i) 1256.64 (ii) 754.3
(iii) 11.94 N (iv) 3000 J

11 (i) 200 N (ii) 3
(iii) 1.5 (iv) 600 J

12 5, 8, 62.5%

13 100 N

14 a. 2.0 m b. 120 J c. 20 J

15 83.3%.

16 6.28; 157; 6.4N

Exercise 8

1 D
2 C
3 E
4 E
5 C
6 D

8 –254.9°C, 10.56 ohm

13 63.4°C

15 35°C.

17 16.6°C

Exercise 9

1 D
2 B
3 B
4 B
5 C
6 A
7 A
8 B
9 C
10 A
11 0.033 kg
12 6.7°C

13 (i) 4200 $Jkg^{-1}k^{-1}$ (ii) 400 $Jkg^{-1}k^{-1}$
(iii) 2100 $Jkg^{-1}k^{-1}$

14 a. 5040 J b. 42 W c. 265
d. 2390 J e. 265 Jk^{-1}
f. 481 $Jkg^{-1}k^{-1}$

17 10.29 g

20 3.03×10^6 J

21 2.29×10^{-2} kg

23 55°C

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26 a. The statement means that 3.4×10^5 J of (heat) energy is absorbed (or given out) when 1 kg mass of ice changes to water (or vice versa) at constant temperature.

27 35.6 K

28 h = 2.86 m, V = 2.525×10^6 J

29 97.5°C

30 4.5×10^4 J, 1.19°CS^{-1}

Exercise 10

1 B
2 B
3 A
4 C
5 B
6 C
7 C
8 A
9 E
10 23°C
11 960 mmHg
12 (i) 23.95 cm (ii) 29.84 cm
13 37.5 cm
14 1366.3 cm^3
18 68 litres
19 1.81×10^5 Nm^{-2}
20 47 per min
21 1.04×10^5 pa
23 594 mmHg

Exercise 11

1 B
2 D
3 C
4 D
5 E
6 C
7 D
8 A
9 D
17 1.5 cm, 1.0 cm, 35.2°
19 4 cm, 20 cms^{-1}, 5 Hz
20 0.2 mm, 2 cm, 0.01 sec, 200 cms^{-1}
22 0.64 Hz; 1.65 s; 1.34 m/s
23 8.0 Hz

Exercise 12

1 C
2 D
3 B
4 D
5 E
6 B
7 E
9 8.8 m
10 10 cm, 5 cm towards the mirror
11 u = 80 cm, v = 26.7 cm
12 Virtual, erect, 9 cm behind mirror, 4.2 cm high
16 5.3 m.

Exercise 13

1 D
2 B
3 A
4 D
5 D
6 C
7 B
8 B
9 D
10 C
11 a. 31°41' b. 1.84 cm
14 7.5 cm, 0.5 cm
16 15 cm on same side as object, h = 2 cm, M = 2, virtual
17 60 cm, 20 cm
18 (i) 0.202 m, 0.01, real
(ii) a. +0.5, +2, +20 D
b. −2.5, −10, 125 D
19 (i) 0.3 m, 0.6 m (ii) 0.1 m, 0.2 m
(iii) 0.12 m, 0.3 m (iv) 0.15 m, 0.3 m
(v) 0.143 m, 0.2 m (vi) 0.33 m, 0.2 m
20 b. 0.75 m
c. 10 cm, 30 cm from object
21 (iii) Face AC.
22 4 cm

Exercise 14

1 A
2 B
3 C
4 C
5 D
6 B
7 D
8 C
11 10.17 cm
12 c. Converging lens, f = 37.5 cm, Power = +2.67 D
d. f = −50 cm, P = −2 D, diverging lengs
13 9 cm
14 24 cm from eye piece, 20 mm long
15 c. 8.33 cm.

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Exercise 15

1 D
2 C
3 C
4 C
5 E
6 E
7 C
8 B and D
9 A
10 A
11 A
12 B
13 D
15 680 Hz
17 333 ms^{-1}, 142 m
18 (i) 333.33 ms^{-1}
(ii) 666.67 m
19 5 revs/s, 300 Hz
20 a. 340 ms^{-1}, 95.8 cm
b. 14.6 cm, 44.5 cm
22 a. $f_0 = 128$ Hz, $f_1 = 384$ Hz, $f_2 = 640$ Hz
c. $f_0 = 256$ Hz, $f_1 = 512$ Hz, $f_2 = 768$Hz
23 72 N
24 a. 1320 b. 0.72
c. 1190 d. 340 e. 1.25
27 0.17 m; 0.51 m; 0.85 m
28 5.4 ms^{-1}
31 (i) 4.2×10^{-4}s
(ii) 1.2×10^{-3} ms^{-1}
(iii) 18.2 ms^{-2}
(iv) 5.0×10^{-11} N

Exercise 16

1 A
2 A
3 E
4 C
5 B
6 B
8 2.5×10^5 Nm^{-2}, 6.18×10^6 Nm^{-2}
10 500 Nm^{-2}, 2500 N
11 0.75

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BOOK 3

Exercise 1

1 D
2 E
3 B
6 1.5 MHz.

Exercise 2

3 3.33×10^{-6} N
4 2.00×10^{20} N
5 6.0×10^{24} kg
6 4.17×10^{23} N
7 6.4×10^{3} km
8 0.8 W_e, 7.8 ms^{-2}
9 (i) 3.48×10^{-47} N
(ii) 1.38×10^{-67} N
(iii) 1.16×10^{-65} N
10 9.8 Nkg^{-1}

Exercise 3

1 D
2 1 F
3 D
4 B
7 0.576 N
9 4×10^{6} NC^{-1}
10 2 NC^{-1} upwards; 3.2×10^{-19} N downwards
11 1800 V
13 6 J
16 4 μf; charge on 8 μf = charge of 12 μf capacitor; 45V
17 107 v/m; 1.6×10^{-12} N; 1.7×10^{18} m/s^2

Exercise 4

1 B
2 C
3 A
4 D
5 B
12 3 Ω, 4 Ω, 1.5 V, 2 Ω
13 $I_3 = 1.04$ A, $I_4 = 0.59$, $I_5 = 0.47$, $V_3 = 3.12$ V, $V_4 = V_5 = 2.36$ V
14 9.0 m
14 3.5 Ω, 1.2 A, 1.8 V, 0.6 A
16 4 V, 1.5 Ω
17 38 Ω
18 400 Ω, 0.1 Ω
19 0.01 Ω, 2 cm
30 6 Ω, 0.83 A, 0.83 V, 1.68 V, 1.01 V.

Exercise 5

1 C
2 A
3 D
4 E
5 E
6 C
9 1.75 A
10 14.54 hrs
11 0.15 A
12 33.33 mins
13 4.2 A

Exercise 6

1 D
2 C
3 D
4 B
5 B
6 C
7 B
8 A
9 B
10 A
19 1.5 N
20 (i) 0
(ii) 4.8×10^{-11} N
(iii) 2.4 N
21 (i) 34° West of North
(ii) 4° East of North
22 4×10^{-12} N
23 9.43×10^{-2} m

Exercise 7

1 E
2 E
3 C
4 C
5 D
6 A
7 B
8 B
9 A
10 A
15 150 turns
16 3.6 A
17 100 V, 2.5 A, 6 kW
22 5 V a.c.
23 20 : 1, 14 mA.

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Exercise 8

1 B
2 B
3 C
4 A
5 C
6 C
7 C
8 A
9 B
10 $X_c = 636.4\ \Omega$, $Z = 1185.3\ \Omega$, $I = 0.076$ A
$V_c = 48.37$ V, Av. power = 5.78 W
11 10.6 A
12 (i) 245 V
(ii) 130 μF
(iii) 15.9 mH
15 633.8 Ω, 80.9°
17 3.18 Hz, 127 Hz
18 36 W
20 385 V, 391 V
21 Amplitude, 56.6 A.
22 $V_C = 244.9$ V
$X_C = 129$; C = 129.97 μf
L = 0.0159 H
23 89.1 Ω, 1.12 A, 89.1 V
24 50 V, 5 A, 70.7sin1000t

Exercise 9

1 C
2 D
3 A
4 E
5 A
6 A
7 *X* and Z are isotopes, 3 protons, 3 electrons, 4 neutrons.

Exercise 10

1 C
2 C
3 C
4 C
5 D
6 E
7 C
8 B
9 A
16 60 mins
17 0.112 mg.
19 $1.28 \times 10^{-4} S^{-1}$

Exercise 11

1 C
2 C
3 B
4 C
8 0.08 Hz
9 3.31×10^{-9} J
10 3.39 eV, 3.39 V
11 6.62×10^{-19} J
14 4.41×10^{-19} J, 2.76 eV; 1.83 eV
15 3.64×10^{-19} J, 2.27 eV; 0.49 eV.
18 6×10^{14} Hz; 3.96×10^{-19} J
0.92×10^{-19} J
19 1.20 x 107 Hz; 990 w
21 (i) 1.6×10^{8} ms^{-1}
(ii) 4.32×10^{-12} m
(d) 8.68×10^{7} ms^{-1}
23 $\dfrac{E_2 - E_1}{F_2 - F_1}$

Exercise 12

1 B and C
2 D
3 C
4 E
8 7.46×10^{-7} m; 2.67×10^{-19} *J*; 3.05×10^{-18} *J*

Exercise 13

1 B and C
2 C and E
3 B and E.

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